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Powder metallurgy of titanium – past, present, and future

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ABSTRACT

Powder metallurgy (PM) of titanium is a potentially cost-effective alternative to conventional wrought titanium. This article examines both traditional and emerging technologies, including the production of powder, and the sintering, microstructure, and mechanical properties of PM Ti. The production methods of powder are classified into two categories: (1) powder that is produced as the product of extractive metallurgy processes, and (2) powder that is made from Ti sponge, ingot, mill products, or scrap. A new hydrogen-assisted magnesium reduction (HAMR) process is also discussed. The mechanical properties of Ti-6Al-4V produced using various PM processes are analyzed based on their dependence on unique microstructural features, oxygen content, porosity, and grain size. In particular, the fatigue properties of PM Ti-6Al-4V are examined as functions of microstructure. A hydrogen-enabled approach for microstructural engineering that can be used to produce PM Ti with wrought-like microstructure and properties is also presented.

Abbreviations: AM: additive manufacturing; ARC: Albany Research Center; BE: blended elemental; BUS: broken-up structure; CCGA: close-coupled gas atomisation; CHIP: CIP-sinter-HIP; CIP: cold isostatic pressing; CP-Ti: commercially pure Ti; DRTS: direct reduction of Ti-slag; CSIR: Council for Scientific and Industrial Research (South Africa); CSIRO: Commonwealth Scientific and Industrial Research Organization (Australia); EIGA: electrode induction gas atomisation; EMR: electronically mediated reduction; FFC: Fray, Farthing, and Chen; GA: gas atomisation; GIF: gaseous isostatic forging; GSD: granulation-sintering-deoxygenation; HAMR: hydrogen-assisted magnesium reduction; HDH: hydride-dehydride; HIP: hot isostatic pressing; HSPT: hydrogen sintering and phase transformation; MA: master alloy; MER: Materials & Electrochemical Research Corporation (US); MHR: metal hydride reduction; MIM: metal injection molding; OM: optical microscope; OS: Ono and Suzuki; PA: pre-alloyed; P/C: performance to cost ratio; PIF: pneumatic isostatic forging; PM: powder metallurgy; PREP: plasma rotating electrode process; PP: post-processing; PS: press and sinter; QIT: Quebec Iron & Titane, Inc. (Canada); SEM: scanning electron microscope; SPS: spark plasma sintering; SOM: solid oxide membrane; THP: thermohydrogen processing; TMP: thermomechanical processing; UFG: ultrafine grain; UGS: upgraded titanium slag; UTS: ultimate tensile strength; USTB: University of Science and Technology Beijing (China); VA: vacuum atomisation; VHP: vacuum hot pressing; WP: wrought process; YS: yield strength

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Nomenclature

α_{GB}	grain boundary α
α_p	primary α
$\sigma_{y,0.2}$	0.2% offset yield strength
E^o	standard electrode potential
%EL	per cent, elongation (tensile test)
%RA	per cent, reduction area (tensile test)
%TD	per cent of theoretical density

Introduction

As the ninth most abundant element in the earth's crust, and fourth among structural metals, titanium has been a fascinating metal to mankind for more

than half of a century. Titanium has three unique properties: it has a high specific strength and the same magnitude of strength as that of steel; it is the most biocompatible metal, making it a common selection for biomedical implants; and it is the most corrosion resistant common metal, making it an ideal choice for maritime applications. Particularly for our modern society and other automobile-dependent nations around the world, titanium would be an ideal structural metal to replace steel in vehicles, but there is one thing that stands in the way: cost. Although markets fluctuate, Ti in general is at least 20 times more expensive

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than carbon steel and approximately 4–5 times more expensive than stainless steel. The traditional production and manufacturing of Ti follows the same general procedures as steel, including primary metal production, melting and pouring of alloy ingots, forging and rolling to produce mill products, and manufacturing of components or structures from the mill products. Owing to its inherent affinity to oxygen and the associated high cost involved in the fabrication of components using Ti, the buy-to-fly (or yield) ratio, using aerospace industry terminology, is typically 12:1 and often more than 20:1 for parts fabricated from mill products, i.e. plates or bar stocks [1,2]. The buy-to-fly ratio for the F-22 fighter jet is 12.2:1, meaning 82% of the titanium became scrap metal [2]. The unfortunate and unavoidable high cost of Ti is exacerbated in almost every step of manufacturing from mineral to products, as illustrated in Figure 1, and the high cost of parts fabricated using the conventional melt-wrought approach is the primary motivation for research and development of powder metallurgy (PM) Ti [3,4].

Given its attractive properties and current cost, it is understandable that reducing the cost of Ti production and manufacturing has been a continuing and primary motivation for Ti research. Powder metallurgy, being a near-net-shape manufacturing technology, has been applied to Ti research and development for more than four decades. A comparison of typical PM titanium processes with wrought processes is given in Figure 1. Powder metallurgy includes the production of powders, compaction and shaping of powders, and sintering, as well as post sintering processes to fabricate ready-to-use components. Because of its capacity to streamline processing and reduce the amount of scrap metal produced, the potential of the PM approach for Ti manufacturing is truly substantial. However, a sobering fact is that the commercial successes of PM Ti are still very limited to date, despite decades of research and development and the investment of millions of dollars. Traditionally, wrought Ti is usually chosen over PM Ti because PM Ti parts

are either unsatisfactory in view of their mechanical properties and performance, and/or due to high cost.

The first cost factor concerning PM titanium is that of powders. Frequently, the cost of powders is cited as a key hurdle that prevents the development of PM Ti materials and products. For instance, the high cost of powders is often given as a limiting hurdle to the development of Ti products using additive manufacturing (AM). Although less expensive sources of Ti powder are possible, none of them are currently available in the market. Powders can be produced from the by-products of Ti sponge production, i.e. sponge fines. Powders can also be made by hydrogenating the Ti sponge, ingot, or mill product, crushing and/or grinding it to the desired powder size, and then dehydrogenating the material to produce what is known as hydride-dehydride (HDH) Ti powder [5,6]. HDH powder can also be made by hydrogenating and dehydrogenating Ti scrap metals.

Ti alloy powders are typically made by a variety of highly advanced atomising techniques, all of which consist of first making metal alloy, melting the material, and then atomising the molten metal by different techniques. Atomised alloy powders in general have significantly higher costs than HDH powder [7,8]. Additionally, a large number of different technologies have been developed over the recent decades to produce Ti metal, often in the form of powder, including widely known techniques such as the Armstrong [9] and the Fray, Farthing, and Chen (FFC) processes [10]. This review will categorise different techniques and analyse their characteristics.

Although the cost of powder is discussed as the main issue, the cost of manufacturing products from the powders is actually more challenging. There are generally two kinds of approaches for making PM titanium products: the blended elemental (BE) method and the pre-alloyed (PA) method [3,11]. The BE method refers to the pressing and sintering of BE powders. Sintering of the compacted powder is usually carried out in vacuum. The PA method refers to sintering PA powders, which are typically made using the gas atomisation (GA) or the

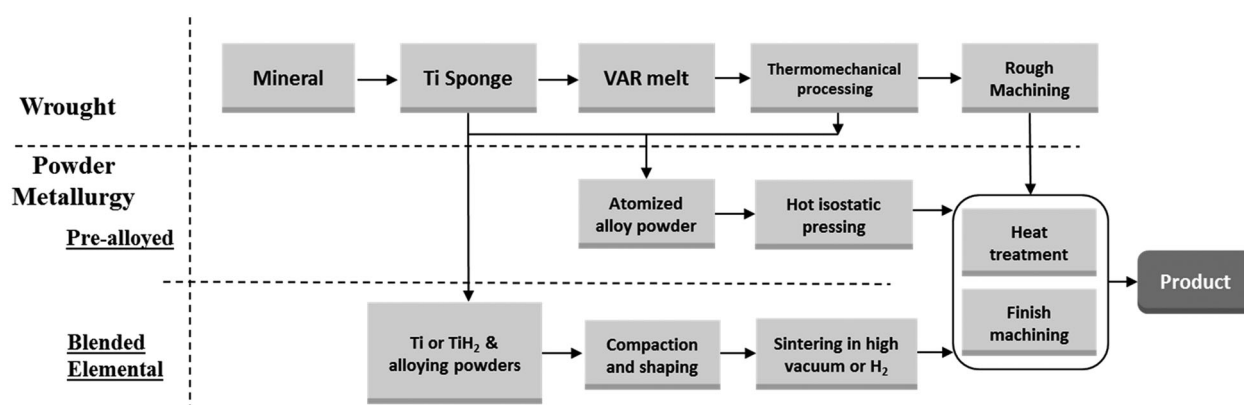


Figure 1. Flow chart of conventional melt-wrought and powder metallurgy approaches for manufacturing Ti products.

plasma rotating electrode process (PREP). Since PA powders have high hardness, and hence poor press-ability if compacted using a conventional uni-axial cold pressing method, parts are usually made by using pressure-assisted consolidation techniques, such as hot isostatic pressing (HIP). Although PA products generally have better mechanical properties than BE products, the costs of PA products are significantly higher; both the process for making PA powders and the process of hot consolidation (e.g. HIP) are very expensive. Therefore, BE is still the preferred cost-effective approach. However, the mechanical properties of BE parts are often unsatisfactory compared to those of equivalent wrought Ti parts, especially the fatigue properties, due to residual porosity, relatively high oxygen content, and relatively coarse microstructures after sintering [12].

In order to improve BE mechanical properties, most of the efforts reported in the literature have been devoted to developing pore-free BE parts. One simple approach to remove residual porosity is to use post-sintering high pressure processes, such as HIP, which can increase the density to greater than 99.8% [3]. Thermo-mechanical processes (TMP), i.e. hot/cold working plus heat treatments may also be used to refine grain size and achieve desired phase composition and morphology. The microstructural evolution of PM Ti materials during TMP follows the same physical metallurgy principles of melt-wrought materials. The trade-off of using post-sintering high pressure processes is, of course, an increase in cost, which partially or completely negates the cost-benefits normally associated with the BE method.

With regard to mechanical properties, a key concern of PM Ti materials is their fatigue resistance [12], and again, the primary factor that determines the mechanical properties of Ti alloys is the residual porosity. The fatigue endurance limit is particularly sensitive to porosity, which causes not only low fatigue strength, but also a wide scatter of fatigue data. The next key factor affecting mechanical properties is the oxygen content. Higher oxygen content causes lower ductility and fracture toughness. An additional factor can be the microstructure. The as-sintered microstructure is generally coarse, and therefore undesired from the perspective of mechanical properties. The correlations of mechanical properties to the microstructure, including porosity and grain sizes, as well as oxygen content, will be reviewed in detail in the latter sections of this article.

In short, although PM of Ti has great potential as a low-cost alternative for Ti manufacturing, a number of issues still restrain the use of PM Ti and prevent its industrial acceptance. The most important mechanical property concerns are the fracture toughness and fatigue performance of PM Ti materials, which are often associated with the oxygen and impurity levels of PM materials that tend to be higher than those of wrought materials. It is also a reflection of the fact that PM Ti materials

generally have some level of residual porosity unless they are consolidated using high-pressure techniques, either during sintering or through post-sintering treatments.

In view of the different choices of processing approaches and corresponding microstructure and mechanical properties, combined with their associated costs, the challenge to PM Ti technology is not merely to achieve a property target, but rather to achieve a property target without increasing cost. High-performance Ti materials, by either wrought or PM processes, are available today at high costs. The research and development of PM Ti should focus on developing methodologies and processes to reduce cost without compromising mechanical properties. In other words, the challenge is to increase the performance to cost (P/C) ratio.

This article provides a comprehensive and critical review of PM titanium technology. The earlier sections of the paper categorise and analyse different powder production technologies, and latter sections continue with the examination of compaction and sintering technologies and the dependence of mechanical properties on microstructural features that result from different processes. Based on the knowledge of processing technology and mechanical properties, we provide possible technical pathways to maximise the P/C ratios.

Titanium powders as products of extractive metallurgy processes

There are many methods for the production of Ti metal powder. These methods can be classified into two categories. In the first category, Ti powder is made as the product of extractive processes that produce primary metal by reducing titanium tetrachloride (TiCl_4) or titanium dioxide (TiO_2). In the second category, Ti powders are made from Ti sponge, Ti alloy ingots, Ti mill products or Ti scrap metals. For each of these two categories, there are a large number of techniques described in the literature; however, only a few have been commercially deployed. There are also a number of published reviews of production methods for titanium [3,7,8]. One of the more comprehensive reviews is provided in a publicly available, but not journal-archived publication by EHK Technologies [13]. As the first, and critical, step of powder metallurgy, this section categorises different techniques, and highlights the basic methodology of selected techniques in each category. Details of each technique are beyond the scope of this article, but they are referenced in the bibliography.

Processes for making titanium powder directly as products of extractive metallurgy processes include those of making Ti from TiCl_4 , purified TiO_2 , and/or upgraded titanium slag (UGS) with more than 90% TiO_2 content. UGS is the product of carbothermal

reduction of titanium ore such as ilmenite. Naturally occurring rutile and synthetic rutile also belong to this category of raw materials. These processes can be further classified into two groups: (1) thermochemical processes, and (2) electrochemical processes. Each of these two categories is discussed as follows.

Thermochemical processes

Thermochemical production of Ti involves the reduction of a precursor with a reductant. There are two main precursors: TiCl_4 and TiO_2 . TiCl_4 is a liquid at room temperature and atmospheric pressure, but it is easily vapourised at moderate temperatures. Using TiCl_4 as the precursor to produce Ti metal has the advantage that all of the undesired impurities in TiCl_4 can be removed by distillation, which enables the production of highly purified Ti metal. However, the purification processes to produce TiCl_4 , including the high-temperature chlorination process from TiO_2 -bearing minerals, involve a series of highly energy-intensive and costly processes [14]. Motivated by the aim to avoid the high-temperature chlorination process, there have been a number of investigations involving the use of commercial TiO_2 , titania slag, synthetic rutile, or UGS, as the precursor material [15–26]. Intuitively, TiO_2 would be a lower cost precursor than TiCl_4 if the high-temperature chlorination step could be avoided. However, the precursor is not the only factor that affects the cost of Ti. The differences in the inherent energies of the reduction of either TiO_2 or TiCl_4 , and the selection of reductant also affect the final cost of Ti metal.

Theoretically, free energy-based Ellingham diagrams for a metal's reaction with oxygen and chlorine (Figure 2(a,b)) are the basis for the selection of reductants. Based on the Ellingham diagrams [27], there are three possible reductants for the reduction of either

TiO_2 or TiCl_4 , namely Mg, Na and Ca. Based on thermodynamic feasibility, Figure 3 shows multiple potential thermochemical routes for the production of Ti metal from the precursors using Mg, Na, or Ca as reductants.

Thermochemical processes based on the reduction of TiCl_4

Processes based on the reduction of TiCl_4 using Mg.

Commercial production of Ti primary metal is usually carried out in either the Kroll or the Hunter processes. The Kroll process is dominant in industry today. The Kroll process relies on the reduction of TiCl_4 using magnesium, while the Hunter process uses sodium to reduce TiCl_4 . The product of traditional Kroll and/or Hunter processes is Ti sponge, which is the standard Ti primary metal that is commercially available today for melting and manufacturing.

In the Kroll process [28,29], as shown in Figure 4, Ti is produced by the reaction of TiCl_4 with Mg. TiCl_4 is produced by chlorination of Ti slag with the assistance of petroleum coke at 800–1000°C in a fluidised bed. However, impurity oxides in Ti slag are also chlorinated, and thus the crude TiCl_4 has to be further purified to produce refined TiCl_4 prior to Mg reduction.

Purified TiCl_4 is fed into the reactor and brought into contact with molten Mg. The Mg reduces TiCl_4 to Ti, and MgCl_2 forms. The Ti nucleates and Ti sponge grow from the bottom, as well as along inner walls of the reactor. As the reaction progresses, MgCl_2 and Mg vapour are recovered through an outlet port. The product of the reduction, which is Ti sponge with considerable MgCl_2 and residual Mg, must be distilled in vacuum (0.1–1 Pa) at a high temperature of 1000°C for days at a time to ensure the removal of residual Mg, MgCl_2 and other volatile species. For a 10 ton batch, the distillation may take 90 h, due to the difficulty of supplying heat to the interior of the

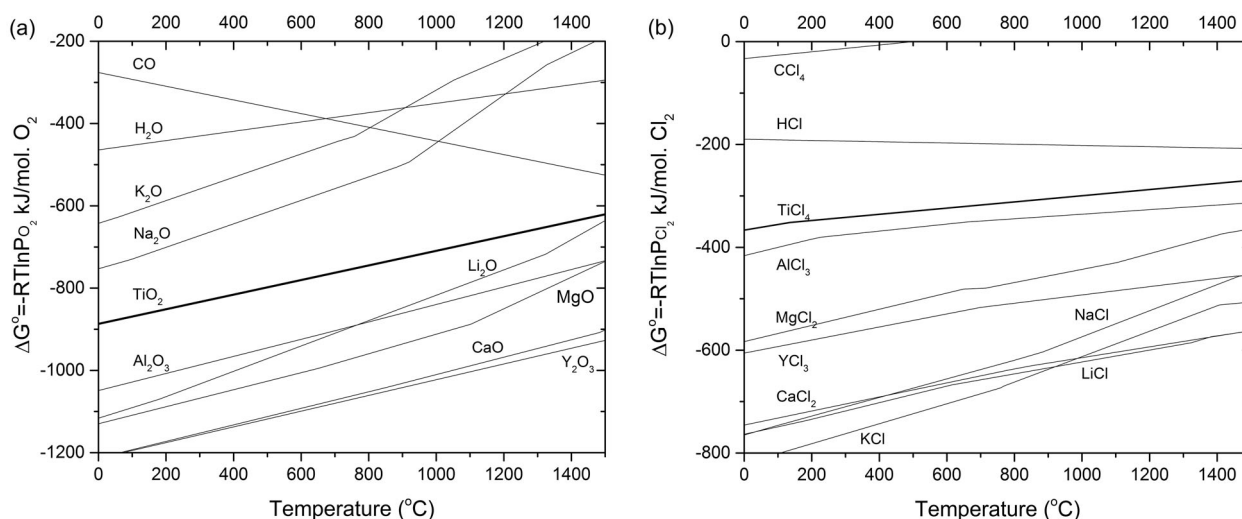


Figure 2. Ellingham diagrams: the formation of (a) oxides and (b) chlorides [27].

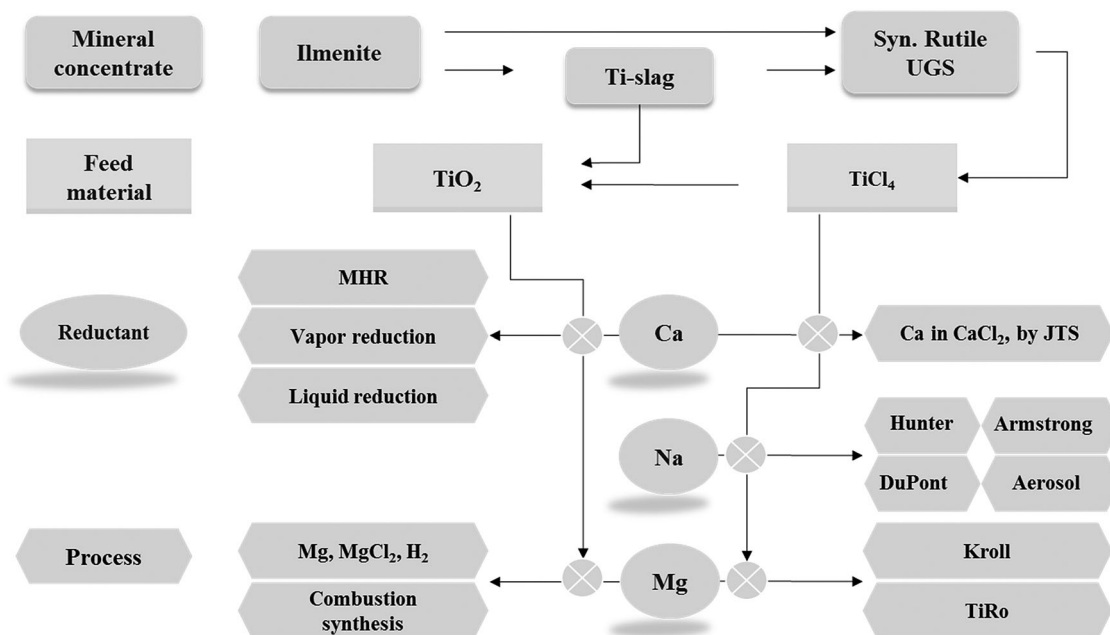


Figure 3. Thermochemical reduction routes to produce Ti metal through various reductants.

container for Mg and MgCl_2 evaporation [30]. Furthermore, the vacuum separation is followed by a long period of cooling. To date, the largest facilities can produce 10 ton of titanium per reaction container; however, one cycle takes more than 10 days, i.e. 1 ton of Ti per day. Based on one estimate, the energy consumption of the distillation can be nearly 70% of the total energy consumption of the entire Kroll process from upgraded Ti slag to sponge metal [31]. This is illuminating with regard to the cost of Ti sponge, and subsequently the cost of powders. It shows that the cost of purifying the metal is one of the key drivers of cost, in addition to the cost of the precursor and reductant.

As mentioned earlier, the majority of Ti primary metals used globally by industry today are produced using the Kroll method. Ti-sponge, however, is not Ti powder. It is not a ‘powder production method,’

although the Kroll process does produce what is classified as ‘sponge fines’ that may be used directly as powder. Metallic Ti sponge (Figure 5(a)) is the primary product of the Kroll process, but this material can also be crushed using mechanical means to produce powders. Ti-sponge is also used as the raw material for making Ti powder using other methods. For example, it is used as the starting material to make HDH powder, which will be discussed in the next section. Therefore, the Kroll process is the starting point for understanding several other powder production processes.

After nearly half of a century of commercial development and optimisation, the Kroll process is firmly established as the industry standard today. However, there have been continuous R&D efforts for decades to reduce the cost of the process. One of the approaches of these efforts has been to develop a continuous process using the same chemistry as that of the Kroll process. TiRo is a prominent example [32], and was developed and reported by the Commonwealth Scientific and Industrial Research Organization (CSIRO) in Australia. The process has two major steps: reduction of TiCl_4 in a fluidised bed by Mg powder, and vacuum distillation to remove the MgCl_2 byproduct and Mg.

In addition to the TiRo process, there are a few other processes that are chemically similar to the Kroll process [14,33–35], all of which attempt to develop a continuous process. Deura et al. reported a technique to produce Ti powder by injecting TiCl_4 gas into Mg through MgCl_2 molten salt in order to spatially separate the Ti product from Mg by molten salt [33]. During this process, the reaction takes place at the interface of the molten Mg and MgCl_2 .

Another continuous process is based on vapour-phase reduction to produce powder [35]. In this

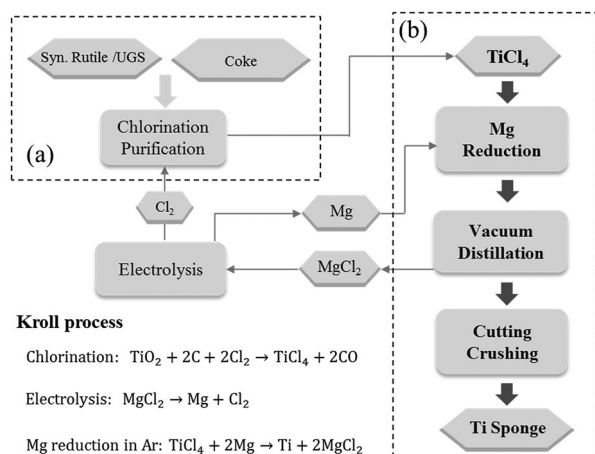


Figure 4. Illustration of the main processing steps of the Kroll process: (a) chlorination and (b) reduction of TiCl_4 .

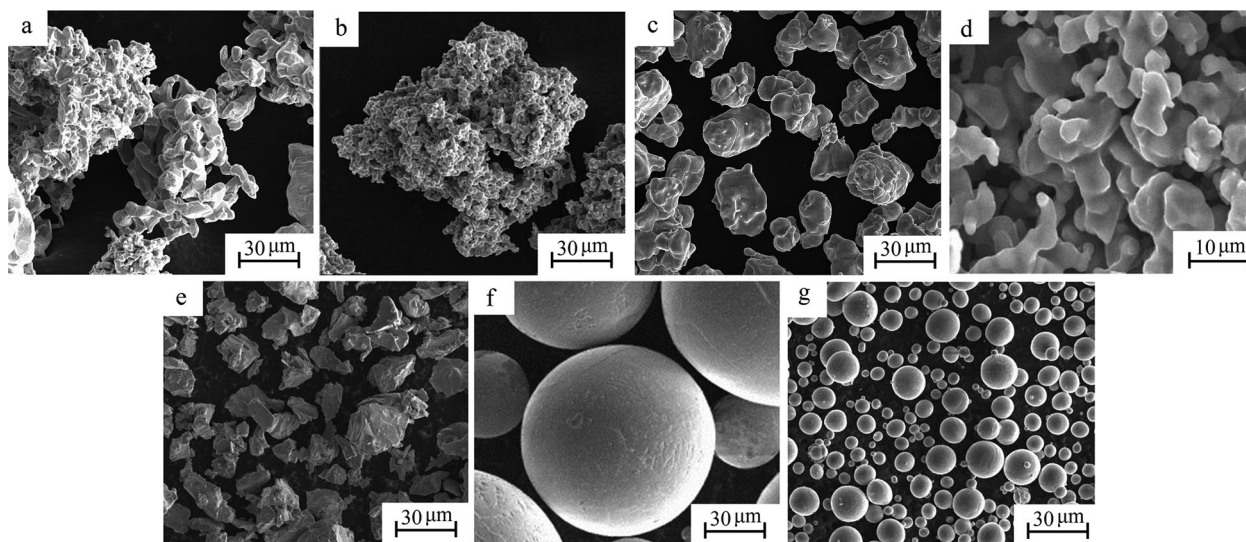


Figure 5. SEM micrographs of CP-Ti or Ti-6Al-4V powders by different processes: (a) Kroll (sponge fine), (b) Armstrong, (c) HAMR, (d) FFC [64], (e) HDH, (f) PREP and (g) plasma atomisation.

process, liquid TiCl_4 carried by argon gas reacts with magnesium vapour that is generated by vaporising Mg wire to form a mixture of Ti, MgCl_2 , and Mg powder. The product powder is removed from the gas stream by an electronic precipitator, and Ti powder product is further separated from Mg and MgCl_2 by either vacuum distillation or leaching. A drawback of this process is that the size of the particles is in the sub-micrometer range, which is too fine for capture and separation. The fine powder also picks up a significant amount of oxygen and nitrogen during subsequent processing.

Yet another continuous two-step process was developed by Van Vuuren et al. [34]. In this process, TiCl_4 is partially pre-reduced by Ti powder or Mg to generate a TiCl_2 -bearing MgCl_2 solution. The TiCl_2 is then further reduced by molten Mg dispersed in the molten MgCl_2 to produce Ti powder. Some of the product is recycled to the TiCl_4 pre-reduction step, and the remainder is withdrawn from the reactor to undergo subsequent steps of sedimentation and distillation. There are also many challenges for this process, one of which is that the particle size of Ti powder produced by this process is very fine, which makes it prone to oxygen pickup [34].

Table 1 compares the major characteristics of different processes based on the reduction of TiCl_4 using Mg, with those of the Kroll process.

Processes based on the reduction of TiCl_4 using Na.

The most well-known process based on the reduction of TiCl_4 using Na is the Hunter process. The Hunter process is similar to the Kroll process in that it is a thermochemical process based on the reduction of TiCl_4 to produce Ti [36]. Similar to the Kroll process, the Hunter process was commercialised during the middle of the twentieth century, and for a considerable period

of time it was one of the two main commercial processes for production of Ti. However, commercial production of Ti sponge using the Hunter process has gradually come to a halt over the years because it became clear that the Hunter process is not economically competitive with the Kroll process. There are many reasons for this, but a primary obstacle is that to produce 1 mol of Ti by the reduction of TiCl_4 , it takes 4 mol of Na, while only 2 mol of Mg are required per mole of Ti. And, the production of Na by electrolysis is at least as costly as that of Mg. These issues make Na processing more expensive than using Mg.

However, the Hunter process is also recognised as a process that can be used to make not only Ti sponge, but also Ti powder [29,37]. During the Hunter process, TiCl_4 and Na are placed in a reactor, either simultaneously, or TiCl_4 is fed gradually into the reactor where Na is already preloaded. The process is typically carried out at temperatures above 800°C , at which Na and NaCl are always in their molten state. Ti forms at the surface of the molten pool, where TiCl_4 gas contacts with Na. Ti crystals then form and settle to the bottom of the liquid pool. Depending on the operational parameters, some of the Ti particles will connect and fuse together to form Ti sponge, while others will settle as Ti powder. In fact, the process operating parameters may be adjusted such that most of the Ti particles are separated from each other; therefore the main product of the process is powder, rather than Ti sponge. The NaCl also provides protection, preventing the Ti particles from being exposed to air, and thus the purity of the powder produced by the Hunter process is often superior.

Typical particle sizes of Ti powder produced using the Hunter process are -60 or -100 mesh. However, there is a substantial fraction (12–19%) of fine powders that will fall through -325 mesh. Oxygen, nitrogen,

Table 1. Comparison of the characteristics of different processes based on the reduction of TiCl_4 with those of the Kroll process.

Category	Process Identifier	Raw material	Reducing agent	Product size and morphology	Reported product chemical composition ^a	Batch or continuous; reactions	Ref.
Reducing TiCl_4 using Mg	Kroll	TiCl_4	Mg liquid	Sponge	O ~ 0.06%	Batch; $2\text{Mg} + \text{TiCl}_4 \rightarrow \text{Ti} + 2\text{MgCl}_2$ (850–940°C)	[28,29]
	TIRO	TiCl_4	Mg powder	Similar spherical shape but sinter to form large chunks; porous particles with d_{50} of 200 μm	O $\geq 0.3\%$, Cl < 0.03%; Largely depends on the purity of the feed material	Continuous; $2\text{Mg} + \text{TiCl}_4 \rightarrow \text{Ti} + 2\text{MgCl}_2(\text{s})$	[32]
	Vapour-phase reduction CSIR-Ti	TiCl_4	Mg vapour	Sub-micrometer, too fine to be captured	Low levels of Mg and Cl, but O $\geq 0.82\%$	Continuous;	[35]
		TiCl_4	Mg	Irregular shape, size ranging from 1 to 330 μm , average particle size 15–20 μm	Cl < 50ppm, N < 50ppm, O > 0.2%	$2\text{Mg} + \text{TiCl}_4 \rightarrow \text{Ti} + 2\text{MgCl}_2$ (1000°C) Continuous; $\text{Mg} + \text{TiCl}_4 \rightarrow \text{TiCl}_2 + \text{MgCl}_2$ $\text{TiCl}_2 + \text{Mg} \rightarrow \text{Ti} + \text{MgCl}_2$	[34]
Reducing TiCl_4 using Na	Hunter	TiCl_4	Na	Sponge and powder, –60 or –100 mesh	Superior pure, purer than that produced by Kroll process	Batch; $4\text{Na} + \text{TiCl}_4 \rightarrow \text{Ti} + 4\text{NaCl}$	[29]
	Armstrong	TiCl_4	Na	Mini sponge, particulates with micro porosity	O: 0.12–0.23%, N: 0.009–0.026%, C: 0.013%, Fe: 0.012%	Continuous; $4\text{Na} + \text{TiCl}_4 \rightarrow \text{Ti} + 4\text{NaCl}$	[38,39]
	ARC	TiCl_4	Na	Powder, small aggregates of small particles (~5 μm)	Oxide layer contributes a lot to the final high oxygen content	Continuous; $2\text{Na} + \text{TiCl}_4 \rightarrow \text{TiCl}_2 + 2\text{NaCl}$ $\text{TiCl}_2 + 2\text{Na} \rightarrow \text{Ti} + 2\text{NaCl}$	[40]

^aReported chemical composition is quoted from the corresponding references. They are not meant to be the limit of the method.

carbon, and iron contents can all meet industry specifications, and are lower than those produced by the Kroll process [29]. The characteristics of the powder, as well as the processes based on Na reduction, are given in Table 1 to compare with processes based on Mg reduction.

Again, similar to the reduction of TiCl_4 using Mg, there have been a number of reports on the development of continuous processes based on the same chemistry as that of the Hunter process, to reduce cost. The Armstrong process is an example. It is one of the few R&D processes that garnered considerable attention and investment, due to its promise to reduce the cost compared to batch processing [38]. Indeed, the main advantage of the Armstrong process is that it is a continuous process, pumping molten sodium to meet gaseous TiCl_4 in the reactor. Figure 6 illustrates the basic aspects of the Armstrong process. The resulting Ti powder and NaCl are removed from the reaction zone by the flowing sodium and the growth of Ti powder, which could trap unreacted raw material and co-products, can be inhibited. After the unreacted liquid Na is filtered out and the residual Na is distilled from the filtrate, the Ti powder is collected in a solution by washing out the salt. The product of the Armstrong process can be characterised as mini sponges [39], i.e. particulates with micro porosity (Figure 5(b)). For subsequent PM processes, powders made from the Armstrong process need to be milled to smaller sizes.

Another process based on sodium reduction of TiCl_4 was developed by the Albany Research Center (ARC) of the US Department of Energy. The ARC process is similar to the Armstrong process in that it is also a continuous process [40], however, the ARC process is a two-step process that first reduces TiCl_4 to TiCl_2 , and then to Ti.

Thermochemical processes based on the reduction of TiO_2

As shown in Figure 3, in addition to the processes described above, which all focused on the reduction of titanium tetrachloride (TiCl_4), there is another sub-category of processes that focus on the reduction of titanium dioxide. These processes are characterised by metallothermic reduction of TiO_2 . Reductants that can be used include Ca and Mg, and thus there are calciothermic and magnesiothermic methods, respectively.

Calciothermic methods are promising, principally because Ca is a very strong reducing agent. Four different forms of Ca have been investigated as options for calciothermic reduction, including CaH_2 (e.g. the metal hydride reduction (MHR)) [41,42], vapour-Ca (e.g. the preform reduction process) [17], liquid-Ca [18], and electronically mediated reduction (EMR) [19,20]. The MHR process dates back to 1945, and the most notable work was reported by Borok [43] in

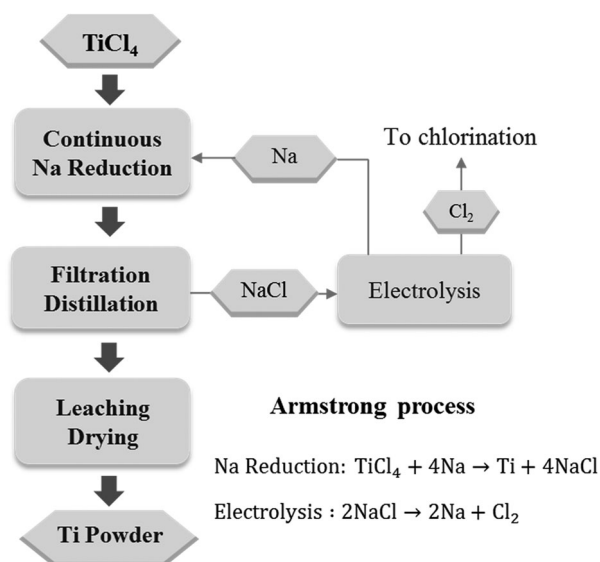


Figure 6. Illustration of the main processing steps of the Armstrong process.

1965 and Froes et al. [44] in 1998. Calcium hydride was used to reduce TiO_2 directly, and reportedly, there is a commercial operation in Russia based on this process. However, concrete information about its commercial reality is not readily available.

The preform reduction method was developed by Okabe et al. in Japan [17]. In this process, reductant Ca is placed underneath but not in contact with TiO_2 . TiO_2 is pre-fabricated in the form of blocks mixed with a flux of either CaO or CaCl_2 , and the preform is then reacted with Ca vapour at temperatures between 800 and 1000°C. The calcium vapour reacts with TiO_2 , leaving Ti and CaO. Fine titanium powder is obtained by leaching the product with acid. The Ca content of the powder is a concern, and the oxygen level was reported to be around 3000 ppm [45].

Suzuki et al. reported a method of calciothermic reduction of titanium oxide in molten CaCl_2 [22]. The TiO_2 was reduced by liquid calcium floating on the molten salt. The CaCl_2 plays the role of transporting Ca to react with TiO_2 , and dissolves the CaO by-product to push the reduction reaction forward. An oxygen level of 1000 ppm was reported after the reduction for 6 h using 5–7 mol-% Ca– CaCl_2 at 900°C by this method [22].

Titanium powder can also be produced directly through the electronically mediated reaction (EMR), using calcium as the reductant [20]. Feed material of TiO_2 or preform and reductant were connected in an otherwise electrically disconnected set-up in a molten calcium chloride salt [17]. An electric current and voltage can be measured between the feed material and the reductant without any externally applied power. The reduction of TiO_2 proceeds without direct physical contact with the reductant. Table 2 compares the characteristics of these processes based on the reduction of TiO_2 .

Magnesiothermic reduction refers to processes that use Mg to reduce TiO_2 . The initial concept of using Mg to reduce TiO_2 dates back to 1964 as reported in a US patent [46]. Rutile was mixed with Mg granules along with MgCl_2 as flux. The reaction mixture was held at about 750°C for a long time in a hydrogen atmosphere to reach nearly complete reduction, and Ti metal powder with an oxygen content as low as 1.7 wt-% was obtained. In more recent decades, a number of other reported research investigations explored this approach using either mixtures of Mg with TiO_2 or using Mg vapour to reduce TiO_2 [24,47,48]. In one approach an exothermic mixture was placed on top of the sample as an igniting agent to stimulate a self-sustaining reduction. The measured temperature for combustion reduction can reach 1600°C or higher. The Mg reduction of TiO_2 at 750°C has also been reported [25], however, an accurate oxygen composition after reduction was not provided.

Thermodynamic analysis showed that there is a lower limit of oxygen content in the powder at approximately 1.9 wt-% when Mg is used as the reductant [46,49]. In fact, this is true for both Mg and Ca with respect to their equilibrium with Ti–O solid solutions. As the Ti–O phase diagram (Figure 7) shows [50], solubility of oxygen in α -Ti can be up to 14.3 wt-%. The lower limit of oxygen content in Ti to which Mg or Ca can remove oxygen from Ti depends on the temperature and oxygen partial pressure, as shown in Figure 8. This figure also shows that the lower limit is much lower when Ca is used as opposed to Mg, which is understandable since Ca is a stronger reducing agent. A new direct reduction of the Ti-slag (DRTS) process using Mg was recently reported by Fang et al. [51]. The Mg reduction is directly carried out on UGS, which is composed of more than 95% TiO_2 . The reduction process is conducted in a hydrogen atmosphere to form titanium hydride (TiH_2) deliberately, which is followed by leaching in aqueous solutions to purify the TiH_2 powder.

An alternative route of this process is to prepare purified TiO_2 first, which is then subjected to a two-step hydrogen-assisted magnesium reduction (HAMR) process (Figure 9) [52,53]. A low-cost method based on alkaline roasting at relatively low temperatures followed by a hydrolysis process is used to produce highly purified TiO_2 . Purified TiO_2 can also be obtained commercially by using the sulphate process or the chloride process [54]. Compared to commercially available TiO_2 pigment, the particle size of TiO_2 produced by the alkaline roasting process is much larger, but costs significantly less than does the chloride process which produces TiO_2 pigment from TiCl_4 [55].

In order to produce Ti powder with extremely low oxygen content, the HAMR process is designed to include the following three key elements: (1) the use

Table 2. Comparison of the characteristics of the processes based on the reduction of TiO_2 .

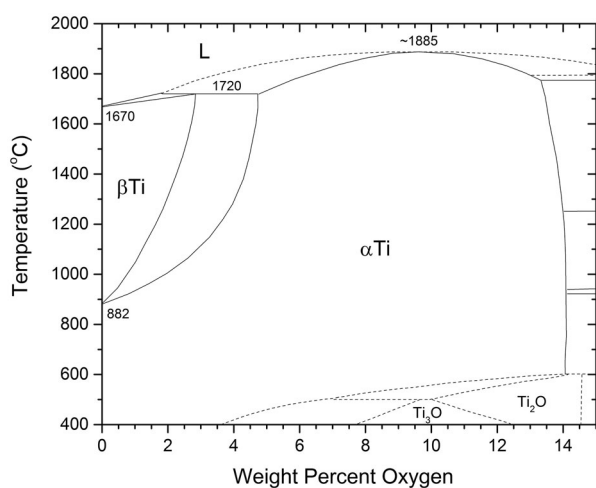
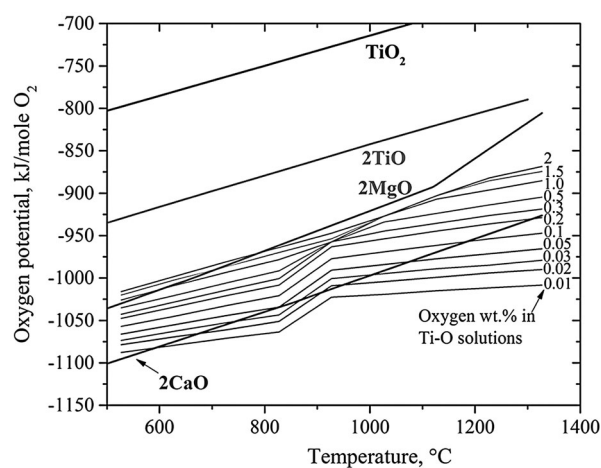
Category	Process Identifier	Raw material	Reducing agent	Product morphology	Reported product chemical composition	Salt	Temperature (°C)	Ref.
Thermochemical	MHR	TiO_2	CaH_2	Irregular, sponge	O: 0.19 wt-%, H: 0.34 wt-%, C: 0.03 wt-%, N: 0.06 wt-%	No salt	1100–1200	[41,42]
	Calciothermic reduction	TiO_2	Ca	Irregular, sponge	O: <0.2 wt-%, Ca: >0.1 wt-%	CaCl_2	900–1200	[22]
	Preform reduction process	TiO_2	Ca	Irregular, sponge	O: 0.2–0.3 wt-%	CaCl_2	>900	[17]
	EMR	TiO_2	Ca	Irregular, sponge	O: 0.15–0.2 wt-%	CaCl_2 or $\text{CaCl}_2 + \text{CaO}$	>900	[19,20]
	Combustion synthesis	TiO_2	Mg + Ca	Irregular, porous	O: 0.2–0.3 wt-%	Ca(OH)_2	850–1000	[24,48]
	DRTS/HAMR	UGS	Mg + Ca	Dense, globular powder	O: <0.15 wt-%	MgCl_2 or $\text{MgCl}_2\text{-KCl}$	< 800	[52,56]
Electrochemical	OS	TiO_2	Electron/Ca	Irregular, sponge	O: >0.15 wt-%	CaCl_2 or $\text{CaCl}_2 + \text{CaO}$	>900	[21,23]
	FFC	TiO_2	Electron	Irregular, porous	O: 0.15–0.40 wt-%	CaCl_2	800–1000	[10]
	MER	TiO_xC_y	Electron	Irregular, porous	O: 0.07 wt-% max, C: 0.08 wt-% max, Fe: 0.2 wt-% max.	Halide eutectic salt	~900	[87,89]
	Chinuka	TiO_xC_y	Electron	Irregular, porous	No information	NaCl-KCl	700–900	[93]
	USTB	$\text{TiO}_x\text{C}_y\text{N}_z$	Electron	Irregular, porous	No information	NaCl-KCl	~800	[94]
	QIT	Ti Slag	Electron	Solid ingot	No information	CaF_2	>1400	[90]
	SOM	TiO_2	Electron	No information	No information	TiO_2 containing $\text{MgF}_2\text{-CaF}_2$ flux	1100–1300	[91,92]

Note: Reported chemical composition is quoted from the corresponding references. They are not meant to be the limit of the method.

of hydrogen atmosphere; (2) the use of molten salt; and (3) the adaptation of a two-step process consisting of the main reduction step and a deoxygenation step. There are advantages of using hydrogen. First, hydrogen helps to destabilise the Ti–O system, increasing the thermodynamic driving force for Mg to react with Ti–O [56]. Another benefit of using hydrogen is the formation of titanium hydride, rather than Ti metal, during the reduction process. Titanium hydride is known to be more impervious to oxidation in air compared to α -Ti, which makes it easier to handle the material after reduction and control oxygen content in the final product.

Another key feature of the HAMR process is the use of molten salt, especially Mg bearing salt such as MgCl_2 . It was found to be necessary to use molten salt to facilitate the reaction and greatly improve the kinetics of the reduction process.

Furthermore, the HAMR process, as shown in Figure 9, employs a deoxygenation step to ensure the oxygen content in the final product powder is sufficiently low, i.e. less than 0.15 wt-% as required by ASTM-B299-13 for general purpose Ti. More details of the deoxygenation process are described in the next section. The combination of the reduction and the deoxygenation steps, (i.e. the use of this two-step

**Figure 7.** Titanium-oxygen phase diagram [50].**Figure 8.** Free energy-based Ellingham Diagram, comparing relative stability of Ti–O solid solutions with MgO and CaO [56].

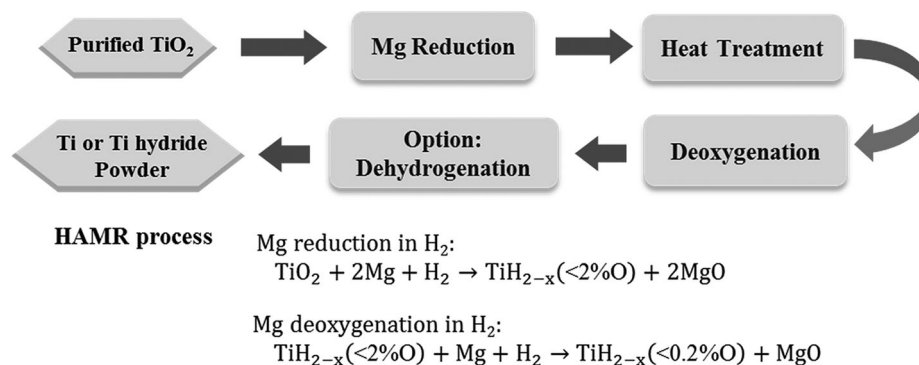


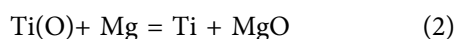
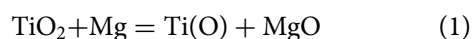
Figure 9. Illustration of the main processing steps of the HAMR process. Purified TiO_2 used in HAMR can be prepared by using the alkaline roasting process, the chloride process or the sulphate process.

approach to remove oxygen from TiO_2) is one of the key factors that make it possible to produce Ti powder with sufficiently low oxygen content. Table 3 shows the chemical composition of the powder produced using the HAMR process, which meets the requirements of ASTM-B299-13. The morphology of HAMR-Ti powder is shown in Figure 5(c). The characteristics of the HAMR process and its products are also included in Table 2 for comparison with other processes based on the reduction of TiO_2 .

Deoxygenation of Ti metal powder

As discussed above, there is a limit to which the oxygen content can be reduced by using Mg as the reducing agent. This limit is thermodynamically shown in Figure 8. The solubility of oxygen in α -Ti can be up to 14.3 wt-% [50]. In order to reduce oxygen content to less than 0.2 wt-%, as required by many professional standards and industrial specifications, the oxygen in the solid solution can be removed by a deoxygenation process. Deoxygenation can be used not only as a part of the metal production process, but also as a method to remove residual oxygen from any Ti powder when the oxygen content is higher than the required specifications.

Conceptually, the reduction of TiO_2 and the removal of oxygen content in Ti can thus be expressed in two separate reactions:



where $Ti(O)$ represents the solid solution of Ti with oxygen. The results of the reduction step (Equation (1)) would only lead to an equilibrium of $Ti(O)$ with the reducing agent, either Mg or Ca. To further reduce

oxygen content, the deoxygenation reaction (Equation (2)) is necessary.

A few calciothermic processes have been reported for deoxygenating titanium and titanium alloys [57–60]. Most of these operate by reacting with molten calcium, through solid–liquid or solid–gas reactions. For example, RMI (now part of Arconic) patented a process that operates above 900°C for reducing oxygen in Ti–6Al–4V powder [59]. Note that in this process, which is also industrially known as the deoxygenation in the solid state process, Ti is in the solid state, while the reducing agent Ca is in the molten state. Okabe et al. [60] tried deoxygenation of titanium in molten $CaCl_2$ at a temperature above 900°C. Because of the high temperature, the evaporation of Ca metal and $CaCl_2$ could not be avoided unless sealed reactors were used. By using these methods, it is also difficult to recover the Ti powder because of sintering between particles and caking of the entire powder mixture with Ca or the melt of Ca/ $CaCl_2$. Oh et al. [58] experimented with solid–gas contact by conducting the reaction at 700–830°C in high vacuum ($\sim 6 \times 10^{-3}$ Pa). However, the significant loss of calcium in vacuum might be a serious problem. The use of high vacuum is also not preferred for large-scale industrial operations.

Recently, Xia et al. developed a low-temperature molten salt deoxygenation process which can reduce the oxygen level in titanium to less than 1000 ppm from a level as high as 14.0 wt-%, which is the solubility limit of oxygen in α -titanium [61,62]. Calcium is used as the deoxygenating agent in this process. The molten salt facilitates the dissolution of solid Ca into the melt below its melting point of 845°C to aid the formation of Ca^{2+} ions, which react at the surface of the particles to cause the deoxygenation. The calcium halide bearing salt creates a low-temperature molten salt condition

Table 3. The chemical composition of the powder produced using the DRTS/HAMR process [52].

	Mg	Al	Fe	Si	Cl	Water	O	N	C	H
HAMR	0.067	0.021	0.044	0.036	0.001	–	0.11	0.019	0.03	0.028
ASTM B299 13	0.5	0.05	0.15	0.04	0.2	0.02	0.15	0.02	0.03	0.03

for deoxygenation. Being able to deoxygenate Ti at moderate temperatures (<800°C) has many advantages. For instance, Ti powders do not sinter to each other during the deoxygenation process as easily as they would if the temperature were greater than 900°C. The Ti powder after being subjected to the deoxygenation process can be readily separated, retaining their original size and morphology. Lower operating temperature also makes it easier to select the reactor materials, which has significant implications for the commercial viability of the process.

In addition to Ca, Zhang et al. reported recently that Mg can also be used to deoxygenate Ti–O solid solution alloys at low temperatures (600–800°C) [56]. Thermodynamically Mg can reduce TiO₂ or α -Ti that contains a significant content of O. When O is dissolved in an α -Ti lattice, forming a Ti–O solid solution, Ti–O can be more stable than MgO, depending on the oxygen content and the temperature. For instances, MgO is less stable than Ti-2 wt-% O at temperatures above ~780°C, or less stable than Ti-1 wt-% O at temperatures above ~600°C, and less stable than Ti-0.5 wt-% O at all temperatures above room temperature, as shown in Figure 8. The new deoxygenation approach is based on the thermodynamic tuning of the relative stability of MgO and Ti–O by introducing hydrogen. It was found that Ti–O can absorb hydrogen to form Ti–O–H solid solutions in a hydrogen atmosphere, and the oxygen potential in the Ti–O–H solid solution is lower than that in the initial Ti–O, making it less stable than MgO. That is, hydrogen can destabilise Ti–O by forming a Ti–O–H alloy, making it possible to reduce O content in Ti–O by using Mg. This thermodynamic destabilisation changes the reaction between Mg and Ti–O to form MgO from a situation where it is thermodynamically impossible to one where it is favoured. Hydrogen in the deoxygenated powder can subsequently be removed via a thermal dehydrogenation treatment, leaving pure Ti or Ti alloy powders with an oxygen content as low as a few hundred ppm. In other words, hydrogen is used as a temporary alloying element to assist the removal of oxygen from Ti or Ti alloy by Mg.

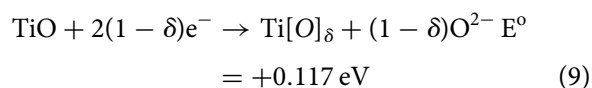
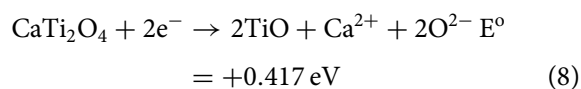
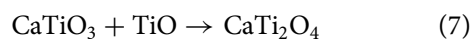
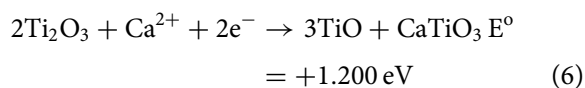
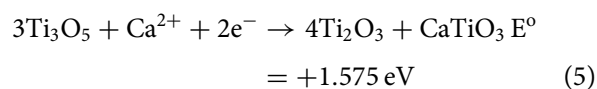
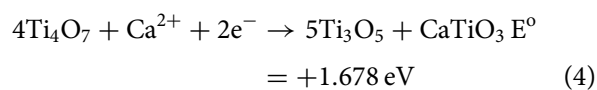
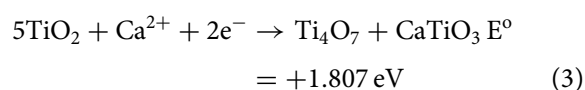
Most recently, an electrochemical technique for deoxygenation of Ti in molten MgCl₂ was reported by Okabe et al. [63]. By applying voltages between a Ti cathode and C anode immersed in molten MgCl₂, the activity of MgO was decreased and the activity of Mg was increased. An example was given showing that the oxygen content in a Ti sample was reduced from 1000 to less than 200 ppm.

Electrochemical processes

Electrochemical methods are another main route for the production of Ti. There are quite a number of electrochemical processes that have been under

development. However, only a brief review of those developments can be described in this section. A more comprehensive review of the production of Ti by electrochemical processes can be found in the literature [64]. Most of the research and development work has focused on the electrolysis of TiO₂, with some work on the electrolysis of Ti_xO_yC, and TiCl₄. Among all reported work on the electrolysis of TiO₂, the most well-known process was first reported by Chen, Fray, and Farthing [10,65], as shown schematically in Figure 10. This process is called as the FFC Cambridge process. It operates in a molten salt medium, normally in the temperature range of 800–1100°C. CaCl₂ is used as the salt, since CaCl₂ can dissolve and transport the oxygen ion, and it is cheaper and less toxic than the alternatives, such as BaCl₂ and LiCl [66].

A typical FFC cell includes a graphite anode and an oxide cathode. Because TiO₂ is an insulator, it was generally believed that it would not be possible to reduce TiO₂ by electrolysis. However, Chen et al. found that when a small amount of oxygen is removed from TiO₂, it forms Magnéli phases (TiO_{2-x}) which are highly conductive [10]. Further electrolysis removes the residual oxygen from the magnelli phases, where the oxygen ions dissolve in the electrolyte, and is then removed as CO₂ or CO. Schwandt et al. studied and reported on the reaction mechanisms [67–69]. They believed that Ca²⁺ from the salt first reacts with TiO₂ to form calcium titanate, CaTiO₃. Suboxides of titanium (Ti₄O₇, Ti₃O₅, Ti₂O₃, and then TiO) are subsequently produced. TiO is eventually formed and is further reduced by electrochemical deoxidation to Ti (O), the solid solution of oxygen in titanium. The overall reaction steps have been reported and outlined by the following equations [69]:



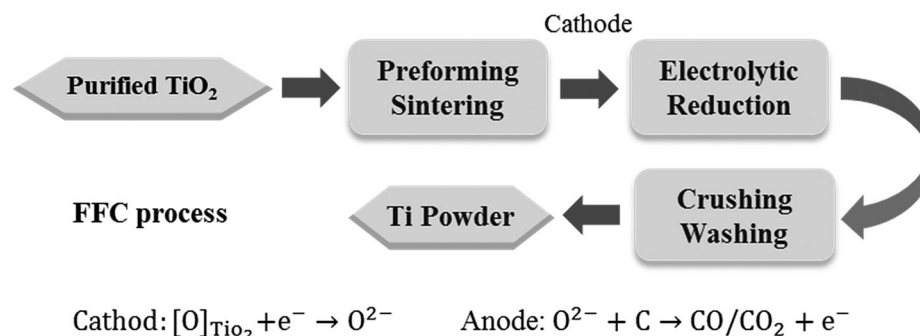


Figure 10. Illustration of the main processing steps of the FFC process.

Figure 5(d) shows that the final product has a porous sponge-like morphology, composed of globular Ti particles attached to each other due to sintering. In the original FFC method, preformed TiO₂ pellets were used as precursors [70,71]. More recently, reports have indicated that free-flowing oxide powders can also be used as the feedstock. Furthermore, it has been reported that it is also possible to reduce granules of synthetic rutile or naturally occurring rutile ore [72]. Because both natural and synthetic rutile contain impurities such as Fe, Al, and Si, it was suggested that a new titanium alloy powder that contains some of those elements can be produced using this method [73].

In addition to making commercially pure titanium (ASTM Grade 4), a wide range of other titanium alloys, such as Ti–Mo [74], Ti–Nb [75], Ti–6Al–4V [76–78], and TiNi [79,80], have also been produced by the FFC process. The characteristics of the FFC process, as well as other research electrochemical processes, are compared in Table 2.

An apparent advantage of the FFC Cambridge method is that it is a one-step process, and it does not use Mg as the reducing agent, which makes it potentially cost effective compared to the Kroll or other processes that use Mg to reduce TiCl₄ or TiO₂. This is because Mg is one of the main cost factors in the Kroll process. As mentioned above, the FFC method can also be used for direct production of alloys, potentially offering cost savings in another manner. However, the FFC process also has challenges, including low current efficiency and the possibility of incomplete/partial reduction of TiO₂ [70,81]. The current efficiency depends on many factors, including the parasitic reactions involving carbon and the formation of calcium, which can dissolve into the salt near the end of electro-deoxygenation. From an end-product quality perspective, there is little published information to date on the quality of the powders produced by using the FFC process. Bertolini et al. [82] reported that powder produced by the FFC process at a multiple kg scale had 0.29 wt-% O, 0.07 wt-% C, 0.014 wt-% Fe, 0.13 wt-% Ca and 0.06 wt-% Cl, provided as an example of the composition of Ti powder made from pure TiO₂ using the FFC process. However, the report does not

necessarily represent the best capability of the process, which is still under development.

Other examples of the electrolytic processing of TiO₂ in published reports include: the OS process, based on the combination of the reduction of TiO₂ by Ca and electrolysis to regenerate Ca in a molten CaCl₂ salt by Ono et al. [21,83–86]; an electrolytic process using carbothermic reduction of TiO₂ compound and Ti_xO_yC as the anode [87–89], the electrolytic process to convert titanium slag to liquid titanium metal [90]; and the solid oxide membrane (SOM) process using a solid-oxygen-ion conducting yttrium-stabilised-zirconia membrane separating the anode from ionic TiO₂-containing flux (MgF₂–CaF₂–TiO₂) [91,92].

Yet another reported work is the electrolysis of titanium oxycarbides (TiO_xC_y) [93] or oxycarbonitrides (TiO_xC_yN_z) [94] represented by the Chinuka process. In this process, oxycarbide is used as the anode material. The consumable anode materials of TiO_xC_y or TiO_xC_yN_z are oxidised to titanium ions and simultaneously CO evolves at the anode; thus titanium ions are reduced to metallic titanium at the surface of the cathode through electrolysis in molten salts.

Titanium powders made from Ti sponge, mill products, or Ti scrap

Today, commercially available titanium powders are mostly made using Ti sponge, Ti ingot, Ti mill products, or Ti scrap metals as the starting material. Especially, commercially pure Ti (CP-Ti) powders are available as HDH powder, while Ti alloy powders, e.g. Ti–6Al–4V alloy, are made using the PREP, or other atomisation methods. Brief descriptions of these methods are as follows.

Hydride–dehydride

To produce HDH powder, either Ti sponge, Ti ingot, mill products, or Ti scrap can be used as the raw material [5,6]. When scrap metals are used, it may be either CP-Ti or Ti alloys such as Ti–6Al–4V, and the scrap must first be cleaned to remove foreign materials.

The hydrogenation is typically accomplished using a batch furnace that can be operated in vacuum and/or hydrogen atmospheric conditions. The Ti metal and metal alloys can be easily hydrogenated under one atmospheric pressure of hydrogen at temperatures below 800°C by forming titanium hydride and alloy hydrides. The hydrogenation process and the equilibrium hydrogen content are a function of both the temperature and pressure of hydrogen. Figure 11 shows the equilibrium pressure of hydrogen as a function of molar content of hydrogen at different temperatures [95]. Because the equilibrium pressure of the Ti–H system increases as temperature increases, the driving force for forming hydride increases as temperature decreases, assuming the hydrogen pressure is held constant. Therefore, in reality, after Ti is heated to a set temperature such as 700°C, the hydrogen content will reach equilibrium at a specific temperature and increase during cooling. Exact hydrogen content of the hydride can, however, vary. Especially in the case of hydrogenated Ti–6Al–4V alloy, hydrogen content is typically less than what is expected based on the Ti–H phase diagram. New hydrogenation methods were developed to make low-cost TiH₂ powder by ADMA Products, Inc., in which hydrogenated titanium is produced during the Kroll process [96–99]. Titanium sponge can be hydrogenated in the same vessel after the reduction/distillation cycle during cooling to form titanium hydride [96]. Titanium hydride can also be produced after the reduction of TiCl₄ by magnesium and hydrogen [99].

One of the advantages of forming hydride as an intermediate step is that the hydride can be easily crushed to different particle sizes ranging from –325 to –60 mesh. A variety of powder-crushing and milling techniques may be used including: a jaw crusher, ball milling, or jet milling. Finer particle sizes are easy to obtain, but rarely used because oxygen content increases rapidly when the powder is finer than 325 mesh. Powder finer than 325 mesh also poses more safety challenges.

After the titanium hydride powders are crushed and classified, they are placed back in the batch furnace under vacuum or argon atmosphere to dehydrogenate and produce Ti metal powder. The powder can be passivated upon completion of both the hydrogenating and dehydrogenating cycles to minimise exothermic heat generated when exposed to air. Figure 5(e) illustrates the typical morphology of HDH Ti powder.

A major advantage of HDH powder is that it is relatively inexpensive. The costs of the hydrogenation and dehydrogenation processes add only a moderate amount of cost to that of input material, and the purity of HDH powder can be very high, as long as the raw material's impurities are controlled. The oxygen content of HDH powder has a strong dependence on the input material, the handling processes and the specific surface area of the powder [5]. The main disadvantages

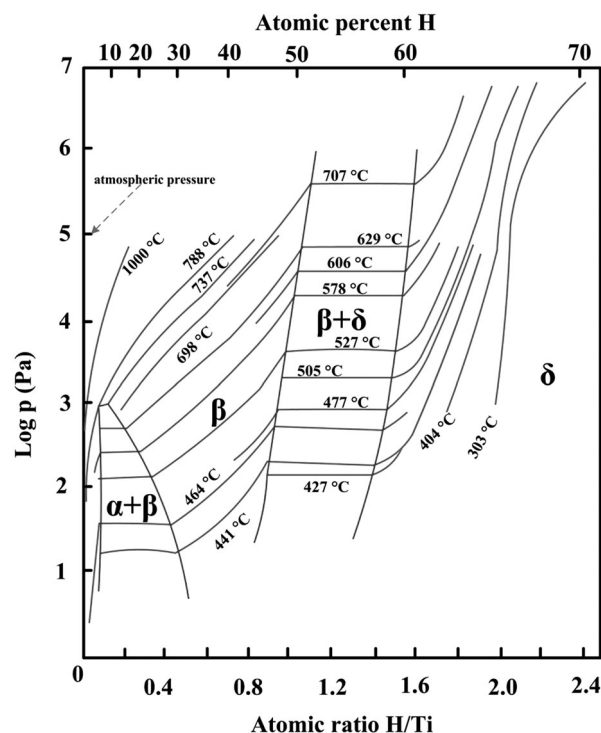


Figure 11. The pressure-composition-temperature curves of the Ti–H system [95].

of HDH powder include: the powder morphology is irregular, and the process is not suitable for making virgin alloyed powders or modification of alloy compositions if the raw material is from scrap alloys.

Atomisation

In general, atomisation processes are used for making alloyed powders, and PA Ti feedstock is generally used as the starting material for atomisation. Atomised alloy powders are typically used for making components using HIP. As mentioned previously, it is generally believed that alloyed powders are not suitable for cold compaction using conventional uniaxial die pressing methods because the inherent strength of the alloyed powders are too high, making it difficult to deform the particles in order to achieve desired green density.

Because almost all atomisation processes are used for producing relatively spherical shaped powders, atomised Ti alloy powders are the most common choice for AM using selective laser or electron beam melting techniques [100,101]. Spherical powders are also required for manufacturing Ti components using metal injection molding (MIM) techniques [102]. Typically, AM and MIM processes require particle sizes of powders to be in the range from a few to 100 µm to ensure good flowability of the powder, which is essential for AM and MIM operations. However, atomisation processes usually produce powders with a wide particle size distribution, from a few to hundreds of micrometers. The yield of powder under –325 mesh (44 µm) is usually low depending on the

specific atomisation technique, but the yield is a major factor contributing to the high cost of usable powders.

There are a large variety of atomisation techniques. Among them, GA, plasma atomisation, and PREPs are used commercially for the production of spherical Ti alloy powders. All atomisation processes consist of three main integrated steps: melting, atomisation, and cooling. Melting can be accomplished by conventional melting techniques such as vacuum induction melting, plasma arc melting, or induction drip melting, where solid bar feedstock is used, or by direct plasma heating. Atomisation is the process during which liquid metal is broken into droplets, which cool during flight in a cooling chamber under inert gas protection. Atomisation is normally accomplished using a high pressure gas, forced through a nozzle. Droplets can also be formed by the spinning of a liquid stream off of a disc, causing molten droplets to form and undergo centrifugal acceleration away from the centre of spin. The droplet will subsequently solidify during flight. Each specific atomisation technique varies in details from others, but they all share the same three main steps as described here. A detailed description of the different techniques is beyond the scope of this article. Rather, we will highlight some key features of different techniques, which are also summarised and compared in Table 4.

Gas atomisation

GA of titanium was originally developed by Crucible Materials Corporation in the 1980s [103,104]. The basic configuration of the GA process is shown in Figure 12(a). GA usually produces powder in a wide size range (up to 500 μm) [105]. Close-coupled gas atomisation (CCGA) is the latest technique, improving the production yield of relatively fine particles (<45 μm) [106,107]. However, there is a risk of yttrium contamination due to the use of laminated composite pour tubes made of Y_2O_3 . Although GA is, generally, a mature technology, there are a few issues worth noting. There are considerable interactions between droplets while they cool during flight in the cooling chamber, causing the formation of satellite particles. Owing to the erosion of atomising nozzle by the liquid metal, there is potential for contamination by ceramic particles, and there may also be argon gas entrapment

within particles. Since molten titanium is very reactive to most common metals and ceramics, electrode induction gas atomisation (EIGA) was developed by ALD Vacuum Technologies to produce ‘ceramic-free’ powder, in which the melt is not in contact with the ceramic lining material or the crucible [108,109]. A PA rod is used as the feed material. In order to minimise possible contamination pickup during atomisation, a gas-atomisation apparatus with a Ti coating on the inner wall of the atomisation chamber and other components in the flow path was designed and developed by FMW Composite Systems, Inc. [110].

Plasma rotating electrode process

The PREP is one of the most recognised techniques for making Ti alloy powders, and avoids the potential contamination issues posed by ceramic nozzles [111,112]. The basic configuration of this process is shown in Figure 12(b). The feedstock must be precisely machined titanium alloy bars, which are used as the anode that rotates at roughly 15 000 rev min^{-1} while being melted by the plasma arc. The rotation of the electrode causes molten droplets to spin away from the anode under centrifugal acceleration. The liquid droplets form spherical shapes spontaneously to minimise surface energy. The cooling rate is typically less than 100°C s^{-1} , and particle sizes of PREP powder typically range from 50 to 350 μm [112], which is ideal for HIP applications. The PREP powder is shown in Figure 5(f).

Plasma atomisation

Plasma atomisation is described here to represent processes that use PA wire as the feed material [113,114], which is a significant cost contributing factor. The Ti alloy wire is melted in a plasma torch, and a high velocity plasma flow breaks up the liquid into droplets which cool rapidly, with a typical cooling rate in the range of $100\text{--}1000^\circ\text{C s}^{-1}$. Plasma atomisation produces powders with particle sizes ranging from 25 to 250 μm . In general, the yield of particles under 45 μm using the plasma wire atomisation technique is significantly higher than that of conventional GA processes. The plasma-atomised powder is shown in Figure 5(g).

Table 4. Key features of different atomisation techniques.

Methods	Feed material	Size range (μm)	Advantage	Disadvantage
GA	Elemental/ ingot/bar	<500	Flexible alloy and feedstock options	Satellites; possible porosity in coarse powder; possible ceramic contamination
EIGA	Bar	<200	Relative high fine powder yield	Possible porosity in coarse powder; relative high Ar flow rate
Plasma atomisation	Wire	<300	Less satellite; relative high fine powder yield	Expensive feedstock wire; possible porosity in coarse powder
PREP	Bar	50–350	High purity; no satellites	Low fine powder yield
Plasma spheroidisation	Powder	>5 (the same size with feed powder)	Relative low cost, high fine powder yield	Subject to availability of low-oxygen feed powder
GSD	Scrap/ elemental	10–100	Very high fine powder yield; minimal satellites, low cost, low oxygen	Not perfect spherical shape; possible porosity

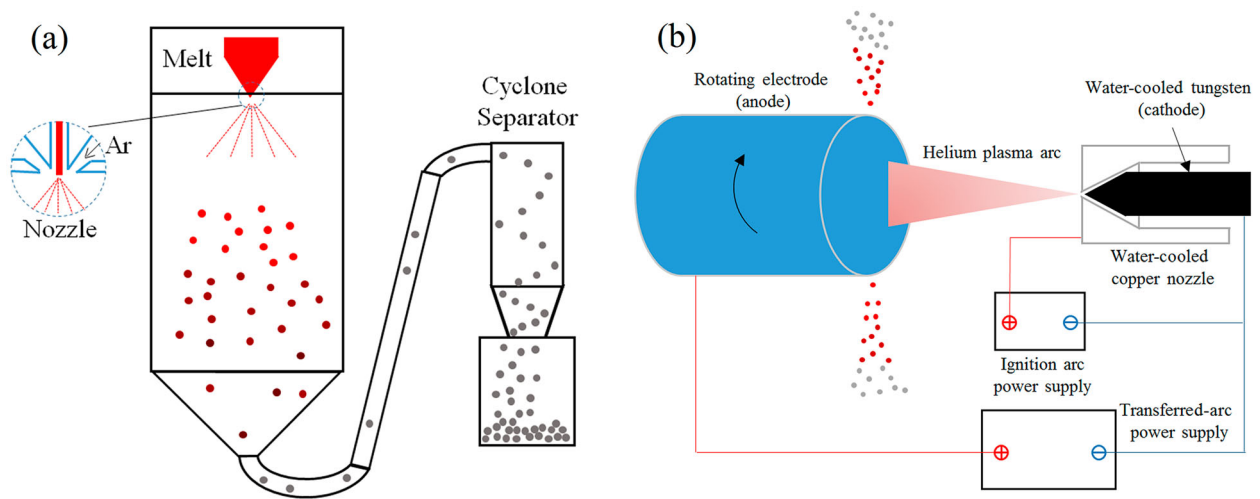


Figure 12. Schematic diagrams of (a) GA process and (b) PREP.

Production of spherical Ti powder

Making spherical powder is not a specific processing technique. However, it is singled out here because it is in high demand for AM using selective laser sintering or electron beam sintering techniques. Titanium alloy, Ti-6Al-4V in particular, is one of the prime examples of AM metals. AM is a manufacturing technique that is most valuable for making complex shaped parts or systems that are custom designed for special applications which do not need large quantities of material or large production runs. These features match well with the characteristics of Ti alloys, because Ti is mostly used for applications with unique and stringent requirements that only Ti can meet. Examples include aircraft components and biomedical implants that are custom made for each individual patient. Spherical powders are also necessary for MIM with Ti alloys. As a forming technique, both AM and MIM of Ti are described later in this article.

Characteristics of powder that are important for AM and/or MIM include particle size and size distributions, morphology, tap density, flowability, uniformity of alloying composition, and low interstitial impurity elements, including oxygen, carbon, and nitrogen. Table 5 lists chemical properties of powder, as required by various industrial standards [115–119].

In general, spherical Ti powders that are available in the marketplace today are made by either atomisation methods or plasma spheroidisation of non-spherical powders. Atomised Ti powder can be made by all of the atomisation techniques mentioned above. However, the published literature suggests that several techniques are primarily being used: PREP, GA, CCGA, EIGA, and plasma atomisation. These techniques are discussed in section 'Atomization'. A key consideration for selecting one technique over another is the yield of the powders that meet the requirements for size and size distributions for AM or MIM. The required

particle sizes are generally in the range of $<45\ \mu\text{m}$ for MIM or AM based on selective laser sintering, and $45\text{--}106\ \mu\text{m}$ for AM based on electron beam processes. One of the key advantages of using the PREP process is to avoid using nozzles that are often made from high-temperature ceramic materials. Ceramic inclusions from the erosion of the nozzles must be avoided for Ti alloys.

There has been some reported research effort towards making spherical Ti powder more affordable. Among these efforts, the plasma spheroidisation of powders is a relatively new but popular technique [120]. Plasma spheroidisation of powders has been applied to a variety of different powders, including refractory metals such as tungsten [121]. During plasma spheroidisation, the metal powder is melted by a plasma torch and forms molten droplets, which solidify to form spherical solid powder before reaching the bottom of the reactor chamber [121]. A unique characteristic of plasma spheroidisation is that the particle sizes do not change during plasma processing. Plasma-spheroidised particles typically have the same nearly perfect round shape as the other atomised powders [6,122]. Another example is a continuous method during which low-cost Ti sponge fines, HDH powder or electrolytically produced Ti and alloy powders are fed through a plasma transferred arc torch to make spherical alloy powder [88]. In addition to spheroidising, or producing, particles in the molten state, there are reports on modifying particle shape in the solid state by mechanical means [123,124]. The flowability of irregularly shaped powders was reportedly improved by removing sharp angles on the particles through high-speed blending or high shear milling.

Recently, a new approach, called granulation-sintering-deoxygenation (GSD), for making spherical Ti powder was developed by Fang et al. [125,126]. Using

Table 5. Chemical compositions of Ti–6Al–4V powder as required by various industrial standards.

	AMS4998	ASTM F2924	ASTM F1580	ASTM F3001 (ELI)	ASTM F2885
Al	5.50–6.75	5.50–6.75	5.50–6.75	5.50–6.50	5.5–6.75
V	3.50–4.50	3.50–4.50	3.50–4.50	3.50–4.50	3.5–4.5
O	0.13–0.18	<0.2	<0.2	<0.13	<0.2
N	<0.04	<0.05	<0.05	<0.05	<0.05
C	<0.10	<0.08	<0.08	<0.08	<0.08
H	<0.012	<0.015	<0.015	<0.012	<0.015
Fe	<0.3	<0.3	<0.3	<0.25	<0.3
Sn	<0.10	<0.1	<0.1		
Mo	<0.10				
Cu	<0.10				
Mg	<0.10				
Zr	<0.1				
Y	<0.2	<0.005	<0.005	<0.005	<0.005
Other elements, total		<0.4			
Other elements, each		<0.1			
Application	Aerospace	AM Power bed fusion	Coating Surgical Implant	AM Power bed fusion	MIM Surgical implant

this approach, spherical Ti powders can be made by forming spherical granules composed of fine Ti particles. The spherical granules can be sintered to obtain solid spherical Ti powder. Figure 13 illustrates key steps of this process.

There are a number of key issues when using this process to make spherical Ti powder. First, to produce spherical powders of less than 45 µm, the particle sizes of the initial fine powder must be less than a few microns. The finer the initial particle size, the better the granules will be with respect to surface finish and sinterability. However, the limiting factor is that the oxygen content, as well as the content of other interstitial elements, in the final Ti powder increases with a decrease in the initial particle size. Thus, there is a balance between the considerations of achieving spherical morphology and minimising the content of interstitial elements in the final spherical powder. Second, the oxygen content of the powder depends strongly on the powder handling and processing steps. Oxygen content will inevitably increase with the granulation, sintering, and other associated processes. Therefore, a deoxygenation step is necessary. Results showed that the oxygen content of the spherical powder can be lowered to 0.1 wt-% or below, which is sufficient for many applications. The third concern is that particles may bond to each other during sintering. Therefore, measures must be taken to prevent or eliminate the sintering of particles to each other. Figure 14 shows the morphology and the cross-section of the spherical powder.

Sintering and consolidation of Ti powders and the resulting microstructures

Ti powder compacts can be sintered using a variety of processes, including pressureless sintering under vacuum or inert gas, as well as pressure-assisted consolidation techniques such as vacuum hot pressing (VHP) and HIP. Of course, the microstructure of the sintered Ti alloy depends on the specific process and the processing parameters employed, as well as the feedstock powder. The following sections will focus on the effects of sintering processes on porosity, purity, and microstructure (grain size and morphology), and analyse how these three factors affect the mechanical properties of sintered Ti alloys. It should be noted that this review does not delve into the sintering mechanisms or kinetics of sintering, but rather the focus is on the effects of PM processing on the microstructure and properties of Ti alloys.

Issues of residual porosity and purity were the primary focus of the earliest research into Ti powder metallurgy. Impurity elements in Ti alloys include: oxygen, chlorine, carbon, and nitrogen. A significant advance in the BE approach has been achieved with

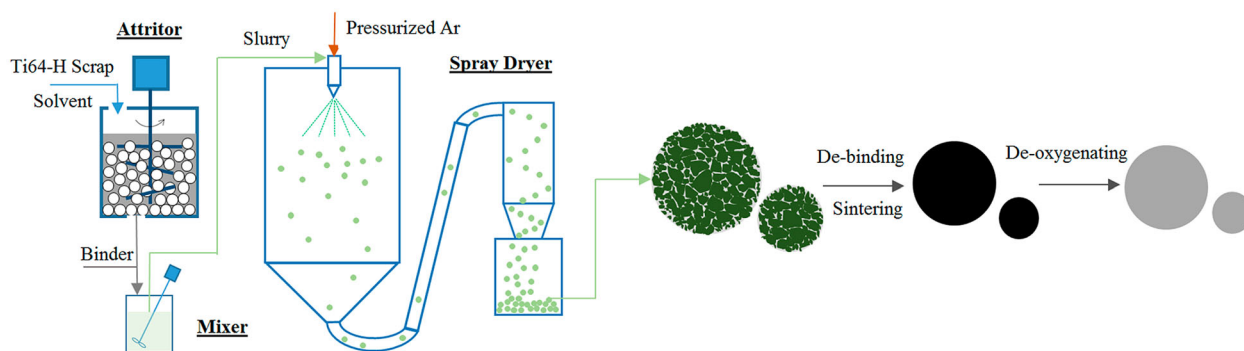


Figure 13. Schematic illustration of the GSD process for making spherical titanium alloy powders from scrap or hydrogenated Ti sponge [125]. (Reproduced with permission.)

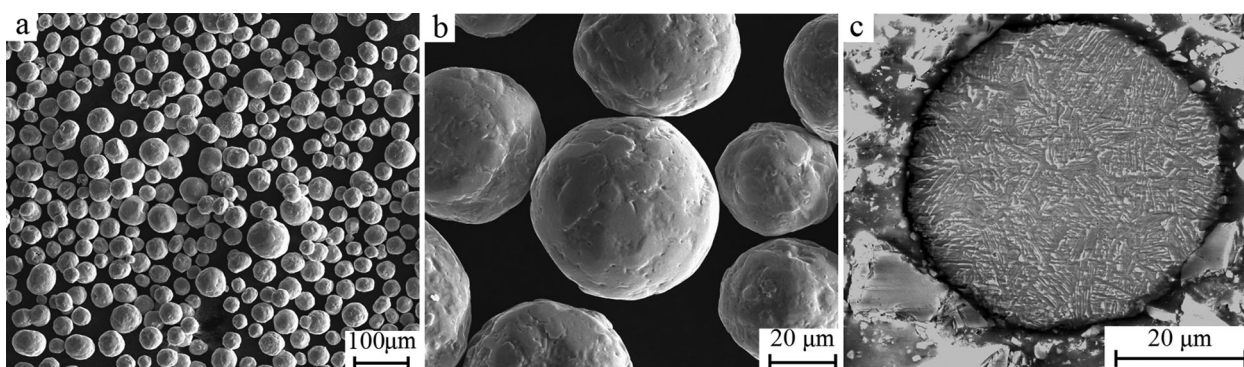


Figure 14. SEM micrographs of spherical Ti-6Al-4V powder by the GSD method [125]. (Reproduced with permission.)

the use of low chloride Ti powder [127]. Chlorine or chloride contents as low as 200 ppm, which is typical for inexpensive Ti powders, have been reported to adversely affect the densification process during sintering [128]. Profound improvement in the as-sintered density has been achieved simply by using powder with less than 10 ppm chlorine [129]. Although the effects of chlorine, carbon, and nitrogen are important, they can generally be controlled during Ti sponge and powder production processes. Additionally, sponge can be produced with very low oxygen content. Nonetheless, oxygen content is the most challenging problem during powder processing and sintering.

As with other reactive metals, a native oxide layer is often present on the surface of titanium powder particles. However, this oxide layer, which tends to retard sintering of aluminium and magnesium alloys, can dissolve at elevated temperatures and does not impede the sintering of Ti [130]. Mo et al. reported that the oxide layer is effectively removed at approximately 700°C [131]. It should be noted that there is some disagreement in the literature regarding the exact temperature, though it is consistently reported that this effect occurs below 1000°C [132,133]. Ti also exhibits comparatively strong diffusion bonding at relatively moderate temperatures. In fact, solid state diffusion bonding at a temperature below the β -transus (995°C for Grade 5 Ti-6Al-4V [134,135]) has been used by Rolls Royce

to join Ti sheet in the production of turbine engine blades [136]. For these reasons, Ti readily densifies at temperatures above 1200°C, though residual porosity remains when sintered using conventional pressureless sintering processes.

Regarding the issue of porosity, it has been demonstrated that the residual porosity may be effectively closed via pressure-assisted consolidation (e.g. HIP) or thermomechanical processing (TMP, e.g. forging) after sintering [129,137,138]. However, incorporating these energy-intensive post-processing steps drives up cost [4].

Additionally, refining the as-sintered microstructure produced by traditional sintering processes is almost impossible because of the lack of stored energy to drive recrystallisation. This is particularly true for $\alpha + \beta$ Ti alloys (e.g. Ti-6Al-4V), which have a tendency to form coarse lamellar microstructures during sintering. The coarse microstructures are detrimental to mechanical properties, especially the fatigue strength, but recent breakthroughs have identified mechanisms to reduce the residual porosity and refine the microstructure of as-sintered Ti-6Al-4V (section 'Sintering of TiH₂').

This article will highlight the microstructure as a function of the process, laying the groundwork for examining the relationships between processing, microstructure, and properties in the subsequent sections.

Pressureless sintering of Ti metal powder

Pressureless sintering includes inert gas and vacuum sintering, which are both reported to have been used for sintering Ti. However, vacuum is far more commonly used than inert gas sintering for Ti.

The first experiments aimed at sintering Ti sponge were performed by Kroll in 1937. During these experiments, Kroll used a low-pressure argon atmosphere (6.7×10^3 Pa) [139]. In the 1940s, additional investigations into the effect of the sintering atmosphere were performed by Dean et al. [140] with the aim of producing bulk Ti samples with improved ductility. In this study, it was found that helium was a suitable protective gas. However, it was also determined that vacuum was required to remove residual hydrogen, as well as magnesium left from the extractive process.

Vacuum has long been considered the ideal sintering atmosphere for Ti, owing largely to the work performed by Dean et al. [140] in the 1940s. However, any process which requires vacuum is inherently limited to batch processing. For this reason, continuous sintering processes under protective gases have long been investigated. All non-noble protective gases (e.g. nitrogen) react readily with Ti and will compromise the purity and mechanical properties of the alloy if used for sintering. With respect to cost, argon is the only feasible option of the noble gases. In fact, argon sintering has been used commercially by DuPont, dating back to the 1950s, to produce PM Ti components [141]. Additionally, Toyota uses argon sintering to produce PM Ti metal matrix composites [142].

Owing to Ti's strong gettering of oxygen and nitrogen, commercial grades of argon may result in unacceptable pickup of these elements during sintering. Methods are available for purifying argon *in situ*, such as flowing the gas over Ti chips or sponge at temperatures over 800°C [143]. Another approach utilises specialised 'OXYNON' sintering furnaces produced by Kanto Yakin Kogyo (Japan). These furnaces use a carbon-fiber sintering belt, designed to remove oxygen from argon during sintering [144]. OXYNON furnaces have reportedly been used to continuously sinter Ti since 2002 [130]. Heaney and German reported interesting results in 2004 from a study in which CP-Ti was sintered using a vacuum furnace versus an OXYNON furnace under argon [145]. During the study, identical sintering experiments were performed in each type of furnace and repeated for three different starting powders. It was reported that the samples sintered in the OXYNON furnace had lower oxygen, nitrogen, and carbon content than identical samples sintered in the vacuum furnace. The lower carbon content is a surprising result, considering the fact that the OXYNON furnace employed carbon hardware at high temperatures. However, the samples were separated from the carbon belt with zirconia plates and

covered with molybdenum sheets. It was determined that the small amount of carbon introduced by the vacuum oil residue in the vacuum furnace was greater than that picked up from the carbon hardware in the OXYNON furnace. Each sample set had similar density after sintering for both the OXYNON and vacuum-sintered samples. The OXYNON-sintered samples exhibited similar ductility to the vacuum-sintered samples in all but one sample, which had significantly lower ductility. However, the poor elongation of this particular sample was attributed to contamination by sodium-reduced titanium powder next to it in the sintering furnace. The vacuum-sintered samples exhibited consistently higher strength for each sample set by approximately 100 MPa. This fact was not discussed by the authors in the context of the sintering atmosphere, though it could be due to the increased interstitial content of the vacuum-sintered samples.

For BE powders of $\alpha + \beta$ alloys, temperatures of 1200°C or higher are usually necessary to both facilitate densification and allow for sufficient homogenisation of the alloying elements [146,147]. For $\alpha + \beta$ alloys, this is well above the β -transus (995°C for Grade 5 Ti-6Al-4V [134,135]). Therefore, at a typical sintering temperature, the microstructure consists entirely of equiaxed β grains on the order of hundreds of microns in diameter and with a homogenous distribution of the alloying elements. As the material is cooled relatively slowly, preferential nucleation of α grains along the prior β grain boundaries results in a continuous layer of 'grain boundary α ' (α_{GB}). As the material is continually cooled, α grains grow into the bulk of the β grains as colonies of parallel plates with a $(110)_{\beta} \parallel (0001)_{\alpha}$ Burgers relationship (Figure 15) [134,148,149]. The size of the colonies, as well as the individual α lamellae, is determined by the cooling rate. Faster cooling forces the nucleation of more α grains, which results in a finer microstructure. The high sintering temperatures and relatively slow cooling rate of conventional pressureless sintering, therefore, consistently result in a coarse lamellar microstructure for PM Ti-6Al-4V (Figure 16(a,b)).

Even with 100% density and low interstitial content, a coarse lamellar microstructure is undesirable with respect to mechanical properties; particularly fatigue performance [150]. The Burger relationship of the α grains with the parent β grains results in a common basal plane for entire α colonies. Since faceted fatigue fracture is most common along the basal plane in these alloys, a fracture along the basal plane can readily propagate across colonies [148]. Therefore, the typical as-sintered PM Ti-6Al-4V microstructure is unacceptable for critical applications, especially those requiring high fatigue performance due to the fact that it contains large α colonies.

The typical as-sintered PM Ti-6Al-4V microstructure is similar to the ' β -annealed' microstructure that results from super-transus heat treatment in wrought

processing (WP) [134]. Traditionally, this coarse lamellar microstructure is refined through plastic deformation during TMP, which produces high dislocation densities that serve as a driving force for recrystallisation. Microstructural evolution, however, can also be driven by introducing temporary alloying elements (i.e. hydrogen), which make new phase transformations possible. These phase transformations are the basis for thermohydrogen processing (THP) and hydrogen sintering and phase transformation (HSPT). This is discussed in detail in section 'Sintering of TiH_2 in H_2 '.

Vacuum sintering is traditionally the most common method of sintering Ti. The typical as-sintered density of PM $\alpha + \beta$ Ti alloys (e.g. Ti-6Al-4V) sintered in vacuum is approximately 98% theoretical density (TD) [11]. Oxygen content in sintered materials is typically greater than 0.2 wt-% [11], depending on the particle size and size distributions [151]. The typical as-sintered microstructure of vacuum sintered Ti-6Al-4V alloy is a relatively coarse lamellar structure [3,11].

Pressure-assisted consolidation

Pressure-assisted consolidation techniques are typically necessary for sintering PA powders. This is because the inherent strength of PA powders makes compaction of

these powders much more difficult. In fact, it has been reported that to produce a green compact of Ti-6Al-4V from -100 mesh powder to 84% TD required 965 MPa for PA powder and only 413 MPa for BE powder [152]. In the same study, it was found that the PA powder required higher sintering temperatures to reach a similar as-sintered density. It was theorised that diffusion of the alloying elements in BE compacts helped densification during sintering. For these reasons, PA powders generally require pressure-assisted consolidation such as HIP or VHP to achieve similar consolidation as pressureless sintered BE powder. PA Ti-6Al-4V sintered via pressure-assisted techniques has also been shown to exhibit improved mechanical properties, particularly fatigue performance, in the as-sintered material when compared with the BE approach [11].

The preferred powder for most pressure-assisted processes is spherical powder that has been produced by GA or the PREP [153]. The improved flowability of these powders is necessary to sufficiently fill complex near-net-shape geometries. As mentioned previously, these powders are generally expensive [105]. Therefore, in order to be commercially viable, any process using PA powder usually needs to be near-net-shape to offset the cost of the powder.

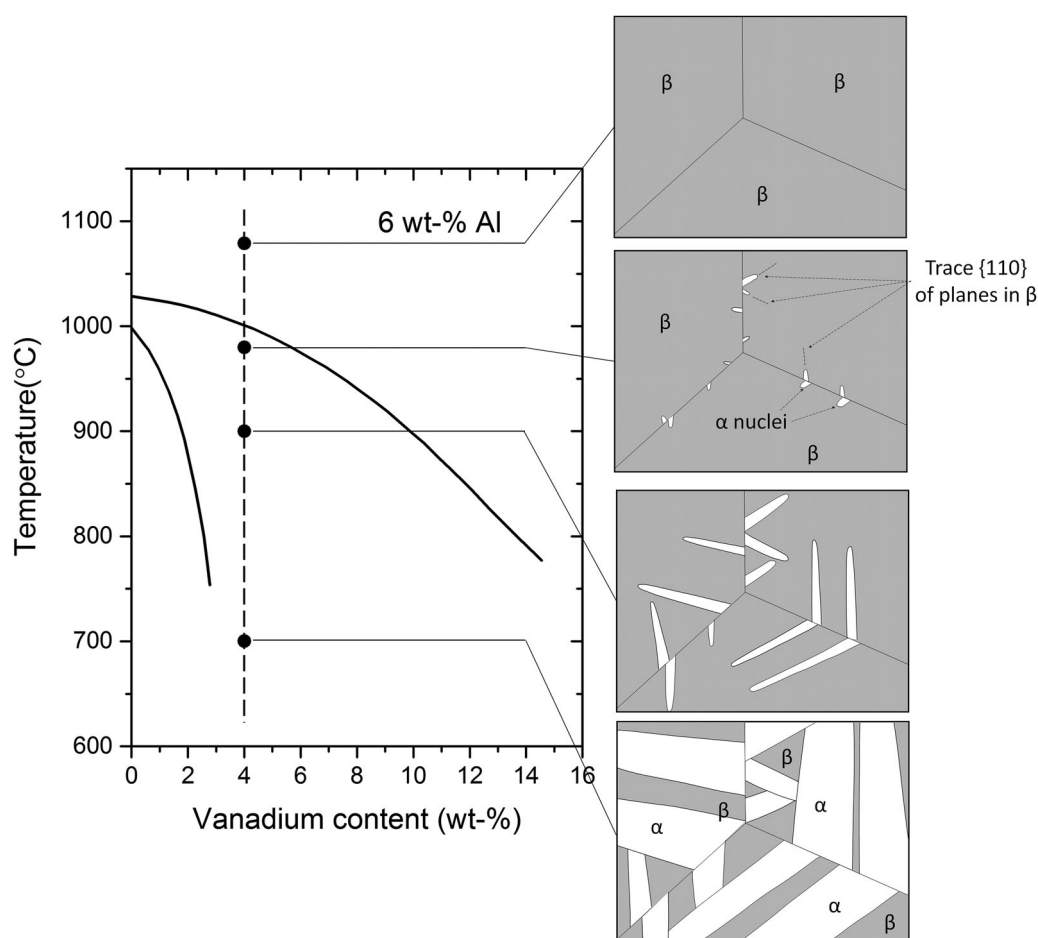


Figure 15. Phase transformations during cooling of Ti-6Al-4V from above β -transus [149]. (Adapted with permission.)

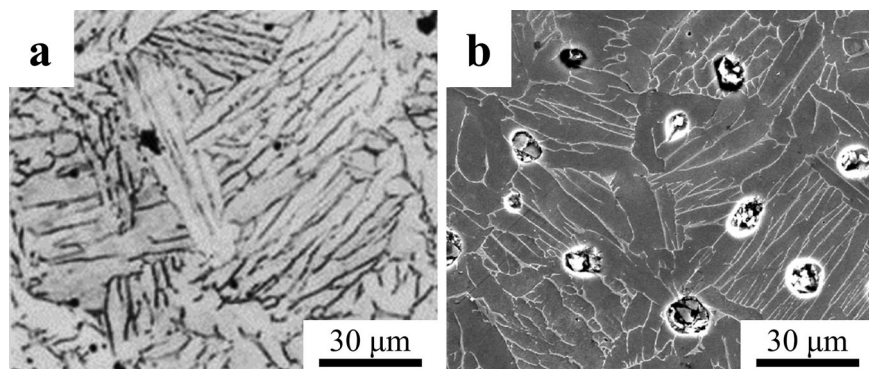


Figure 16. Micrographs of press and sintered Ti–6Al–4V: (a) OM, α is light phase and β is dark phase [11], and (b) SEM, α is dark phase and β is bright phase. (Reproduced with permission.)

Hot isostatic pressing

HIP is the most common method used for consolidating PA Ti powders [11,105]. Initial incarnations of the HIP process for producing Ti alloys utilised ceramic molds prepared similarly to investment casting molds [154,155]. During this process, the shaped ceramic molds were filled with titanium powder and placed inside of a steel can. The steel can was then packed with spherical alumina, evacuated, sealed, and placed inside the HIP. While this provided a relatively simple method for directly producing a shaped part via HIP, it was found that foreign ceramic particles introduced by this process were detrimental to fatigue performance [156]. Therefore, the ceramic mold process has been largely abandoned. Currently, a shaped metal that can be filled with the powder and placed directly inside the HIP is used.

HIP is generally conducted at temperatures between 850 and 1200°C. The isostatic nature of the deformation in HIP densifies the material at lower temperatures than pressureless sintering. Therefore, consolidation can be achieved at temperatures below the β -transus, thereby avoiding the coarse lamellar transformed β microstructure that is the result of conventional sintering processes (section ‘Pressureless sintering of Ti metal powder’). When consolidated via HIP below the β -transus, Ti–6Al–4V tends to have a microstructure with a mixture of elongated α platelets and apparently equiaxed α grains (Figure 17(a,b)) [105,157]. It should be noted that while these microstructures seem to show two different α morphologies, they are unlike the bi-modal microstructures typically seen in WP. The equiaxed grains in a bi-modal microstructure produced via WP tend to be located at the triple points of the α colonies [134]. However, in the HIP microstructures shown here, grains with similar morphology appear in groups.

After HIP processing, the material may be forged, or processed with another form of TMP, to refine the microstructure and improve the mechanical properties [158]. Of course, this option is only viable for the production of mill products meant for subsequent working

(e.g. rolling, extrusion, etc.), as the significant plastic strain required to drive recrystallisation is not near-net-shape compatible. Additionally, as mentioned above, TMP is inherently energy-intensive and subject to lower yields. Therefore, including TMP after HIP processing could significantly increase the embodied energy (total energy per mass of product produced) and decrease the performance-to-cost ratio of the final component [4].

The microstructure of the material after HIP is strongly dependent on the temperature used during HIP processing. Zhang et al. reported on samples that were consolidated via HIP for 4 h at 880, 930, and 1020°C [157]. While the samples consolidated via HIP at 880°C exhibited the finest microstructure, the best ductility (18–21%EL) and fatigue performance (500 MPa at 10^7 cycles) was reported for the samples consolidated at 930°C [157]. Fractography of a tensile sample consolidated at 880°C showed crack propagation along the prior particle boundaries. Therefore, despite the finer microstructure, the decreased mechanical properties of the samples consolidated at 880°C are likely the result of arrested diffusion and poor inter-particle bonding due to the low temperature. This demonstrates that an important trade-off between microstructural evolution and diffusion during sintering must be considered with regard to mechanical properties. When the sample was consolidated at 1020°C, which is above the β -transus, a coarse fully lamellar structure was produced (Figure 17(c)). As discussed in section ‘Pressureless sintering of Ti metal powder’, any time Ti–6Al–4V is heated above the β -transus and cooled slowly, it is effectively β -annealed, producing this microstructure [134]. Therefore, performing HIP above the β -transus effectively eliminates any microstructural benefits that are gained by pre-alloying the powder.

In the literature, various microstructures in PA Ti–6Al–4V obtained by varying the processing parameters, such as temperature and pressure, are reported. PA powders often exhibit a martensitic microstructure within the particles due to rapid cooling

during powder production (e.g. PREP, GA, etc.) [157]. Thus, the microstructures of PA Ti-6Al-4V after HIP are characteristic of a phase transformation from martensite α' to $\alpha + \beta$ [159]. The as-sintered microstructure after HIP, however, significantly changes with the relationship of the HIP temperature to the β -transus [160]. Performing HIP at a temperature below the β transus leads to a bi-modal microstructure of equiaxed α and transformed β (lamellar α colonies). Zhang et al. proposed that the formation of equiaxed α is caused by the deformation at the inter-particle interfaces during the HIP process, which leads to localised recrystallisation [157]. This leads to grouping of the equiaxed grains along the prior particle interfaces, as shown in Figure 17(b). As mentioned above, this microstructure is different from the traditional bi-modal microstructure produced by working and duplex annealing wrought Ti-6Al-4V, where the equiaxed grains would be located at the triple points of the lamellar colonies. Additionally, characteristics of the starting PA powder also affect the resulting microstructure of the alloy. Higher strain energy stored in the powder can result in a greater degree of recrystallisation and a more equiaxed microstructure, which is preferred for ductility [150].

Vacuum hot pressing

Early work on VHP of PA Ti powders dates back to at least the 1980s [161,162]. However, recent work has been reported on VHP to produce Ti alloys from both PA [163–165] as well as BE powders [163–166]. These studies investigated CP-Ti [165,167], as well as Ti-6Al-4V [163], Ti-3Al-2.5V [166], and Ti-6Al-7Nb [164] alloys.

The four samples shown in Figure 18 are Ti-6Al-4V produced by Bolzoni et al. [163] from a study to compare the microstructure and mechanical properties of Ti-6Al-4V produced from BE and PA powders using VHP. As shown in Figure 18(a), VHP at 900°C for 1 h of BE powder leads to an inhomogeneous microstructure, due to insufficient diffusion of the alloying elements. Therefore, when using BE powders, this

method requires sintering temperatures of at least 1100°C to produce a homogenous microstructure. However, when temperatures exceeding 1100°C are used, VHP results in a coarse lamellar microstructure very similar to pressureless sintering or β -annealing (Figure 18(b)). Any type of pressure-assisted consolidation, of course, will result in better densification at lower temperatures. However, with regard to the morphology of the microstructure, there is no apparent benefit of using VHP versus conventional pressureless sintering for BE powder, as higher temperatures are still required to homogenise the alloy.

Figure 18(c) shows the microstructure of Ti-6Al-4V produced in the same study by VHP of PA powder at 900°C for 1 h. This experiment resulted in a fine equiaxed microstructure. Because the powder was PA, the microstructure is homogenous, as opposed to the corresponding sample pressed from BE powder at the same temperature (Figure 18(a)). Additionally, the sample had the best flexural strength (~ 1600 MPa) and deflection ($\sim 2\%$) of all the VHP samples presented, as determined by 3-point bend testing. It should be noted that the oxygen content was nearly 0.5 wt-% for this sample, which would account for the high strength and low ductility. For comparison, Figure 19(d) was hot pressed in vacuum at 1300°C for 30 min. As would be expected, performing VHP over the β -transus resulted in a coarse lamellar microstructure for both the BE and PA samples.

CIP-sinter-HIP process

The CIP-sinter-HIP (CHIP) process is a multi-step PM process in which a green body/preform is produced via cold isostatic pressing (CIP), conventionally sintered under vacuum, and then hot isostatically pressed to close the remaining porosity [158]. A schematic of this process is given in Figure 19. The CHIP process has enjoyed significant commercial attention. In fact, this process is currently reported to be the sole qualified PM production route by the Boeing Material Specification for PM Ti-6Al-4V [168]. Additionally, CHIP has been identified as a possible 'green'

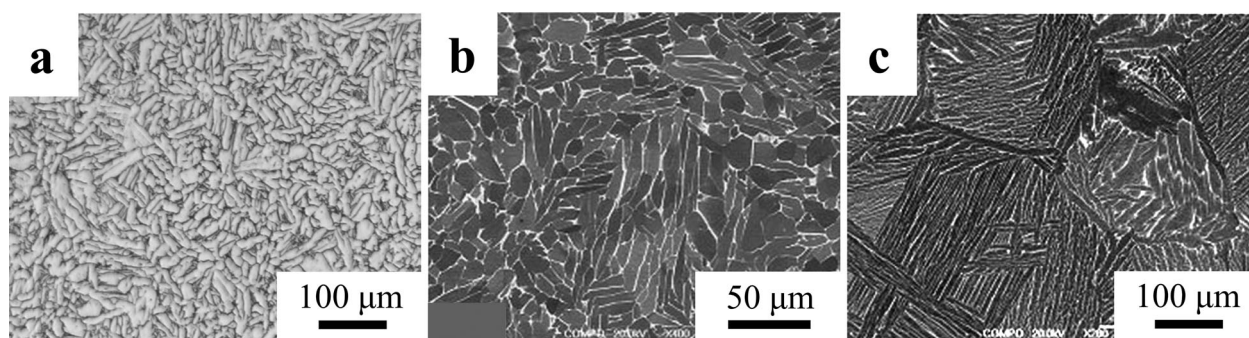


Figure 17. Microstructure of Ti-6Al-4V produced from PA powder using HIP: (a) optical micrograph of GA powder pressed at 954°C [105], (b) SEM (back scattered) micrograph of PREP powder pressed at 1020°C [157], and (c) SEM (back scattered) micrograph of PREP powder pressed at 1040°C [157]. (Reproduced with permission.)

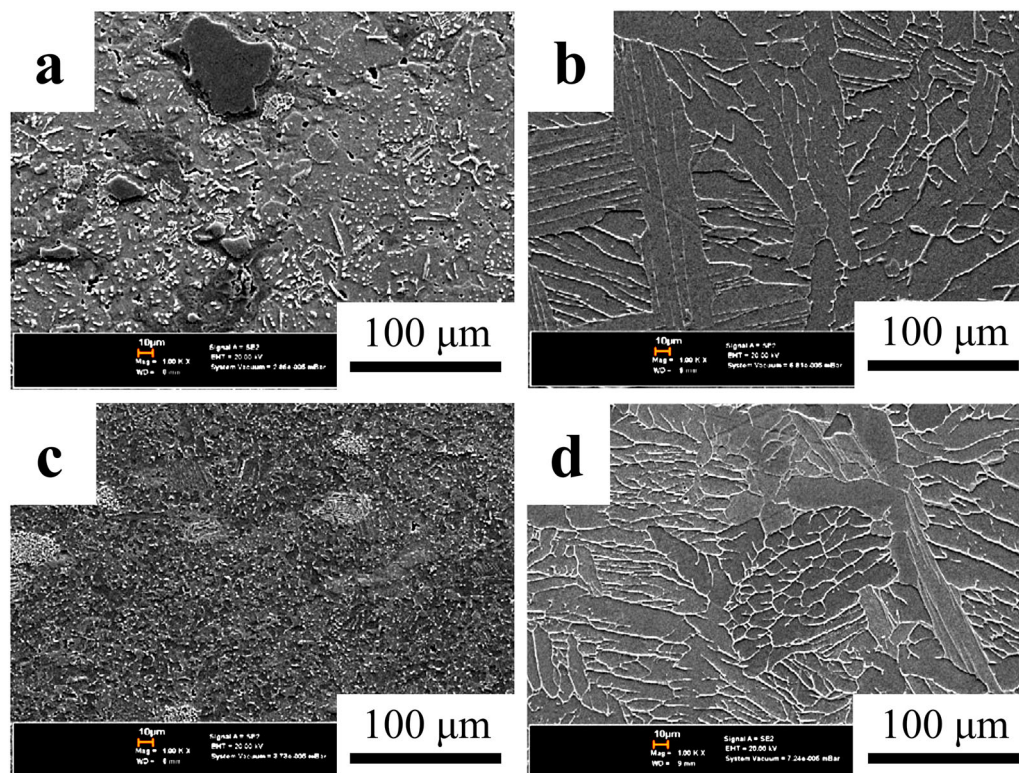


Figure 18. SEM (secondary electron) micrographs of Ti-6Al-4V produced via VHP of blended (BE) and PA powders: (a) and (b) BE pressed at 900°C for 1 h and 1300°C for 30 min, respectively, (c) and (d) PA pressed at 900°C for 1 h and 1300°C for 30 min, respectively [163]. (Reproduced with permission.)

manufacturing technology for sustainably producing Ti alloys [158,169].

As mentioned, HIP is utilised solely to close the residual porosity after vacuum sintering during the CHIP process. Therefore, the microstructure after HIP is similar to that achieved in conventional pressureless (vacuum) sintered compacts of BE powder. An optical micrograph of Ti-6Al-4V alloy produced via the CHIP process is shown in Figure 19 [158]. As seen, the microstructure is comparable in size and morphology to the lamellar structure produced via conventional pressureless sintering. As also shown in Figure 19, CHIP may also be used with subsequent TMP, such as extrusion and forging [158]. In addition to serving as a forming process, TMP can break up the coarse lamellar grains and facilitate the formation of a refined equiaxed or bi-modal microstructure through recrystallisation of the α grains. However, as pointed out earlier, TMP processes are not near-net-shape compatible, are inherently energy-intensive and, therefore, would increase the cost of the final product.

CHIP, as with other pressure-assisted consolidation, is shown to effectively produce fully dense PM Ti alloys. However, when BE powders are used, sintering temperatures well above the β -transus are required to homogenise the alloying elements. Therefore, the microstructures of PM Ti-6Al-4V produced via CHIP are similar to those available with conventional pressureless sintering. If more expensive PA powders

are used, lower sintering temperatures may be used, enabling the formation of different microstructures.

Sintering of TiH_2

As discussed in section 'Hydride-dehydride', the HDH process is a well-known, commercial process for Ti powder production. HDH takes advantage of the brittleness of Ti hydride to improve comminution behaviour in powder production [7]. However, hydrogenated Ti powder may also be used in PM consolidation processes. The use of hydrogenated Ti powder instead of metallic Ti powder has been shown to result in reduced oxygen content in the sintered alloy (see section 'Vacuum sintering TiH_2 ') [170,171]. Additionally, the brittleness of Ti hydride results in a tendency of the powder to crush during cold compaction [172,173]. This has been shown to result in better green density over Ti powder of similar particle size and morphology when using the same compaction pressures. Paramore reported that the relative green densities for compacts produced from -325 mesh ($<44 \mu m$) powder of TiH_2 and Ti yielded theoretical densities of 83 and 66%, respectively, when compacted at 276 MPa, and 87 and 73% when compacted at 414 MPa [174]. The relative densities reported in this study took into account the different theoretical densities of TiH_2 ($3.75 g cm^{-3}$) and Ti ($4.51 g cm^{-3}$). Furthermore, improved densification during sintering has

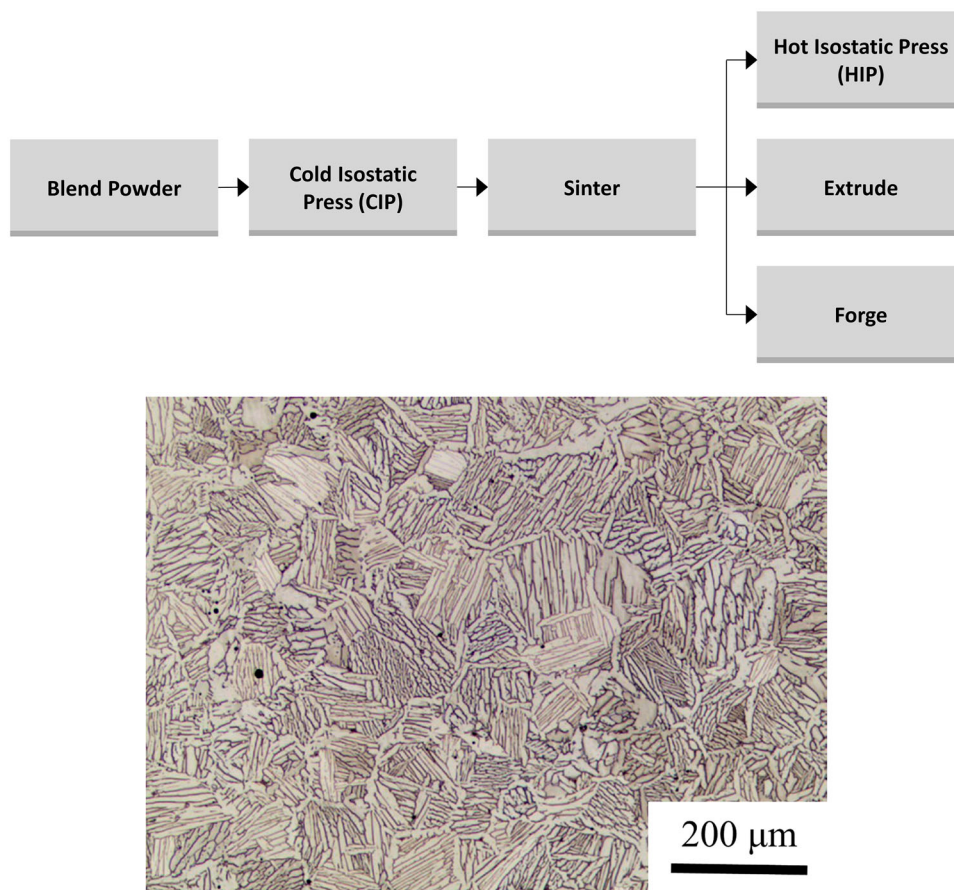


Figure 19. Flow chart of the generalised CHIP process and a representative micrograph showing the microstructure of Ti-6Al-4V after the HIP step [158]. (Reproduced with permission.)

been reported when hydrogenated Ti powder is used (see section ‘Vacuum sintering TiH_2 ’) [175]. Finally, because of the reversible nature of the hydrogenation reaction, hydrogen is easily removed from Ti by annealing under vacuum or inert gas [176,177]. For these reasons, there is a long history of using hydrogenated Ti as the feedstock material for PM processes.

The first patent to use Ti hydride powder in a sintering process, filed by Gregory, dates back to the 1960s [178]. This patent describes a method in which Ti hydride powder is used in the production of niobium-based superconductors. Shortly afterwards, a process to produce Ti alloys and metal matrix composites using Ti hydride powder was developed at the United States Army Materials and Mechanics Research Center [179,180]. This process was called ‘decomposition powder metallurgy’ and was first reported in 1970 by Greenspan et al. as a method to produce Ti alloys and metal matrix composites via hot pressing [179]. The first press and sinter process to produce Ti alloys from Ti hydride was patented by Obara et al. in 1976 [181]. In this process, sintering was achieved under vacuum or inert gas atmospheres. Following these pioneering publications and patents, a wide variety of processes have been published and/or patented that use hydrogenated Ti powder. These processes include both pressure-assisted (i.e. hot pressing or HIP) as well as press and

sinter methods, and have been used to produce a range of products including Ti alloys [173,182–188], sputtering targets [189], decorative Ti products [190], and shape memory alloys [191].

Vacuum sintering TiH_2

Since 2000, a significant amount of work regarding vacuum sintering of hydrogenated Ti powder has been reported by Ivasishin et al. [147,170,175,192–196] and Savvakini et al. [197]. It has been reported that a blend of Ti hydride with 10 wt-% 60Al/40V master-alloy powder produced Ti-6Al-4V with 98.5–99.5% TD in the as-sintered state, versus 90–95% when metallic Ti powder was used [193]. Additionally, it has been reported that using Ti hydride produced samples with an as-sintered oxygen content of 0.21 versus 0.39 wt-% when metallic Ti powder was used [198]. This work has led to a series of patented processes for the production of hydrogenated Ti powder, as well as vacuum sintering of these powders to produce Ti alloys and metal matrix composites [96–98,171,199,200].

The sintering profile used to produce Ti-6Al-4V from hydrogenated powder in these studies utilised times and temperatures similar to sintering metallic powder (1200–1350°C for 4 h) [195,201]. Therefore, as expected, this sintering process produces Ti-6Al-4V with a lamellar structure, very similar to that

achieved via a typical press and sinter process of Ti blended with elemental or master-alloy powders. Figure 20 is a micrograph showing the as-sintered microstructure of Ti-6Al-4V produced by vacuum sintering Ti hydride powder blended with 60Al/40V master-alloy [201]. This microstructure consists of α colonies in the order of 100 μm across with individual lamellae in the order of 10 μm wide and 100 μm long.

Sintering of TiH_2 in H_2

Since 2011, a process called HSPT has been developed by Fang, Sun, and Paramore et al. [4,172,174,202–213]. HSPT is a multi-step pressureless sintering process, in which Ti hydride or partially hydrogenated Ti powder is blended with master-alloy or elemental powders and sintered under a dynamically controlled hydrogen atmosphere. After sintering, residual hydrogen is removed by annealing under vacuum or inert gas (Figure 21). HSPT takes advantage of the phase transformations in the (Ti-alloy)-H systems to simultaneously refine the microstructure during the thermal cycle. A detailed description of the HSPT processing steps [207] as well as an improved (Ti-6Al-4V)-H phase diagram [210] and descriptions of the underlying mechanisms for microstructural refinement [211] are available in the literature.

The HSPT process has been specifically developed as a low-cost process to produce high-performance Ti alloys with wrought-like microstructures and mechanical properties. As such, this research has been focused on employing commercially viable low-cost PM processing, including:

- Low-cost BE powder
- Conventional cold compaction and pressureless sintering ('press and sinter')
- Only low-cost and near-net-shape compatible processing steps (no TMP such as forging and extruding)

Figure 22 shows the microstructure of Ti-6Al-4V produced via HSPT with and without subsequent heat treatments, as detailed in the literature [208]. An SEM micrograph of the as-sintered microstructure produced via HSPT is shown in Figure 22(a), while Figure 22(b) is an optical micrograph of the same material. In contrast with other Ti PM processing technologies, the HSPT process produces an ultrafine grain (UFG) microstructure with α colonies measuring several microns across and individual α grains measuring several microns long with submicron widths. Additionally, while traditional β -phase sintering produces a coarse lamellar α/β microstructure, the β phase in the HSPT material is finely divided throughout the microstructure. This is a key feature of the HSPT microstructure, as this allows for subsequent pressureless heat

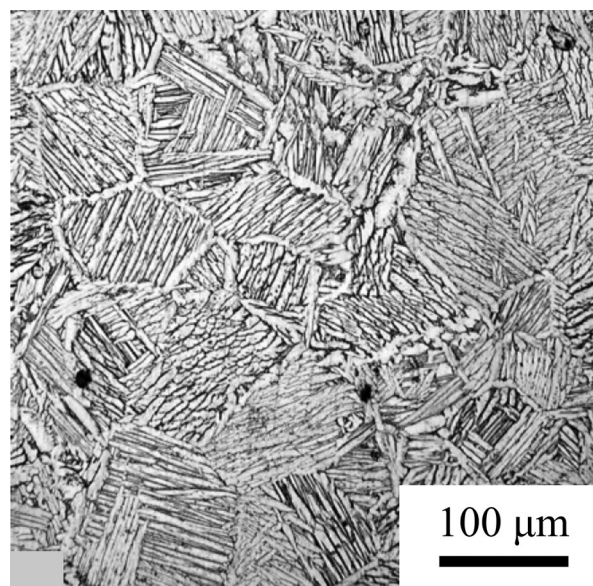


Figure 20. Optical micrographs of Ti-6Al-4V produced via vacuum sintering of TiH_2 blended with master alloy [201]. (Reproduced with permission.)

treatment steps to produce very fine wrought-like microstructures without mechanical working.

By utilising simple heat treatments on the as-sintered material, very fine wrought-like bi-modal (Figure 22(c)) and globularised/equiaxed (Figure 22(d)) microstructures are possible [208,209]. In either microstructure, the globularised α grains are on the order of 5 μm in width. It should be emphasised that in WP or traditional powder metallurgy, producing fine bi-modal or equiaxed microstructures requires energy-intensive TMP to first break up the coarse lamellar microstructure and produce plastic strain to serve as a driving force for recrystallisation, which produces the equiaxed α grains in these structures [134]. However, it has been proposed that the very fine morphology of the as-sintered HSPT microstructure allows for the formation of globularised α grains without recrystallisation [174,208,209]. Detailed discussions on the specific heat treatments used and the mechanisms behind these phenomena are available in the literature [174,207–209,211,213].

Owing to the microstructures, HSPT Ti-6Al-4V has been shown to have wrought-like mechanical properties as well [174,208,213]. As-sintered UFG HSPT Ti-6Al-4V typically has strength exceeding 1 GPa and ductility exceeding 15%EL. The properties may be application-tailored via simple heat treatments to increase the ductility beyond 20%EL or strength exceeding 1100 MPa. Additionally, HSPT has been shown to have fatigue properties competitive with wrought Ti-6Al-4V [174,213,214].

Special hot consolidation techniques

In addition to the traditional processes detailed in the previous sections, specialised consolidation and

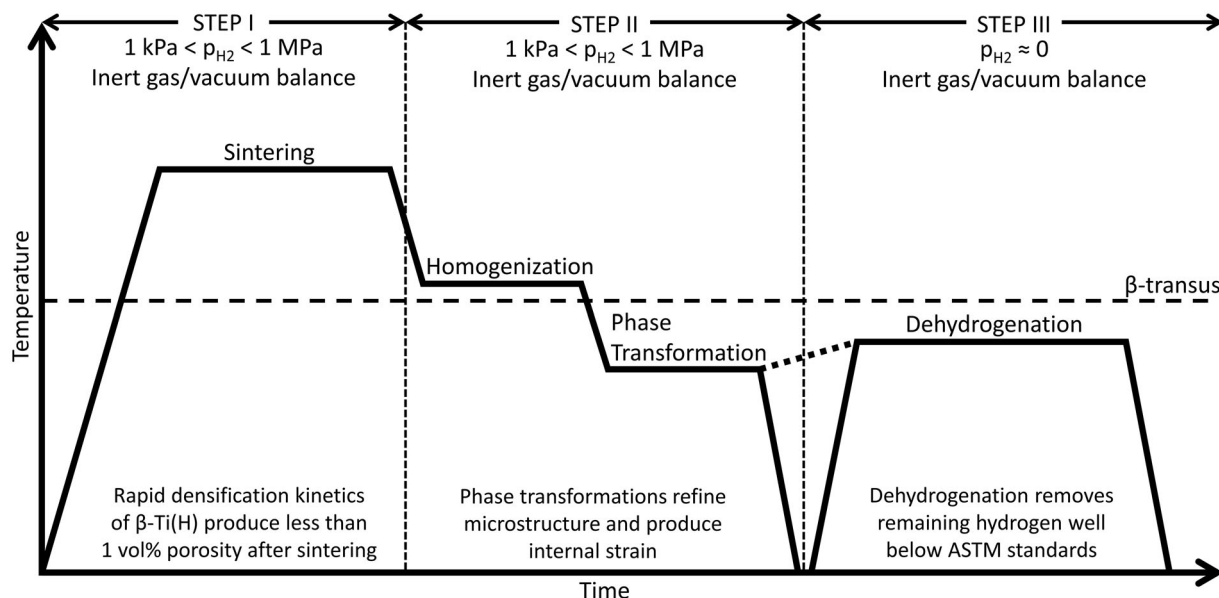


Figure 21. Schematic representation of the HSPT sintering profile.

sintering processes for the production of Ti alloys have also been reported in the literature. These processes are typically pursued to address the shortcomings of Ti PM, such as coarse microstructures from high sintering temperatures, residual porosity, poor alloying element homogeneity, and inability to cost-effectively produce certain part geometries (sheet and plate).

Microwave sintering

There have been several studies on the heating susceptibility of Ti powders to microwave radiation [215–219], the densification and microstructure of microwave-sintered Ti [218–228], and the resulting mechanical properties [218,220]. However, the heating of Ti by microwaves tends to be erratic [229]. For this reason, several studies have utilised microwave susceptors, such as SiC sample holders, to produce predictable and consistent heating of Ti powders [218–220,222,224–226]. It has also been reported that TiH₂ is capable of being heated by microwave radiation without susceptors [230,231] to produce small CP-Ti samples with improved density and mechanical properties compared to those produced from metallic Ti powder [228].

When microwave susceptors are used, the densification of Ti alloys via microwave heating appears very similar to that achieved via conventional pressureless sintering [218,225]. One study reported the production of 99% TD Ti–6Al–4V by using CIP on PA powder at 690 MPa and microwave sintering at 1300°C for less than 1 h [225]. However, it has been reported that elevated sintering temperatures (~1300°C) are required to facilitate diffusion and produce a homogenous microstructure in Ti alloys [229]. Therefore, microwave sintering typically results in the coarse lamellar structure that is common in other PM Ti–6Al–4V processes (Figure 23(a)) [218]. Figure 23

(b,c) shows CP-Ti that was sintered using microwave energy from CP-Ti and TiH₂ powder, respectively. Sintering of CP-Ti, regardless of starting powder, produces an equiaxed microstructure, which is also typical for this alloy.

Ti alloys have been reported to be susceptible to interstitial contamination from volatiles produced from the insulator/susceptor materials during microwave sintering [229]. However, the sample purity and mechanical properties can be improved by using Ti sponge as a getter to protect the samples during microwave sintering [218]. By using Ti sponge to protect the samples, CP-Ti was produced with tensile properties that fall within the range of the ASTM B348 standard [232] for Grade 2 and Grade 4 CP-Ti. Ti–6Al–4V was produced with the ultimate tensile strength (UTS) of approximately 970 MPa and approximately 7.5%EL, which are below the ASTM B348 standard for Grade 5 Ti–6Al–4V. However, better mechanical properties were reported in this study for microwave-sintered Ti–10V–2Fe–3Al (980 MPa UTS and 17.6% EL for the most ductile sample).

Spark plasma sintering

Spark plasma sintering (SPS), which is also known as the field-assisted sintering technique, plasma-activated sintering, pulsed electric current sintering, or plasma-pressure-compaction, is a sintering technology that utilises Joule heating via a pulsed electric current to achieve densification. Similar to VHP and HIP, SPS uses simultaneous heating and pressure to facilitate improved consolidation. Additionally, it has been theorised that the Joule heating could lead to further improved densification via localised plastic flow at the necks of connected particles during sintering [233,234].

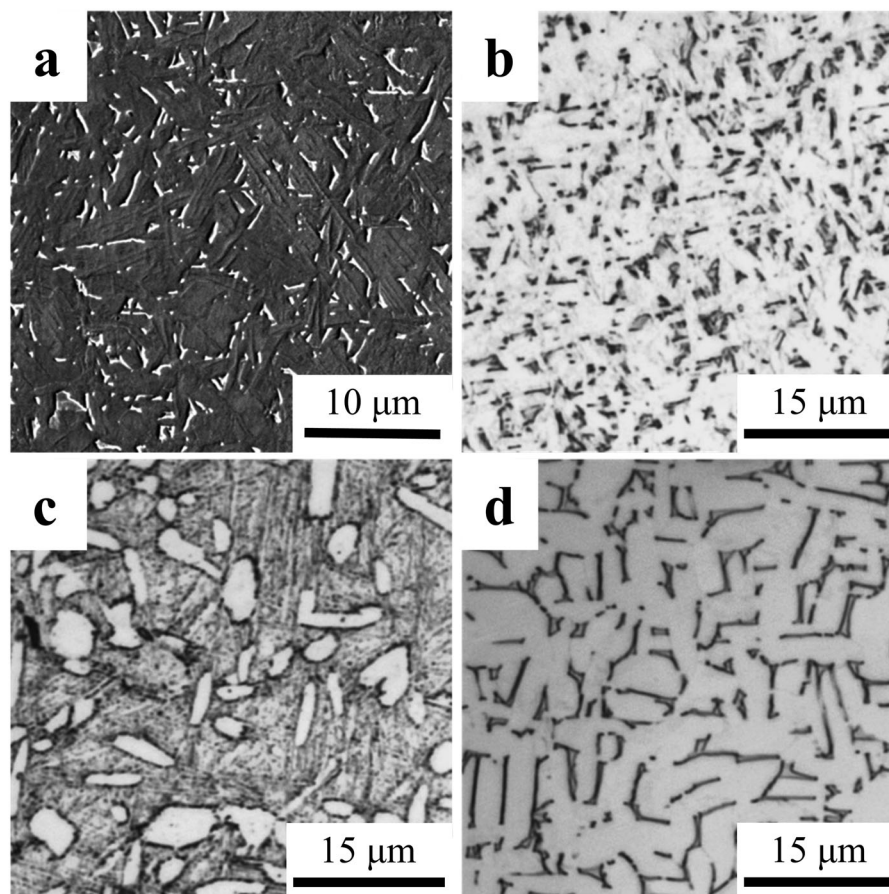


Figure 22. Micrographs of Ti-6Al-4V produced via HSPT: (a) SEM of as-sintered UFG lamellar structure, (b) OM of as-sintered UFG microstructure, (c) OM of heat-treated bi-modal microstructure, and (d) OM of heat-treated globularised microstructure [209]. (Reproduced with permission.)

It has been reported that essentially full density CP-Ti can be achieved from 45 μm HDH powder at approximately 750°C using a 50°C min⁻¹ heating rate and 100 MPa of applied pressure [235]. The mechanical properties that can be achieved using SPS of CP-Ti powders are also very promising. One study reported the ability to produce nanostructured Grade 2 CP-Ti with 840 MPa tensile strength and 27%EL via SPS of cryogenically milled CP-Ti powder [236]. These mechanical properties far exceed those designed for wrought CP-Ti [232].

Figure 24(a) shows the microstructures of Ti-6Al-4V produced via SPS of a powder blend (i.e. BE powder) of HDH Ti (48–75 μm) and 60Al-40V master-alloy [237]. This sample was sintered via SPS at 900°C for 5 min with a heating rate of 20°C min⁻¹ under 30 MPa of pressure. At this time, temperature and pressure, appreciable densification is achieved. However, this sample had an inhomogeneous microstructure [237]. Figure 24(b) shows the same blended powder sintered at 1100°C for 15 min with the same heating rate and applied pressure. For comparison, a sample prepared with the same SPS parameters, but from PA Ti-6Al-4V powder, is shown in Figure 24(c) [237]. All three samples exhibited appreciable densification. However, when a sintering temperature below

the β -transus is used, regardless of whether BE or PA powder is used, an inhomogeneous microstructure is produced. This is indicative of insufficient diffusion of the alloying elements during sintering at lower temperatures.

From current literature, SPS appears to be promising for CP-Ti. However, high temperatures are still required to produce a homogeneous microstructure in Ti alloys due to the necessity for diffusion of the alloying elements. Therefore, this process will normally result in the same relatively coarse lamellar microstructure observed from traditional sintering of Ti-6Al-4V. However, it is important to note that the temperature required for sufficient diffusion of alloying elements during SPS is apparently lower than that required for other pressure-assisted sintering technologies, such as VHP. It has been theorised that the Joule heating caused by SPS may have some beneficial effect on the diffusion of alloying elements during Ti-6Al-4V sintering from blended powders [238].

Powder rolling

Certain geometries, such as sheet, strip, and plate, are not suitable for production via traditional powder metallurgy consolidation techniques. Therefore, the process of powder rolling has been developed to

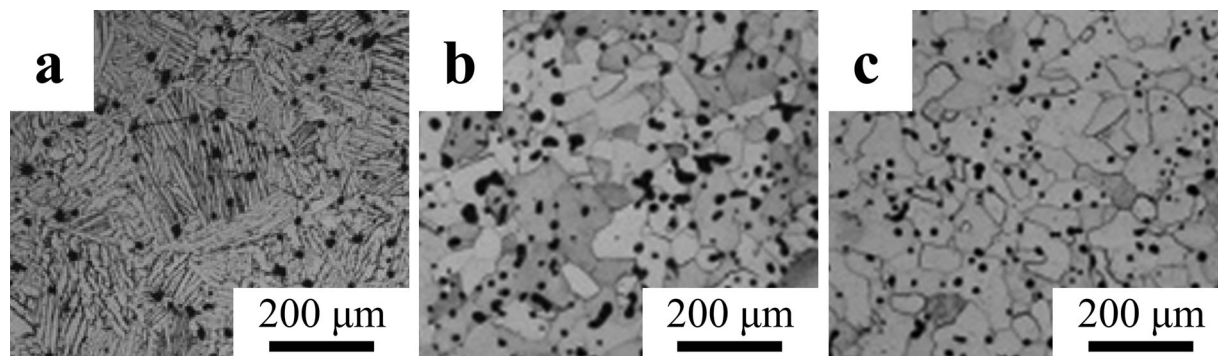


Figure 23. Microstructure of Ti alloys sintered by microwave radiation: (a) Ti-6Al-4V produced from blended Ti and master alloy powder sintered at 1300°C for 2 h [218], (b) CP-Ti produced from Ti powder sintered at 1300°C for 1 h [228], and (c) CP-Ti produced from TiH₂ powder sintered at 1300°C for 1 h [228]. (Reproduced with permission.)

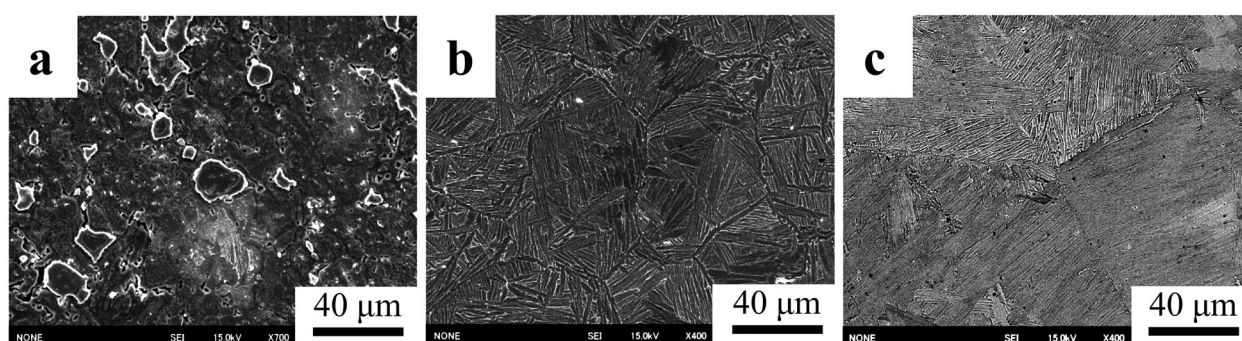


Figure 24. SEM micrographs of Ti-6Al-4V produced via SPS: (a) blended master alloy powder at 900°C for 5 min, (b) blended master alloy powder at 1100°C for 15 min, and (c) PA powder at 1100°C for 15 min [237]. (Reproduced with permission.)

address this limitation. This process has been studied for several alloy systems, including Ti, since 1950 [239]. As such, the ASM handbook on powder metallurgy has a section dedicated to powder compaction by rolling [240]. DuPont developed a commercial process for producing Ti strip via powder rolling in the 1960s [141]. However, despite exhibiting promising mechanical properties, the DuPont strip Ti exhibited poor weldability due to chloride impurities in the final product.

In one study, a green part was produced by direct powder rolling low-cost elemental Ti powder followed by vacuum sintering [241]. In this study, green CP-Ti strips measuring up to 1.5 mm thick and up to 300 mm wide were produced by direct powder rolling. The green parts were sintered under vacuum at 1000–1250°C for 1–4 h. After sintering, the strips were rolled again and annealed to produce Ti sheet measuring approximately 0.5 mm in thickness. Promising mechanical properties were reported, which were dependent primarily on the annealing treatment used after sintering.

Another process has been under recent development by CSIRO in Australia for the production of thin gauge Ti alloy sheets from PA or BE powders [242–245]. The CSIRO process utilises several steps

to produce fully dense Ti alloy sheets. These steps include direct powder rolling, heating under inert gas for hot rolling, mill annealing, final cold rolling, and annealing (Figure 25) [245]. After annealing, the strips are generally treated to remove a thin layer (0.05 mm) from each surface. These steps can be accomplished in a manner that allows for a continuous production process, if implemented commercially.

The microstructural evolution of CP-Ti during the CSIRO processes is shown in Figure 26. It has been reported that typical irregularly shaped HDH CP-Ti powder can be formed into CP-Ti sheet with mechanical properties approaching those available via WP and with similar purity (0.12 wt-% O, 0.013 wt-% N, and 10 ppm H) [245]. This particular study reported a UTS of 517–520 MPa and ductility of 20–27%EL, depending on the testing direction. Greater strength was exhibited when tested perpendicular to the rolling direction, while greater ductility was exhibited when tested parallel to the rolling direction.

Forging

Forging has been used in PM Ti as a means to close porosity or refine the coarse microstructure through TMP. In fact, it has been traditionally thought that achieving superior mechanical properties of PM Ti

alloys would require some form of TMP to be competitive with wrought [246]. During powder forging, a green preform is prepared via die pressing or CIP. The preform may be forged directly, or first sintered prior to die forging. Ideally, the preform will have a geometry that is conducive to producing the desired final net shape after forging. The forging step serves to close residual porosity and refine the microstructure. Forging of PM components has been shown to significantly improve the mechanical properties of PM parts, particularly the fatigue performance. Therefore, this process is currently used commercially for the production of high-strength iron alloy parts for drive train components (e.g. connecting rods) for automobiles. This process has also been investigated for the production of Ti alloy connecting rods [247]. Additionally, powder forging has been patented by Toyota for the production of Ti metal matrix composite engine valves [142,248,249]. Another study has investigated the process of producing billets for extrusion from TiH_2 powder via vacuum sintering [250].

Zhang et al. investigated the ability to produce Ti-6Al-4V by preheating a green part of BE or PA powder to 1200°C and immediately forging. While the microstructure produced by this method was much finer than that typical of traditional sintering, the forgings exhibited cracking, especially at the edges, which appeared to negatively impact ductility. However, two other studies used traditional sintering to produce the preform, followed by forging and heat treatment with better results [247,251]. Figure 27 shows the microstructure of a PM connecting rod of Ti-1.5Fe-2.25Mo that was first sintered at 1300°C before forging and heat treatment. As seen, the as-sintered sample has the coarse lamellar microstructure that is typical of PM $\alpha + \beta$ Ti alloys (Figure 27(a)). However, after forging, the microstructure is significantly refined in all areas of the connecting rod (Figure 27(b)), with the finest

microstructure found in the shank (Figure 27(c)) [247]. Similar results were observed in a study on PM Ti-10V-2Fe-3V sintered at 1200°C followed by forging and heat treatment [251].

Gaseous isostatic forging

Gaseous isostatic forging (GIF) technology, also known as pneumatic isostatic forging (PIF), is a post-sintering process to increase the density of a sintered article [252–255]. GIF is similar to HIP, in that it uses isostatic pressure at elevated temperatures to close pores. However, unlike HIP, GIF utilises a separate furnace to heat the samples before pressurisation [256]. The fact that the pressure chamber is not heated during GIF allows for maximum pressures above 400 MPa, which are significantly higher than what is typically possible with HIP. Furthermore, because the heating and pressurisation are separate processes, the apparatuses used for this process are also much less complicated and do not require expensive materials and designs capable of handling high pressure at elevated temperatures. Additionally, the use of a separate furnace means the samples can be heated rapidly and then quickly pressurised and depressurised, allowing for much faster production rates. This also means that the processed part is at high temperature for only a short time. Therefore, as the name suggests, this process is much more like a forging process, though it is isostatic and is, therefore, better suited for processing of near-net-shape PM parts with complex geometries. This process was first patented in 1998 [257] and has been commercialised to close porosity in parts produced by casting [258] or PM processing [257].

As reported in the literature [257], Ti alloys are typically heated to a temperature between 845 and 900°C during GIF. After heating, the material is loaded into the pressure vessel. Typically, a maximum pressure between 300 and 415 MPa can be achieved with a

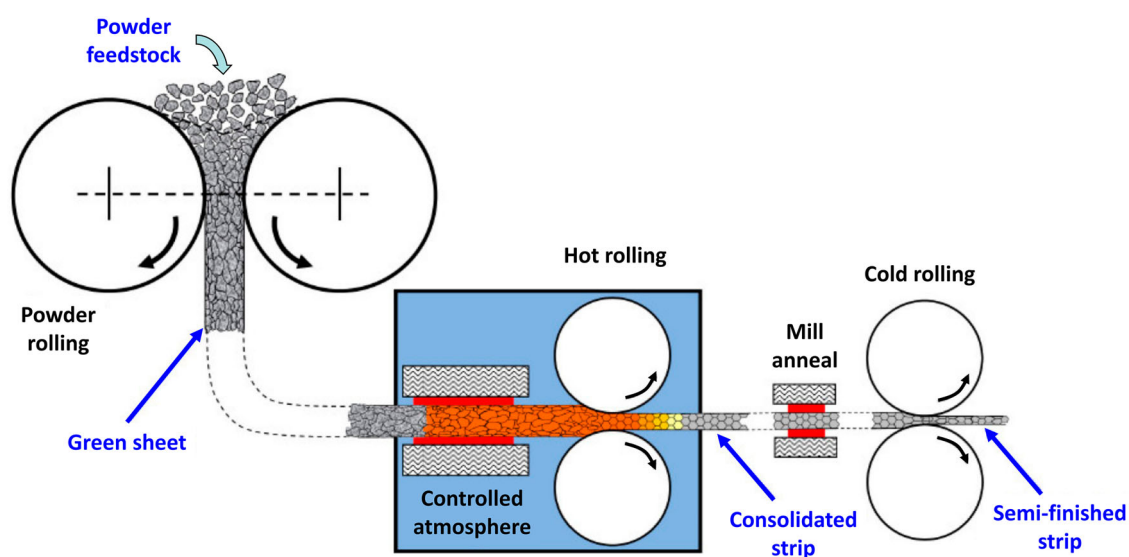


Figure 25. Schematic diagram of CSIRO process for producing CP-Ti strip from metal powder [245]. (Reproduced with permission.)

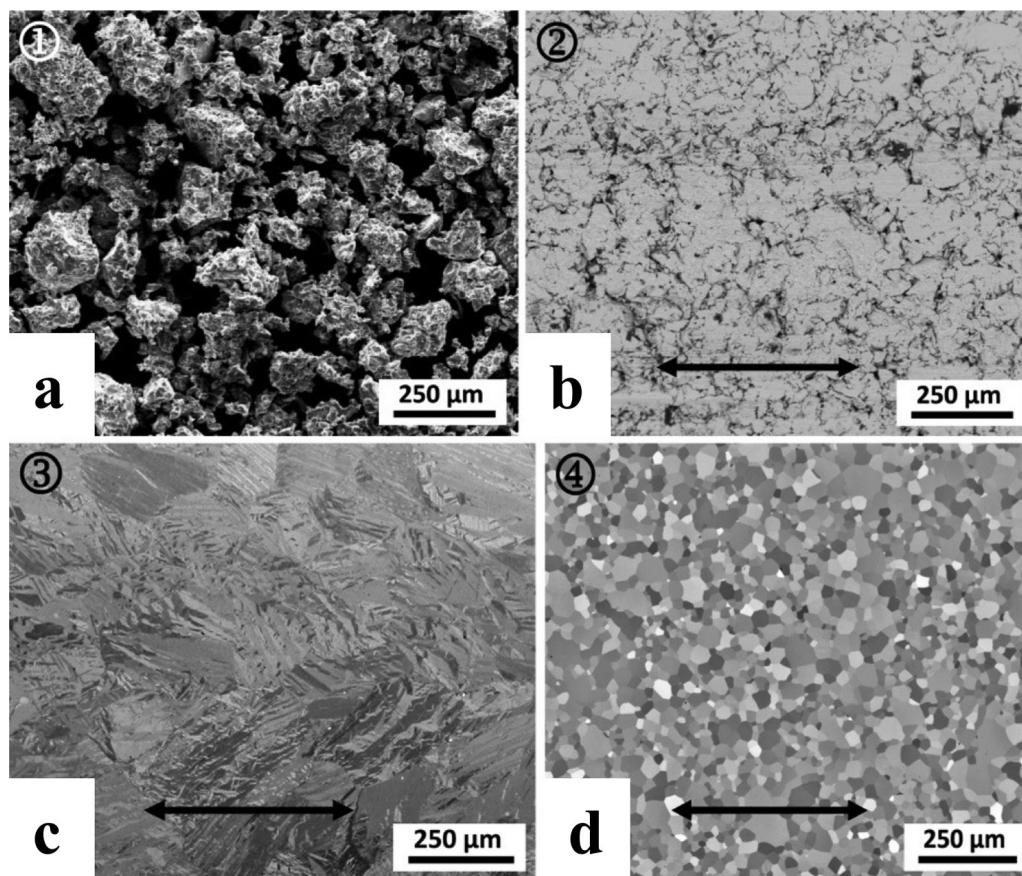


Figure 26. Microstructural evolution of CP-Ti during the CSIRO process (arrows indicate rolling direction): (a) HDH CP-Ti powder feedstock, (b) as direct powder rolled, (c) as hot-rolled, and (d) fully recrystallised structure after annealing [245]. (Reproduced with permission.)

pressurisation rate as fast as 28 MPa s^{-1} . The initial rapid pressurisation has been called an ‘air hammer’, which is achieved by opening a valve between the pressure chamber and a pre-pressurised gas reservoir. Super-transus ($>1000^\circ\text{C}$) GIF treatments of Ti–6Al–4V have also been reported [252]. A simple schematic showing the primary steps of the GIF process is shown in Figure 28.

The primary advantage of GIF in PM processing is to close the porosity of the as-sintered material in a truly low-cost and near-net-shape compatible method that is capable of high production rates. Owing to its

isostatic nature, GIF does not produce shear, meaning it is not capable of refining the microstructure due to recrystallisation. Figure 29 shows the microstructures of PM Ti–6Al–4V that have been traditionally β -forged (Figure 29(a)) versus GIF (Figure 29(b)) [252]. As shown, the GIF microstructure is unchanged by this process and has the as-sintered coarse lamellar microstructure that is typical of PM Ti–6Al–4V. On the other hand, the traditionally β -forged microstructure is significantly finer than the GIF microstructure, as dynamic recrystallisation during hot working above the β -transus temperature refines the microstructure.

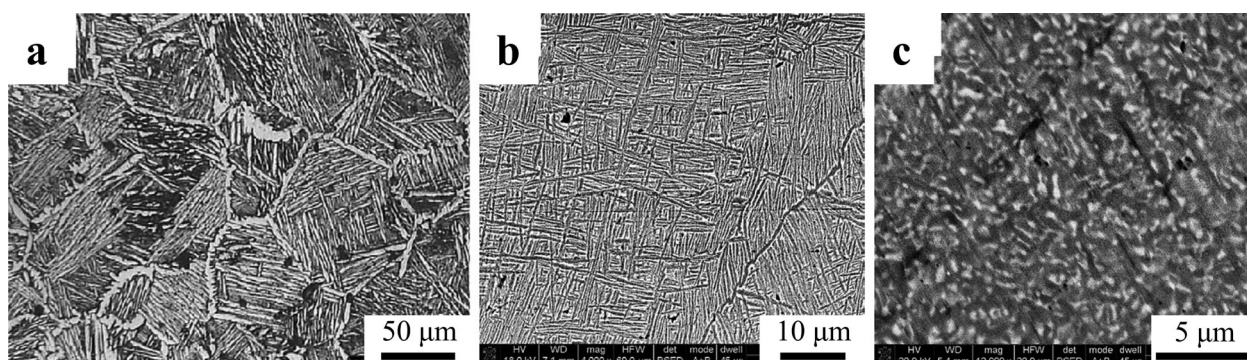


Figure 27. Microstructures of Ti–1.5Fe–2.25Mo connecting rod produced by powder forging: (a) OM of as-sintered material, (b) SEM of forged and heat-treated fork, and (c) SEM of forged and heat-treated shank [247]. (Reproduced with permission.)

Special forming and consolidation technologies

In addition to traditional powder metallurgy methods, PM Ti can be manufactured by special forming and consolidation technologies. Two important near-net-shape technologies, MIM and AM, are briefly reviewed with respect to Ti manufacturing in the following sections.

MIM of Ti

During MIM, a binder consisting of waxes and thermoplastics is mixed with metal powder and formed in a process similar to that used in the plastics industry. The binder must produce sufficient flowability of the mixture at the injection temperature, as well as produce sufficient green strength upon cooling. The MIM process consists of the following steps: preparing feedstock, injection molding, de-binding (solvent and thermal), and sintering. Compared to conventional PM methods, the MIM technology has advantages for mass production of parts with complex shapes and relatively small size, but it has dimensional limitations as well as a risk of contamination and distortion due to the adding and the removal of a large amount of binder. The application of the MIM method to Ti was

first reported by Kaneko et al. in Japan in the late 1980s and early 1990s [259–261]. As a result of consistent development for more than two decades, many Ti components produced via MIM are commercially available today. Additionally, several comprehensive review papers and book chapters have been published on MIM of Ti in the past decade [262–266].

As with other PM processes, the three critical factors which determine the mechanical properties of MIM parts are interstitial impurity concentrations, sintered density, and microstructure. A higher density can be obtained when finer Ti powder is used. On the other hand, finer powders face a higher risk of contamination, especially by oxygen and nitrogen, during handling and sintering due to a larger specific surface area. Therefore, $<44\text{ }\mu\text{m}$ (-325 mesh) spherical Ti powder is usually considered an acceptable compromise with regard to sintered density and interstitial concentrations [187,265]. As discussed in section ‘Atomization’, spherical Ti powder produced by GA or PREP methods is expensive. This consideration has motivated research into the viability of using less expensive raw materials, such as HDH Ti [267–270] or TiH_2 [177,191,260,267,271–273]. The irregular particle shape of HDH Ti or TiH_2 powder generally results

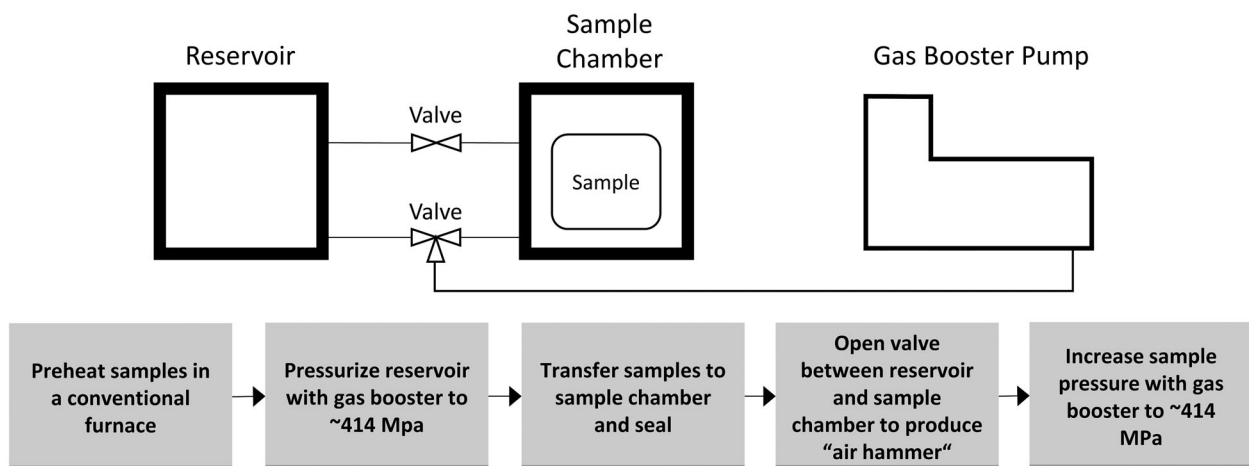


Figure 28. Schematic representation of a GIF process.

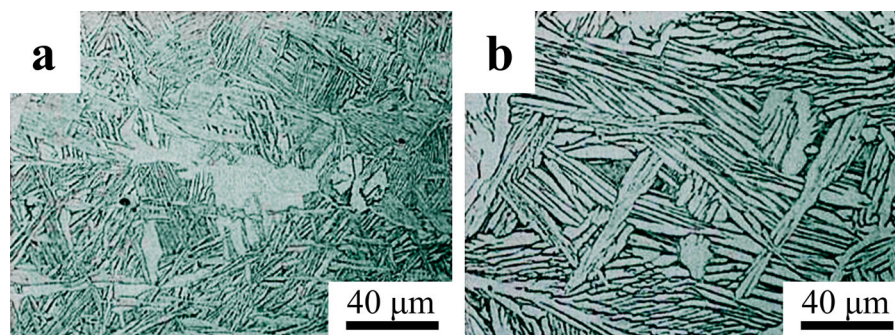


Figure 29. Microstructure of PM Ti-6Al-4V after post-sintering forging treatments: (a) traditional β -forging, and (b) pneumatic isostatic forging (PIF/GIF) [252]. (Reproduced with permission.)

in poor flowability of the powder/binder mixture during injection molding. However, it has been proposed that mixing HDH powder with atomised powder could be an effective approach to balance the cost and rheological factors [266,274].

The pickup of interstitial elements is inevitable to some degree during any MIM process. Typically, oxygen pickup is about 0.08 wt-% during MIM of Ti, but it can be reduced to 0.03–0.05 wt-% by minimising oxygen exposure of the powder during all production steps [265]. Therefore, the oxygen content of the starting powder must be sufficiently low to account for this pickup and meet the desired purity of the final parts [232]. Another important challenge for MIM of Ti is controlling the carbon contamination from the binders. Typically, the carbon pickup can be controlled to less than 0.05 wt-% during the entire MIM process [266]. Therefore, Ti carbide is typically absent from Ti–6Al–4V and CP-Ti produced via MIM, as long as the final carbon concentration is kept below 0.08 wt-%. However, the carbon concentration is of particular importance for β -Ti alloys. For example, as the concentration of molybdenum (a common β stabiliser) is increased from 0 to 15 wt-%, the solubility of carbon decreases from 800 ppm (0.08 wt-%) to 60 ppm [275,276]. Therefore, the formation of Ti carbide has been observed by Zhao et al. [277,278] and Yan et al. [275] in β alloys with trace carbon concentrations produced via MIM.

The as-sintered density of MIM Ti–6Al–4V is generally in the range of 96–97%TD with reasonable purity. However, higher relative densities (>98%) can be achieved by using finer powder, though doing so compromises purity [265]. HIP is an effective way to close residual porosity after MIM and thus improve both tensile and fatigue properties. However, as mentioned previously, doing so increases the cost of the final product. Conversely, GIF, as discussed in section ‘Gaseous isostatic forging’, may be a low-cost alternative to HIP for increasing the density of MIM Ti alloys. Additionally, shot peening has been reported as an economic way to improve the mechanical performance of MIM Ti parts by eliminating the sites of surface crack initiation [279,280].

The main alloys for MIM Ti parts are CP-Ti and Ti–6Al–4V. However, research has increasingly been reported on MIM of β -Ti alloys for potential medical applications [275,281–284]. MIM has also been investigated for producing Ti aluminides for high-temperature application, [285–290]. Additionally, the possibility of using MIM with particulate space holders as a means to produce porous Ti implants has been explored in recent years [291–293].

AM of Ti

AM, or 3D printing, is ideal for low-volume production of complicated components [294]. Therefore, this

technology is very well suited for many complicated and low-volume Ti components [295]. The topic is discussed briefly in this paper, as it holds unique promise for PM Ti. However, the reader is referred to the references cited in this section for detailed discussions on this topic.

AM of metals most commonly utilises selective sintering/melting via a laser or electron beam to build each part layer, either via a powder bed (powder bed fusion) that is replenished between layers or direct feeding of metal to the heat source (directed energy deposition) [295]. Another technology, called binder jetting, uses an inkjet print head to deposit binder on a powder bed in consecutive layers to produce a green part [296]. This technology requires subsequent debinding and sintering. However, using AM to produce a green part can take advantage of microstructural evolution during sintering, can use powders that are difficult to selectively sinter, and produce truly isotropic properties, which is difficult with selective sintering due to the complex thermal histories of each layer.

Owing to the promise of AM for small-volume production, significant research focus and attention has already been given to AM Ti–6Al–4V. In fact, an Aerospace Materials Specification for AM Ti–6Al–4V has been released for direct metal deposition of Ti–6Al–4V [297]. This specification requires an UTS for AM Ti–6Al–4V of 889 MPa and a ductility of 6%EL, taking into account the anisotropic properties. Despite being anisotropic, the currently reported mechanical properties of direct metal deposition of Ti–6Al–4V and other AM processes are generally promising. Post-processing such as HIP is required for most AM processes to produce Ti–6Al–4V with ductility beyond 10%EL. However, the reported tensile strengths in the as-printed Ti–6Al–4V range from just over 1000 MPa to just over 1200 MPa, with slight decreases in strength after HIP [295,298–300].

On a per-pound basis, AM is relatively expensive. However, this cost is easily offset by the ability to rapidly prototype unique parts, thereby eliminating setup and tooling costs, and the ability to produce near-net-shape products. For AM, spherical PA powders are typically preferred for their improved flowability. However, recent work has demonstrated the ability of AM to be used with lower-cost powders in some instances [301,302].

Mechanical properties of PM Ti

As discussed in section ‘Introduction’, PM Ti processing generally uses either BE or PA powder. As will be shown, PA tends to produce significantly improved mechanical properties compared to the BE method. However, PA powders tend to be expensive and require high-pressure consolidation processes. As discussed in section ‘Sintering of TiH₂’, significant breakthroughs

have been achieved using TiH_2 in place of Ti powder in BE processing. Therefore, the mechanical properties that have resulted from these innovations will be discussed separately from the approaches using traditional Ti powder.

In this section, the mechanical properties will be discussed in relation to porosity, purity, and microstructure of the sintered alloys. Additionally, this section will focus primarily on the mechanical properties of Ti-6Al-4V, which can be engineered to have a wide range of properties. Therefore, this alloy is ideal for critically reviewing how processing affects the mechanical properties of PM Ti. Finally, the following discussion on the mechanical properties of PM Ti is divided into two sub topics: static tensile properties (section ‘Static properties’) and fatigue properties (section ‘Fatigue properties’).

Static properties

WP is able to produce Ti-6Al-4V with a wide range of tensile strengths and ductility, owing to the range of microstructures that are possible via TMP and subsequent heat treatments. For example, quenching Ti-6Al-4V from the β phase field will produce a martensitic microstructure with very high strength and limited ductility [303], while producing a microstructure with coarse α grains will produce increased ductility at the expense of tensile strength [304]. Additionally, interstitials such as oxygen can significantly strengthen the alloy, though an excess will compromise ductility [134,150,305]. By the ASTM B348 standard, the acceptable oxygen content for wrought Ti-6Al-4V is below 0.2 wt.% [232]. However, recent work has determined that higher levels may be acceptable with respect to ductility of PM Ti alloys (see section ‘Static properties of PM Ti-6Al-4V using BE powders’).

The mechanical properties of PM Ti-6Al-4V, as reported in the literature, are summarised in the property maps given in Figure 30 and compared with literature-reported values for wrought Ti-6Al-4V [3,129,137,138,154,157,160,161,174,175,192,193,204,207,212,303–343]. The type of powder used, the sintering process, and any post-processing will affect the resulting mechanical properties. Therefore, the property maps have been broken into three different figures, representing the mechanical properties reported when BE Ti powder, PA Ti powder, and BE TiH_2 powder are used. Additionally, each property map shows several fields, corresponding to the type of sintering used with each type of powder and whether or not post-processing was used, such as HIP, TMP, and/or heat treatments. The effect of the feedstock and processing on the density, purity, and microstructure and the resulting tensile properties will be discussed in the following sections. While this review will provide a concise overview of the tensile properties reported for

PM Ti-6Al-4V, the authors refer the reader to a comprehensive review recently published by Kumar and Chandran [344] for a more detailed discussion on this particular topic.

Static properties of PM Ti-6Al-4V using BE powders

Conventional cold compaction pressureless sintering of PM Ti-6Al-4V produced from BE Ti powder tends produce samples with relatively high amount of porosity, compared with the other processing routes discussed in this section. Both the tensile strength and ductility are strongly dependent on the density of the material. Therefore, as shown in Figure 30(a), PM Ti-6Al-4V produced from BE Ti powder has the poorest mechanical properties, particularly when no additional processing is utilised to increase the density of the alloy. In fact, much of the data reported for this process fails the ASTM B348 standard for Grade 5 titanium produced via WP (895 MPa UTS and 10%EL) [232]. Kumar and Chandran found that when comparing reported densities to tensile properties, there was more scatter among samples with higher relative density [344]. They attributed this finding to the occurrence of large rouge pores in the microstructure of some samples despite the relatively high density.

Applying pressure-assisted compaction such as HIP after sintering (e.g. the CHIP process discussed in section ‘CIP-sinter-HIP process’) has been reported to increase the as-sintered density from 98 to 100% TD, with a corresponding slight increase in ductility from 15 to 16%EL and a 13–14 MPa increase in strength [158]. Additionally, studies have reported success in increasing the sintered density of PM Ti-6Al-4V by cold working the BE powder compacts before sintering [308,317]. However, due to the deformation of the green part, such a process would not be feasible for near-net-shape production of complex geometries, so the applicability of such a process for PM is limited.

Chlorine content also has important implications for the tensile properties of BE Ti-6Al-4V. As opposed to oxygen, chlorine, which is often leftover in sponge as chlorides from the Kroll or Hunter process, has no documented positive effects on the tensile properties of PM Ti-6Al-4V and can have a deleterious effect on densification [128]. In fact, one paper reported a maximum as-sintered density of 95% TD when using sponge fines, which resulted in ductility of only 6% EL [310]. Conversely, in a later study, 98% TD was reported in the as-sintered state using extra low chlorine Ti powder produced by the HDH process, which resulted in an increase in ductility to 13%EL [314].

By the ASTM B348 standard, oxygen is typically limited to 0.2 wt-% in wrought alloys to avoid loss in ductility [232]. However, as shown in Figure 31, recent studies have determined that oxygen actually has a negligible effect on the ductility of PM Ti-6Al-4V up to

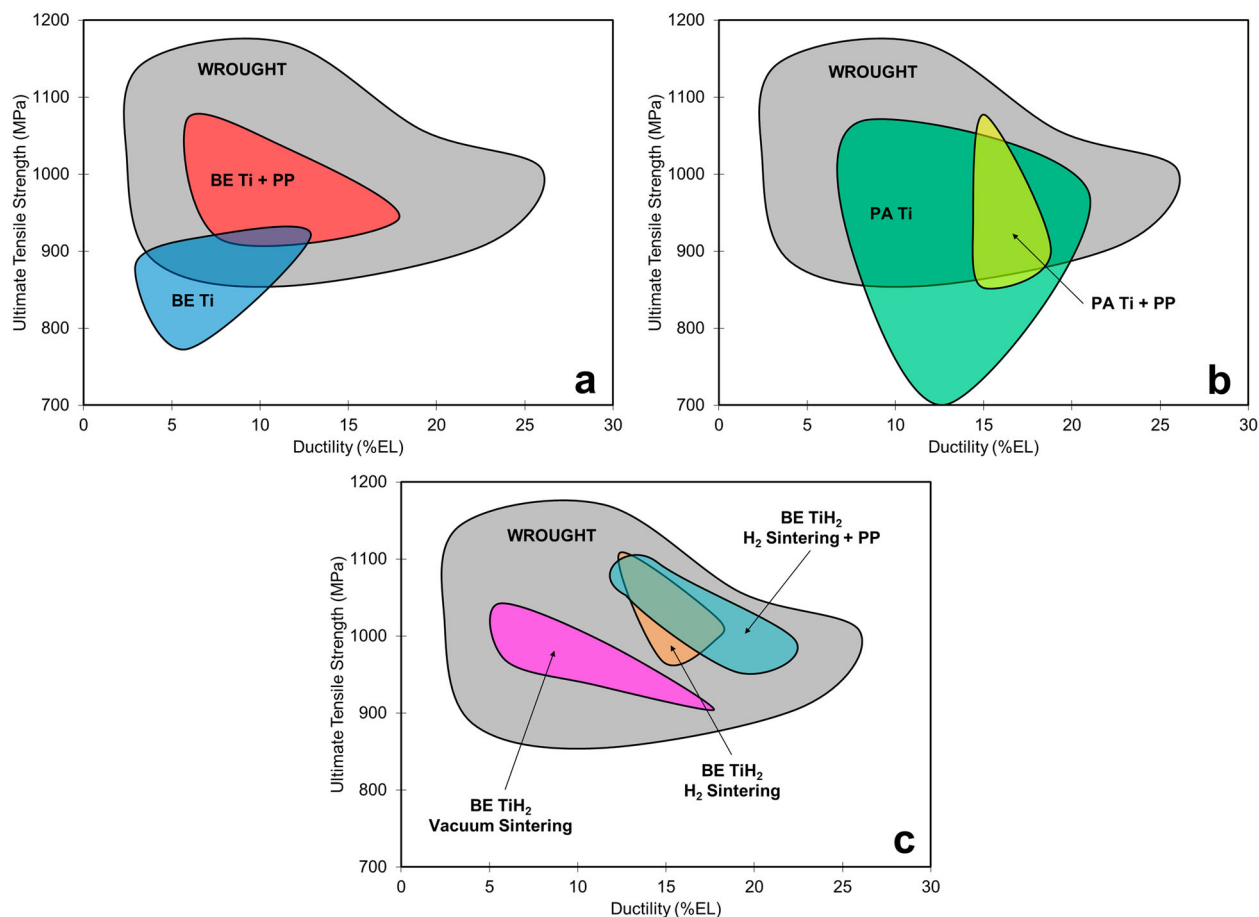


Figure 30. Mechanical property maps showing fields for the various PM processing routes versus WP: (a) using BE Ti powder with conventional sintering in the as-sintered state versus with subsequent post-processing (PP), (b) using PA Ti powder with pressure-assisted sintering in the as-sintered state versus with subsequent post-processing, (c) using BE TiH₂ powder with vacuum sintering versus H₂ sintering (section 'Sintering of TiH₂') as well as H₂ sintering with subsequent post-processing [3,129,137,138,154,157,160,161,174,175,192,193,204,207,212,303–343].

about 0.33 wt-% [345–348], which is a 65% increase over the traditionally accepted limit. Beyond the critical oxygen level of 0.33 wt-%, a sharp decrease in tensile ductility is observed with increased oxygen content. The data in these papers were generated from powder injection molding studies, though the consolidation method should be of little consequence. Oxygen is well-known to be an alloying element with strong solution strengthening behaviour of Ti alloys which is also reflected in the data shown in Figure 31. Furthermore, oxygen content is directly related to the particle size (i.e. surface area) of the feedstock powder [5], which, in turn, has a strong effect on the as-sintered density of PM Ti. That is, finer powders have greater specific surface area/energy, meaning densification occurs more readily. The general trend for the dependence of the as-sintered oxygen content and density of PM Ti–6Al–4V on the specific surface area (particle size) of the starting powder is highlighted in Figure 32 [151]. These particular data were generated by milling TiH₂ in heptane for different lengths of time, blending with master alloy, and consolidating using cold compaction and pressureless sintering. However, the general trend between particle size, oxygen concentration, and

densification is consistent with essentially all PM methods. With all of these considerations, the ability for PM Ti alloys to have higher oxygen concentrations (up to 0.33 versus 0.2 wt-%) without sacrificing ductility may create the possibility for greater solution strengthening and improved as-sintered density and, therefore, ductility. As such, these findings have potentially significant implications for PM Ti alloys.

Owing to the strong strengthening behaviour of oxygen, intentionally introducing oxygen into PM Ti alloys has been investigated as well [349–351]. Sun et al. [351] reported that by allowing CP-Ti powder to pick up oxygen during warm compaction, elevated oxygen levels could be achieved that produced significantly improved tensile properties. It was reported that CP-Ti with 0.64 wt-% oxygen had a tensile strength of 974 MPa and ductility of 26%EL. However, all the samples tested in this study were extruded to a ratio of 37. Therefore, it is possible that the promising properties reported in this paper are reliant on post-sintering TMP. The same group has used a similar method to strengthen Ti–6Al–4V with oxygen for aerospace and defense applications [350]. The elevated oxygen levels were produced by mixing Ti–6Al–4V and TiO₂

powders, which were subsequently spark plasma sintered (see section ‘Spark plasma sintering’) and then extruded to form the test material. The results obtained from oxygen strengthened Ti-6Al-4V were less impressive than CP-Ti. The strongest sample had a UTS of 1286 versus 1101 MPa for standard Ti-6Al-4V sample (17% increase), while the same two samples had a decrease in ductility from 21.2%EL for the standard sample to 8.5%EL for the high oxygen sample (60% decrease). One sample in the study exhibited a slight increase in both strength and ductility, though the remaining samples had a loss in ductility that was consistently more dramatic than the increase in strength.

Because of the effects that excessive oxygen or chlorine can have deleterious effects on mechanical properties, several studies have endeavored to utilise oxygen and chlorine scavenging alloying elements to improve tensile properties. In these studies, rare earth compounds, such as yttrium hydride (YH_2) [352,353], lanthanum hydride (LaH_2) [352], and lanthanum

boride (LaB_6) [342], have been added to Ti alloy powder compacts. It has been reported that these additions can successfully getter oxygen and chlorine, with a corresponding increase in tensile ductility. However, excessive additions can also lead to a decrease in tensile properties beyond a critical concentration of the gettering element. It has also been reported that, under certain circumstances, less expensive alloying elements may have beneficial effects on PM Ti alloys [354]. For example, Yan et al. [355] postulated that the reaction kinetics between Ti and Fe are accelerated when TiH_2 powder is used as the feedstock in a PM process. Therefore, it was proposed that this phenomenon could enable the use of Fe to scavenge oxygen in PM Ti alloys via the formation of submicron $\text{Ti}_4\text{Fe}_2\text{O}$ particles.

The microstructure of BE Ti-6Al-4V is typically a coarse lamellar structure, which results from supertransus sintering. This microstructure limits the mechanical properties that are possible, though the effect is greater on fatigue performance than static properties. However, because of this, even when post treatments are used to close porosity and the purity is effectively controlled, Ti-6Al-4V produced from BE Ti still has a limited range for tensile properties if the microstructure is unchanged. The as-sintered microstructure may be refined via TMP, which breaks up the coarse structure and drives the formation of a finer and more equiaxed grains through recrystallisation. Additionally, TMP will effectively increase the density of the material, which also positively affects tensile properties. However, TMP is energy-intensive and incompatible with near-net-shape manufacturing, which drives up the cost of the final product and is generally unsuitable for PM processing.

In short, for BE Ti-6Al-4V, the clearest trend among the available data for static properties is the effect of porosity on tensile ductility. The effect of purity is also important, as oxygen can act as a strong solution strengthener, though it is shown to compromise ductility beyond a critical level. The as-sintered microstructure of BE Ti-6Al-4V, in terms of grain size and morphology, cannot be controlled with traditional sintering processes. Therefore, this parameter cannot be readily optimised with respect to mechanical properties without incorporating costly post-sintering processing.

Static properties of PM Ti-6Al-4V using PA powders

Similar to BE processed Ti-6Al-4V, the strength and ductility of Ti-6Al-4V produced from PA powders will suffer if the relative density is below 99% [330]. As discussed in section ‘Pressure-assisted consolidation’, PA powders require some form of pressure-assisted sintering such as HIP or VHP to produce

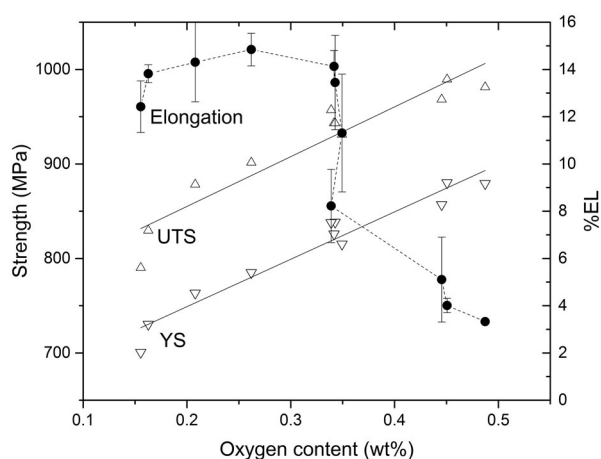


Figure 31. Relationship between strength, ductility, and oxygen content of Ti-6Al-4V produced by MIM [346].

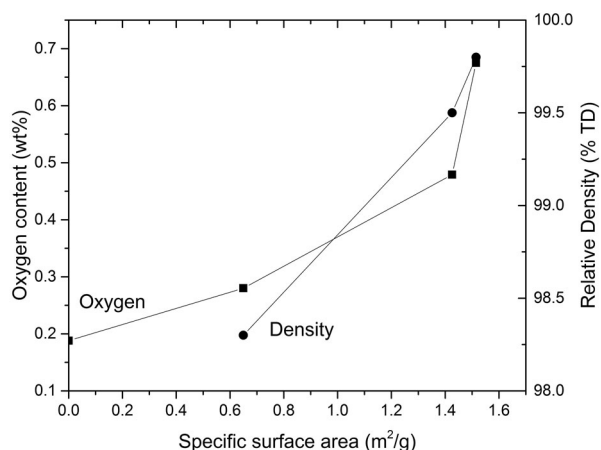


Figure 32. Effect of specific surface area (particle size) on the as-sintered oxygen content and density of Ti-6Al-4V produced from TiH_2 powder after milling in heptane for different times [151].

reasonable densities. These processes are generally conducted at pressures beyond the flow stress of Ti-6Al-4V. As such, porosity tends to be less of an issue with PA Ti-6Al-4V versus BE processing. In general, PA processing can produce better tensile properties than BE processing. However, as shown in Figure 30(b), there is a wide range of tensile properties reported for Ti-6Al-4V produced from PA powder. For example, ductility as low as 7.5%EL has been reported for fully dense PA Ti-6Al-4V [356]. One possible cause for relatively poor properties in PA Ti-6Al-4V despite full density is incomplete diffusion bonding across closed pores [344], which would be due to insufficient temperature or time during the consolidation process. Of course, purity and microstructure play important roles in the tensile properties of these alloys as well, which is discussed below.

Owing to their high cost and the nature of processing required to produce PA powders, oxygen and other interstitial element contents are generally optimised for PA powders. Therefore, this parameter tends not to play a significant role among the static properties of PA Ti-6Al-4V as reported in the literature.

Microstructure is an important controlling feature in fully-dense PA Ti-6Al-4V samples. As discussed in section 'Pressure-assisted consolidation' the microstructure of PA Ti alloys is largely controlled by the processing parameters used during consolidation (e.g. temperature). Therefore, PA processing allows for the production of a range of microstructures and corresponding mechanical properties. If a temperature beyond the β -transus is used during the consolidation process, a coarse lamellar microstructure and the corresponding mechanical properties will be produced. Therefore, if a finer and more equiaxed microstructure is desired, it is necessary to optimise the process to use a temperature below the β -transus while still producing a fully density and full bonded product.

As is evident in Figure 30(b), post-processing such as TMP and heat treatment does not significantly alter the strength and ductility available with PM Ti-6Al-4V produced from PA powders. It was pointed out in section 'Hot isostatic pressing' that 100% TD and a range of microstructures can be achieved in as-HIPed PA Ti-6Al-4V alloy. Therefore, if the primary consolidation process is optimised appropriately, additional processing steps will have a marginal effect on tensile strength and ductility combination. However, secondary processing steps have been used to enhance the fatigue properties of these alloys (section 'Fatigue of PM Ti-6Al-4V alloys using PA powders').

In summary, PA powders combined with pressure-assisted consolidation can produce superior tensile properties versus Ti-6Al-4V produced from BE Ti powders. However, it is important to remember that PA powders are generally up to an order of magnitude

more expensive than BE powders. Furthermore, the need for applied pressure during consolidation significantly complicates the sintering process and makes it more energy-intensive. As such, PA processing would be better suited to processing that is less restrictive with regard to cost and productivity.

Static properties of PM Ti-6Al-4V using BE TiH₂ powders

As discussed in section 'Sintering of TiH₂', both the density and purity of PM Ti-6Al-4V is improved when TiH₂ powder is used in a BE process versus Ti powder. As would be expected and as shown in Figure 30(c), this translates to improved tensile properties as well. Vacuum sintering of BE powders using TiH₂ has reported tensile strengths as high as 1050 MPa or ductility as high as 17%EL for lower strength samples [201,202,341,357]. It has been suggested that by optimising the variables, such as initial powder size, compaction pressure, and sintering temperature and time, the tensile properties can be further improved for vacuum sintering of PM Ti-6Al-4V produced using BE TiH₂ powders [340].

Despite the improved density and purity, conventional vacuum sintering of BE TiH₂ powders produces a coarse lamellar microstructure in the as-sintered state similar to BE Ti powders. However, it has been recently shown that an H₂ sintering process called 'hydrogen sintering and phase transformation' (HSPT, see section 'Sintering of TiH₂ in H₂'), produces an ultra-fine grain microstructure in the as-sintered state. The as-sintered microstructure may also be modified via simple heat treatments to produce a range of wrought-like microstructures via simple heat treatments. It should be noted that the heat treatments utilised in HSPT Ti-6Al-4V are performed without TMP. A low-cost high pressure process called GIF (see section 'Gaseous isostatic forging') has been used to successfully close the remaining ~1 vol-% porosity that remains after the pressureless sintering process. As reported in the literature, the various microstructures that are possible with the HSPT process produce Ti-6Al-4V with tensile strengths in the range of 970–1100 MPa and ductility in the range of 13–22%EL (Figure 30(c)) [174,204,207,212,213,358,359]. Additionally, recent publications have reported globularised microstructures with tensile ductility exceeding 22%EL while maintaining strength level around 1000 MPa [174,213].

Using TiH₂ instead of Ti in a BE process has several apparent advantages over other processing. These studies report tensile properties that are on par with wrought Ti-6Al-4V. However, they do not require expensive PA powder or pressure-assisted sintering. Additionally, when H₂ sintering is used (i.e. HSPT), a range of wrought-like microstructures and

application-tailored mechanical properties are possible without requiring TMP. This is an important capability, as the popularity of wrought Ti-6Al-4V is due largely to its ability to meet the specific property requirements of a wide range of applications. Therefore, it is possible that these processes could be developed into economical and truly near-net-shape compatible PM processes as a feasible alternative to WP for producing high-performance Ti alloys.

Fatigue properties

The relatively low fatigue strength of PM Ti parts is a key issue that hinders the adoption of PM Ti, especially for aerospace and automobile applications. Therefore, this section focuses on evaluating the fatigue performance of Ti-6Al-4V alloys made from different types of powders. The goal of the review is to illustrate the powder, processes, and microstructural improvements that are necessary to achieve high fatigue strength in PM Ti-6Al-4V alloys, comparable to that achieved via WP.

Fatigue of PM Ti-6Al-4V alloys using BE powders

A compilation of currently available fatigue data of Ti-6Al-4V alloys [306,314,316,360] made by cold pressing and vacuum sintering of BE powders is presented in Figure 33. The data on fatigue behaviour after some post-sintering treatments are also included for comparison. The processing conditions, microstructural features, and tensile properties of the PM materials from which the fatigue data in Figure 33 are generated are listed in Table 6. In this paper, the common convention is used, defining the fatigue endurance limit as the cyclic stress maximum at which a specimen survives 10^7 cycles.

Porosity has the most significant effect on fatigue performance. In PM Ti alloys with density <99% of the TD, the tensile and fatigue strength levels are lower [316,361] than acceptable, as mechanical properties strongly degrade when the levels of porosity increase. Therefore, the fatigue data presented in Figure 33 have been restricted to materials with density >99% TD. On this basis, it can be seen in Figure 33 that the fatigue strength levels for the BE method alloys have a range from 300 MPa to approximately 400 MPa.

HIP has proved to be an effective way to obtain components at or near 100% density. As seen in Figure 33, the lower-cycle fatigue strengths are improved when HIP is used to close porosity. However, the coarse lamellar microstructure in PM Ti-6Al-4V cannot be avoided in conventional BE sintering processes [313,360]. Thus, even if the porosity is eliminated by HIP or some other process, the coarse lamellar colonies initiate fatigue cracks with relative ease, leading to a generally poor high cycle fatigue strengths for BE Ti-6Al-4V.

The typical BE Ti-6Al-4V sintering process produces a coarse lamellar microstructure, which leads to early crack nucleation in fatigue, even when porosity is absent [148]. This type of microstructure also has the lowest fatigue strength level of all the microstructures possible in wrought and heat-treated Ti-6Al-4V. Chait and DeSisto [362] showed that the high cycle fatigue strength correlated well with the size of the α phase, more specifically with the width of α -platelets in the transformed β microstructure. Therefore, the fatigue strength of BE Ti-6Al-4V may be further improved by refining the microstructure, though this would typically require costly post-processing that is not near-net-shape compatible, such as TMP.

Eylon et al. [360] performed HIP and heat treatment on BE Ti-6Al-4V compact. However, the fatigue strength of these materials is much lower than that obtained by others using comparable processes. Although the volume fraction of porosity was low (>99.5%TD), it is possible that large rogue pores or other coarse microstructural features were responsible for the poor fatigue strength. Cao et al. have theorised that despite the overall density of the material [363,364], the fatigue strength of titanium alloys can be severely limited by the occurrence of a single pore in the microstructure. The size, geometry, and location of the pore are all reported to play significant roles in the resulting fatigue performance of the material. This is because these features determine under what loading the pore will initiate a fatigue crack, which will ultimately lead to the failure of the material.

Fatigue of PM Ti-6Al-4V alloys using PA powders

The use of PA powder to produce PM Ti parts can be quite effective, because of two advantages: (i) alloying of elements are accomplished in the liquid state before powder atomisation and (ii) improved densification by pressure-assisted consolidation techniques such as HIP. Direct consolidation by HIP at a temperature below or above β -transus has been used in industry to produce near fully dense PA Ti-6Al-4V. The fatigue data of Ti-6Al-4V alloys made from PA powders, as reported in past studies, are compiled in Figure 34 and Table 7 [154,343,360]. Porosity is less of a concern with PA Ti-6Al-4V, owing to the fact that consolidation must be performed via pressure-assisted consolidation techniques such as HIP, effectively producing densities close to 100%.

The microstructure of PA Ti-6Al-4V is invariably finer than that obtained from BE processing when the HIP temperature and time is well optimised [12,154]. Because of the improved density and finer microstructure, the fatigue strengths of as-sintered Ti-6Al-4V alloys made from PA powders consolidated using pressure-assisted sintering techniques, like HIP, are generally much higher than those of as-sintered samples obtained from BE powders. This can be

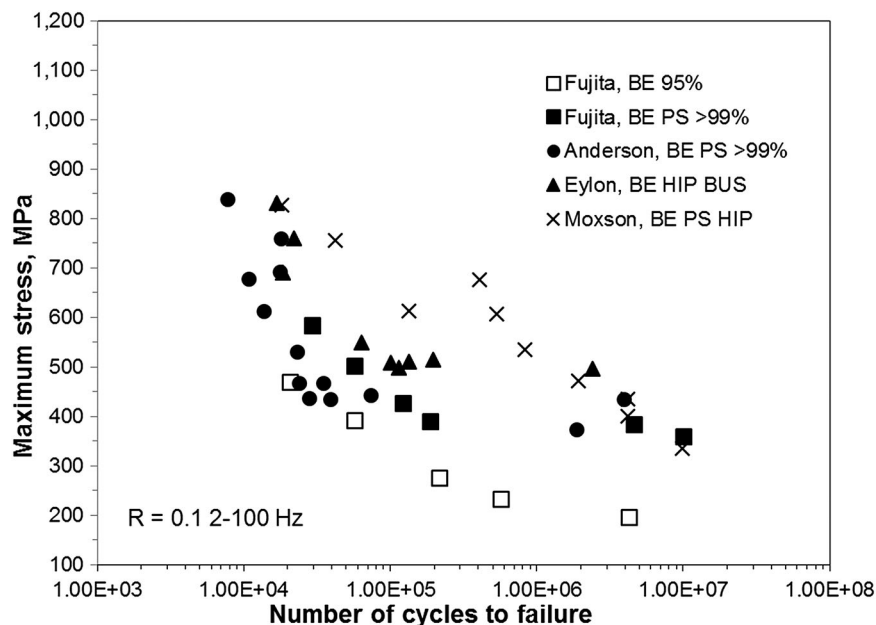


Figure 33. S–N fatigue data of PM Ti–6Al–4V alloys made from BE powders [306,314,316,360]. (PS: press and sinter, HIP: see section ‘Hot isostatic pressing’, BUS: broken-up structure.)

Table 6. Examples of published processing conditions, microstructure, and tensile properties of PM Ti–6Al–4V alloys using BE powders.

Source	Powder	Consolidation	Heat treatment	Microstructure	$\sigma_{y,0.2}$ (MPa)	UTS (MPa)	%EL	%RA	%TD
Fujita [316]	HDH, <149 μm	Cold press VS 1260°C 4 h	–	Lamellar Colonies and α_{GB} $\sim 50 \times 10 \mu\text{m}$ α	809	926	19	31	99.6
Andersen [306]	Sponge fine <149 μm	Cold press 620 MPa VS 1260°C 4 h	–	Equiaxed α $\sim 30 \mu\text{m}$ α	866	933	12.5	20	>99.0
Moxson [314]	HDH <177 μm	CIP 410 MPa VS 1260°C 4 h HIP 955°C 207 MPa	–	Lamellar Colonies $\sim 100 \times 10 \mu\text{m}$ α	865	965	12.9	31.6	–
Eylon [360]	Sponge fine	Cold press 420 MPa VS 1260°C HIP 930°C 105 MPa 5 h	1025°C WQ 815°C 24 h	Coarser α Platelets Broken up	834	937	11	26	>99.5

Note: VS: vacuum sintering; WQ: water quench.

understood by direct comparisons of data between Figures 33 and 34.

Heat treatments have also been attempted, in order to further improve the fatigue strength in PA Ti–6Al–4V. However, as shown in Figure 34, the improvement is not as significant as that seen in BE approaches. This is probably because the as-sintered properties of BE Ti–6Al–4V are lower than those of PA Ti–6Al–4V prior to heat treatment.

Thermomechanical processes such as isothermal forging, rolling, and swaging have been used with the intent of refining the microstructure [154,312], which could lead to further increases in the fatigue strength. Wirth et al. [154] used a swaging process on compacts obtained by HIP of PA powders, followed by two different heat treatment procedures to get PM Ti–6Al–4V alloys with equiaxed α and lenticular α microstructures (Table 7). This study produced Ti–6Al–4V with a 10^7 endurance limit of 750 MPa, which is the best reported for PM Ti–6Al–4V to date. Traditional

ingot-melt Ti–6Al–4V samples were subjected to identical mechanical working and heat treatment, which produced nearly identical fatigue performance. Therefore, the performance of the PM Ti–6Al–4V was proposed to result from improved crack propagation resistance of the microstructure. This was attributed to the uniform precipitation of α phase inside the β grains as well as in the regions near to the prior β grain boundaries [154,365]. This data indicates that an effective method to produce wrought-like fatigue performance is to apply adequate cold working before HIP to induce uniform recrystallisation of the microstructure.

Fatigue of PM Ti–6Al–4V alloys using BE TiH₂ powders

Similar to static properties, reducing as-sintered residual porosity and improving purity, by using TiH₂ in a BE process results in improved fatigue properties. Table 8 is a compilation of the processing

conditions and tensile properties of BE Ti-6Al-4V made from TiH₂ using conventional cold compaction and pressureless sintering as well as the HSPT process (see section ‘Sintering of TiH₂ in H₂’). The lateral bending fatigue properties of Ti-6Al-4V produced by conventional pressureless (vacuum) sintering were investigated by Ivasishin et al. in an electrodynamic fatigue testing system under conditions of resonance vibrations. In this study, the bending fatigue strength was reported to be 450–500 MPa at 10⁷ cycles for specimens with rounded edges, and about 350 MPa for specimens with sharp edges [194]. Joshi et al. [341] conducted rolling and heat treatment on Ti-6Al-4V produced by the same vacuum sintering process to close the residual porosity and obtained a finer microstructure. Rotating beam fatigue (RBF) testing was used to evaluate Ti-6Al-4V samples after the TMP, which reported a fatigue strength of 500–550 MPa at 10⁷ cycles. Traditional axial (tensile) fatigue data is not available for BE Ti-6Al-4V produced from TiH₂ and conventional pressureless sintering.

As mentioned previously, the HSPT process produces Ti-6Al-4V with a significantly refined microstructure. Cao et al. [214] evaluated the fatigue performance of HSPT produced materials under different processing conditions, using cyclic axial fatigue testing, as defined by the ASTM standard [366]. Figure 35 shows that the 10⁷ cycle fatigue strength of as-sintered HSPT Ti-6Al-4V is in the range between 450–500 MPa. The fatigue strength at 10⁷ cycles increases to approximately 550 MPa after GIF is applied, which produces density >99.9%. Figure 35 illustrates that GIF greatly increases the fatigue strengths for cycles less than 5 million. As the

endurance limit is approached, the S-N curves for the different conditions converge. It was proposed by Cao et al. that the rapid decline of fatigue strength near the endurance limit is due to the existence of grain boundary α along the prior β grain boundaries. However, it is evident that the HSPT Ti-6Al-4V fatigue strength is far superior to that of traditional BE Ti-6Al-4V and is competitive with wrought or PA Ti-6Al-4V when the GIF process is used, especially below 5 million cycles. Since the fatigue data for the vacuum-sintered Ti-6Al-4V reported by Ivasishin et al. [194] and Joshi et al. [341] were not measured using axial fatigue testing, it is not possible to make a valid comparison between the two types of BE processing of TiH₂.

As mentioned in section ‘Sintering of TiH₂ in H₂’, the as-sintered ultrafine grain microstructure produced by the HSPT process may be further modified via simple heat treatments to produce globularised and bi-modal microstructures [208,213]. The samples with microstructures that result from these heat treatments were tested for fatigue performance by Paramore et al. [174,213]. Before heat treatment, the HSPT-produced samples were processed via GIF to close porosity (section ‘Gaseous isostatic forging’), as porosity would obscure the effect of the microstructure on fatigue performance. In these studies, it was found that the heat-treated microstructures increased fatigue strength compared with the as-sintered via HSPT and as-processed via GIF samples. As shown in Figure 36, the globularised microstructure has a 10⁷ cycle fatigue strength of 575 MPa, while the bi-modal microstructure has a 10⁷ cycle fatigue strength of 600 MPa. These data show that an endurance limit that is

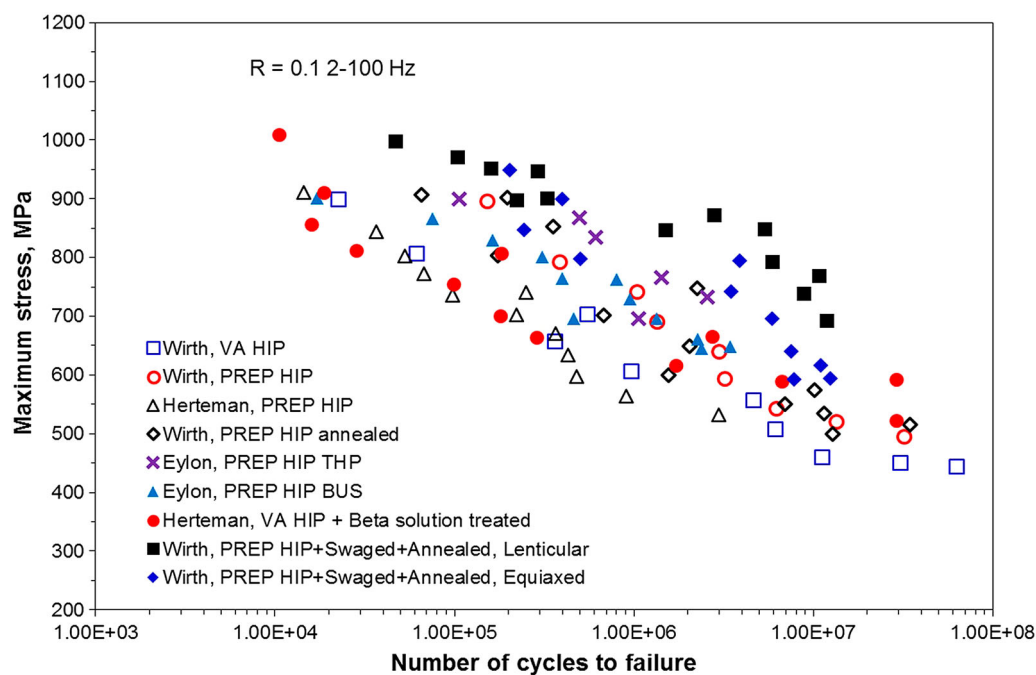


Figure 34. S-N fatigue data of PM Ti-6Al-4V alloys made from PA powders [154,343,360]. (VA: vacuum atomisation, THP: THP, BUS: broken-up structure.)

Table 7. Examples of published processing conditions, microstructure, and tensile properties of PA Ti–6Al–4V alloys.

Source	Powder	Consolidation	TMP	Heat treatment	Microstructure	$\sigma_{y,0.2}$ (MPa)	UTS (MPa)	%EL	%RA
Herteman [343]	PREP <425 μm	HIP 925°C 1050 MPa 5 h	–	–	Lamellar Colonies $\sim 80 \times 10 \mu\text{m } \alpha$	860	937	17	42
Wirth [154]	PREP 63–350 μm	HIP 925°C 198 MPa 3 h	–	–	Lamellar Colonies $\sim 30 \times 5 \mu\text{m } \alpha$	900	967	14.5	41.4
Wirth [154]	VA	HIP 925°C 198 MPa 3 h	–	–	–	871	934	11.9	26.4
Herteman [343]	VA 100–630 μm	HIP 950°C 1000 MPa 5 h	–	975°C, 1h WQ, 700°C, 2h, AC	Martensite No GB- α	1020	1095	9	21
Wirth [154]	PREP 63–354 μm	HIP 925°C 198 MPa 3 h	–	1010°C 0.05h WQ 900°C 2h AC	Fine Lamellar	932	1036	13	29.1
Eylon [360]	PREP	HIP 925°C 105 MPa 5 h	–	1025°C WQ 815°C 24 h	Coarser α Platelets Broken up	910	1000	15	23
Eylon [360]	PREP	HIP 925°C 105 MPa 5 h	–	1025°C 0.3hr WQ Hydrogenate 600°C De-H 760°C	Finer α Platelets Broken up	965	1048	8	17
Wirth [154]	PREP 63–354 μm	HIP 930°C 193 MPa 2.5 h	Extrude 1020°C Swage 900°C	830°C 2h AC	Fine Equiaxed α	931	1048	16.7	38.2
Wirth [154]	PREP 63–354 μm	HIP 930°C, 193 MPa 2.5 h	Extrude 1020°C Swage 900°C	1010°C 0.05h WQ 900°C 2h AC	Lenticular α	990	1088	12.7	28.3

Note: VA: vacuum atomisation; WQ: water quench; AC: air cool; De-H: hydrogenation.

competitive with high-performance wrought Ti–6Al–4V can be achieved via truly low-cost and near-net-shape compatible BE powder metallurgy without requiring costly post-processing.

Summary

The review presented above shows that there are a large variety of techniques for making titanium powder, as well as for forming parts from these powders. Among the powder production processes, many are in various developmental stages. The motivation for developing new processes is often firmly rooted in the ambition of attaining a low-cost alternative to the Kroll process for Ti primary metal production. The status of the industry, including the research community, is that no technique is available to date for meeting both cost and quality goals matching those of the Kroll method. A number of developmental methods have garnered considerable interest and scale-up efforts, including the Armstrong process and the FFC Cambridge process. The Armstrong process is an example of using a continuous process for the reduction of TiCl_4 using Na or Mg. Although continuous processing has the potential to lower the processing energy costs, the combined cost of the feedstock for reduction (TiCl_4 or TiO_2) and the reducing agent (Mg, Na, or Ca) plays an even more important role in determining the cost of Ti metal production. The FFC Cambridge process is an example of the various electrochemical methods that convert TiO_2 to Ti metal in molten salt. The development of the FFC and other electrochemical methods is an ongoing effort. Challenges to improve energy efficiency and scale-up are yet to be overcome.

A recently developed method, i.e. the HAMR process, is based on the destabilisation of the Ti–O system

using hydrogen as a temporary alloying element during the magnesiothermic reduction of TiO_2 . The published data of the chemical compositions, especially oxygen content, of the powders produced by the HAMR process indicates that it is a very promising method.

Titanium powders that are commercially available today are all made from titanium sponge, Ti ingot, Ti mill products, and/or Ti scrap metals. There are mainly two types: HDH powder and atomised powder. Atomisation is the process of choice when alloyed powders are required. However, atomised powders are considerably more costly than HDH powders, which is partially because the yield of particle sizes that are marketable is generally low. Despite these issues, atomising of Ti alloys has seen continued technology advancements with respect to increasing the yield of fine powders and minimising the contamination from nozzle materials by eliminating the need for nozzles. Owing to the advent of AM using metals, the development and utilisation of atomisation processes for making spherical Ti alloy powders have found a new and burgeoning frontier.

In terms of converting these powders into functional components, a long list of processes has been applied to the consolidation and sintering of Ti and Ti alloy powders, including both pressure-less and pressure-assisted sintering processes. Owing to the strict requirements for low oxygen, sintering under vacuum is the most popular process. Understandably, pressure-assisted techniques such as HIP yield both full density and good mechanical properties, although accompanied by relatively high cost. Regardless of which traditional sintering processes is used, as-sintered Ti–6Al–4V always has a coarse columnar microstructure because sintering is almost always carried out at a temperature above the β transus temperature. Although heat treatments can be

Table 8. Examples of published processing conditions, microstructure, and tensile properties of BE Ti–6Al–4V using TiH₂ powders.

Source	Powder	Consolidation	Post/heat treatment	Microstructure	$\sigma_{y0.2}$ (MPa)	UTS (MPa)	%EL	%RA	%TD
Ivasishin [194]	BE: 100 μm TiH ₂ 65 μm 65Ti–35Al 65 μm 25Al–75V	Press 700 MPa, VS 1350°C	–	Coarse Lamellar Colonies	920–960	1020–1040	5.3–5.7	23–25	98.3
Cao [214]	BE: <44 μm TiH ₂ <44 μm 60Al–40V	CIP 350 MPa HSPT 1200°C 4 h 650°C, 4 h De-H 750°C 12 h	–	UFG	955	1033	13.7	26.0	98.5
Cao [214]	BE: <37 μm TiH ₂ <44 μm 60Al–40V	CIP 350 MPa HSPT 1200°C 8 h 650°C 4 h De-H 750°C 12 h	–	UFG	946	1032	15.6	31.6	99.2
Cao [214]	BE: <37 μm TiH ₂ <44 μm 60Al–40V	CIP 350 MPa HSPT 1200°C 4 h 650°C, 4 h De-H 750°C 12 h	GIF 850°C	UFG	952	1023	18.2	42.1	>99.9
Paramore [174,208]	BE: <37 μm TiH ₂ <44 μm 60Al–40V	CIP 350 MPa HSPT 1200°C 4 h 650°C 4 h De-H 750°C 12 h	GIF 850°C HT 954°C 1 hr FC Age 550°C 6 hr FC	Fine Globularised $\sim 5 \mu\text{m}$ a_p	923–931	993–1002	21.0–22.3	45.0–47.6	>99.9
Paramore [174,208]	BE: <37 μm TiH ₂ <44 μm 60Al–40V	CIP 350 MPa HSPT 1200°C 4 h 650°C, 4 h De-H 750°C 12 h	GIF 850°C HT 954°C 1 hr WQ Age 550°C 6 hr FC	Fine Bi-modal $\sim 4 \mu\text{m}$ a_p UFG acicular α	984–1019	1058–1101	13.1–16.2	27.3–34.5	>99.9
Paramore [174,208]	BE: <37 μm TiH ₂ <44 μm 60Al–40V	CIP 350 MPa HSPT 1200°C 4 h 650°C, 4 h De-H 750°C 12 h	GIF 850°C HT 1050°C 1 hr AC Age 550°C 6 hr FC	Lamellar Colonies $\sim 30 \times 2 \mu\text{m}$ α	902–920	999–1015	15.7	25.4–26.0	>99.9

Note: VS: vacuum sintering; De-H: dehydrogenation, HT: heat treatment, FC: furnace cool, WQ: water quench, AC: air cool.

applied, the microstructure and properties of conventionally sintered PM Ti cannot match those of wrought alloys because the TMP processes that wrought alloys are subjected to provide the driving force for microstructural augmentation. However, the recent development of the HSPT process in combination with the subsequent standard heat treatment has overcome that impasse, i.e. the PM Ti sintered via the HSPT process can be heat treated to yield microstructures that are equivalent to those of wrought alloys.

Technically, there are three key factors that affect the mechanical properties of PM Ti alloys: porosity, oxygen content, and microstructure. As discussed in section ‘Mechanical properties of PM Ti’, the alloy must have no less than 98% of TD to achieve generally acceptable static tensile properties. However, anything less than 99% will limit the fatigue performance. Furthermore, to achieve the fatigue strength of high-performance wrought Ti–6Al–4V (endurance limit >500 MPa) requires $\sim 100\%$ density. When TiH₂ is used as the raw material and dissolved hydrogen is present during β -phase sintering, >99%TD is possible via pressureless sintering. However, closing the remaining <1 vol.-% porosity requires some form of pressure-

assisted consolidation. HIP is the traditional method to close the remaining porosity, though the GIF process appears to be a promising low-cost alternative.

It is important to note that the size of pores is critical to fatigue performance. Crack initiation often occurs at the edge of the largest pores present. The critical size of the pores that will dominate fatigue performance is dependent on the microstructure’s ability to resist crack propagation. Therefore, even in the presence of pores, fatigue limiting cracks can initiate by the fracturing of large α grains as well as trans-granular fracturing across entire α colonies, owing to the shared slip systems of the adjacent lamellar grains. As such, producing a fine microstructure is paramount for achieving high-performance fatigue strength. Furthermore, refining grain size has shown improved fatigue performance despite the presence of moderately sized pores. It has also been shown that PM Ti–6Al–4V processed using the HSPT technology has the potential to rival that of wrought Ti–6Al–4V by producing a fine wrought-like microstructure both in the as-sintered state and after heat treatment.

Controlling oxygen content is also a critical issue for PM Ti. Finer powders tend to produce better

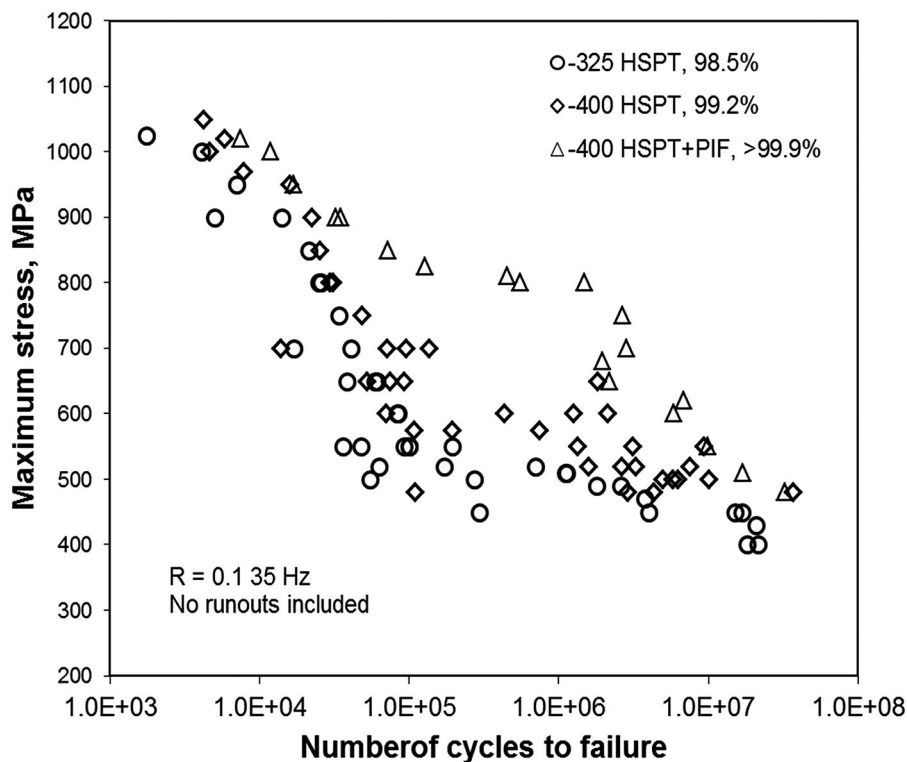


Figure 35. S–N fatigue data of PM Ti–6Al–4V alloys made from BE TiH₂ powders [214]. (HSPT: see section ‘Sintering of TiH₂ in H₂’, GIF: see section ‘Gaseous isostatic forging’.)

densification, but are also vulnerable to contamination due to increased surface area. Using hydrogenated Ti powder can improve the final purity, due to the

decreased reactivity of the hydride. By the ASTM standard, 0.2 wt-% is the maximum allowed content for Grade 5 Ti–6Al–4V. However, recent work has

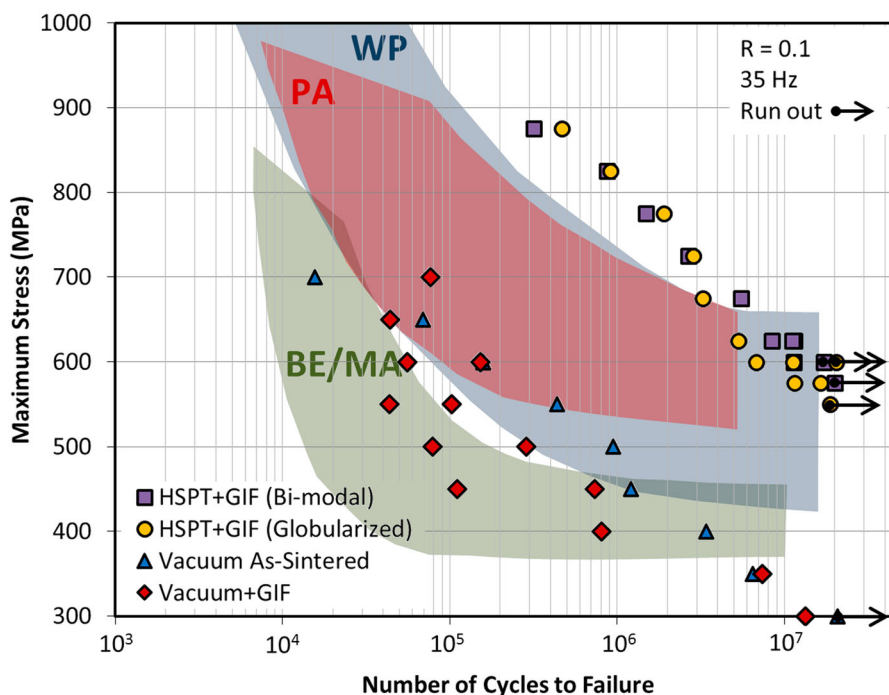


Figure 36. S–N curves for Ti–6Al–4V produced via HSPT (see section ‘Sintering of TiH₂ in H₂’) followed by GIF (see section ‘Gaseous isostatic forging’) and heat treated to produce ‘globularized’ and bi-modal microstructures. S–N curves of vacuum sintered Ti–6Al–4V produced from identical powder with and without subsequent GIF processing is also shown as a comparison to the HSPT process. For reference, these data are overlaid on the fatigue life scatter-bands (prepared from data in the ASM Handbooks [11]) for Ti–6Al–4V produced via ingot metallurgy/WP with a mill-annealed microstructure, PA powder metallurgy, and BE/master-alloy powder metallurgy [213].

indicated that oxygen plays a minimal role on tensile ductility up to 0.33 wt-%. There is no direct study to date on how the fatigue properties vary as a function of oxygen content, provided other microstructural features are identical.

The primary purpose of using PM processes, as opposed to traditional wrought processes, is to reduce the cost of manufacturing and, therefore, expand the use of Ti in industrial and consumer applications. However, the commercial applications of PM Ti are, incongruously, very limited to date because of high cost. The cost of PM Ti is often higher or equivalent to that of wrought Ti when it is held to the same performance standards, because additional high-cost processing steps and expensive feedstock are required to produce satisfactory microstructures and mechanical properties. In other words, the lack of an adequate performance-to-cost ratio is the main issue hindering the adoption of PM Ti in industry. In general, PM Ti made with the BE approach is truly low cost; however, it has unsatisfactory mechanical properties. Conversely, PM Ti made with the PA approach has adequate mechanical properties; however, it does not have significant cost advantages, if any, over wrought Ti alloys. A new process, namely the HSPT process, has shown the potential to maximise the performance-to-cost ratio by producing wrought-like Ti–6Al–4V using press-and-sinter and conventional heat treatment processes.

The goal of any PM Ti process is to produce Ti alloys with the mechanical properties necessary for critical applications while maintaining the low-cost benefit of PM. In order to remain truly low-cost, PM Ti must adhere to the following guidelines:

- Use low-cost powder as raw materials and employ a BE approach.
- Use low-cost compaction techniques to retain the ability to produce near-net-shape products.
- Use pressureless sintering techniques and avoid costly high-pressure consolidation processes.
- Obtain wrought-like microstructures through conventional heat treatment and avoid the need for TMP, which is not compatible with near-net-shape manufacturing and is inherently costly.

Finally, in short, titanium alloys made by using the conventional powder metallurgy approaches have not met the performance-to-cost ratios that are necessary for it to substitute for wrought titanium alloys, and thereby have widespread commercial success. The emerging technologies for making low-cost powders and the sintering processes to create wrought-like microstructure and mechanical properties without using wrought processes hold the promise to advance the PM of Ti to a new level. However, significant challenges remain for these emerging technologies to

succeed, challenges which require integrated investments of effort in relation to end-use engineering, manufacturing, and materials science.

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References

- [1] Seong S, Younossi O, Goldsmith BW, et al. Titanium-industrial base, price trends, and technology initiatives. Santa Monica (CA): RAND Corporation; 2009; [cited 2017 Jul 26]. Available from: www.rand.org/content/dam/rand/pubs/monographs/2009/RAND_MG789.pdf
- [2] Aerospace workshop planning session-summary report. Oak Ridge National Laboratory; 2010. [cited 2017 Jul 26]. web.ornl.gov/sci/manufacturing/docs/Aerospace_Workshop_Report.pdf
- [3] Froes FH, Eylon D. Powder metallurgy of titanium alloys. *Int Mater Rev*. 1990;35(3):162–184.
- [4] Fang ZZ, Sun P. Pathways to optimize performance/cost ratio of powder metallurgy titanium – a perspective. *Key Eng Mater*. 2012;520:15–23.
- [5] McCracken C. Production of fine titanium powders via the hydride-dehydride (HDH) process. *PIM Int*. 2008;2(2):55–57.
- [6] Barbis DP, Gasior RM, Walker GP, et al. Titanium powders from the hydride-dehydride process. In: Qian M, et al., editors. *Titanium powder metallurgy: science, technology and applications*. Oxford: Butterworth-Heinemann; 2015. p. 101–105.
- [7] McCracken CG, Motchenbacher C, Barbis DP. Review of titanium powder-production methods. *Int J Powder Metall*. 2010;46(5):19–26.
- [8] Froes FH. Titanium powder metallurgy: a review – part 1. *Adv Mater Process*. 2012;170(9):16–22.
- [9] Armstrong DR, Anderson RP, Jacobsen LE. Preparation of alloys by the Armstrong method. US Patent 7,041,150 B2. 2006.

- [10] Chen GZ, Fray DJ, Farthing TW. Direct electrochemical reduction of titanium dioxide to titanium in molten calcium chloride. *Nature*. 2000;407(6802):361–364.
- [11] Eylon D, Froes FH, Abkowitz S. Titanium powder metallurgy alloys and composites. In: Lee PW, et al., editors, *ASM handbook*. Materials Park (OH): ASM International; 1998. p. 874–886.
- [12] Wang H, Fang ZZ, Sun P. A critical review of mechanical properties of powder metallurgy titanium. *Int J Powder Metall*. 2010;46(5):45–57.
- [13] Summary of energing titanium cost reduction technologies. EHK Technologies; 2004.
- [14] Hartman AD, Gerdemann SJ, Hansen JS. Producing lower-cost titanium for automotive applications. *JOM*. 1998;50(9):16–19.
- [15] Xu B, Yang B, Jia J, et al. Behavior of calcium chloride in reduction process of titanium dioxide by calcium vapor. *J Alloys Compd*. 2013;576:208–214.
- [16] Nersisyan H, Won H, Won C, et al. Direct magnesiothermic reduction of titanium dioxide to titanium powder through combustion synthesis. *Chem Eng J*. 2014;235:67–74.
- [17] Okabe TH, Oda T, Mitsuda Y. Titanium powder production by preform reduction process (PRP). *J Alloys Compd*. 2004;364(1–2):156–163.
- [18] Kikuchi T, Yoshida M, Matsuura S, et al. Rapid reduction of titanium dioxide nano-particles by reduction with a calcium reductant. *J Phys Chem Solids*. 2014;75(9):1041–1048.
- [19] Abiko T, Park I, Okabe TH. Reduction of titanium oxide in molten salt medium. *Proceedings of the 10th World Conference Titanium ‘Ti-2003 Science and Technology’*; 2003 Jul 13–18; Hamburg, Germany. Wiley-VCH.
- [20] Park I, Abiko T, Okabe TH. Production of titanium powder directly from TiO_2 in CaCl_2 through an electronically mediated reaction (EMR). *J Phys Chem Solids*. 2005;66(2–4):410–413.
- [21] Suzuki RO, Ono K. OS process – a new calciothermic reduction of TiO_2 in the molten CaCl_2 . *Proceedings of the 18th Annual ITA Conference*; 2002 Oct 6–8; Orlando, FL. International Titanium Associate; p. 1–24.
- [22] Suzuki RO, Inoue S. Calciothermic reduction of titanium oxide in molten CaCl_2 . *Metall Mater Trans B*. 2003;34(3):277–285.
- [23] Suzuki RO. Direct reduction processes for titanium oxide in molten salt. *JOM*. 2007;59(1):68–71.
- [24] Won C, Nersisyan H, Won H. Titanium powder prepared by a rapid exothermic reaction. *Chem Eng J*. 2010;157(1):270–275.
- [25] Eshed M, Irzh A, Gedanken A. Reduction of titanium dioxide to metallic titanium conducted under the autogenic pressure of the reactants. *Inorg Chem*. 2009;48:7066–7069.
- [26] Zheng H, Ito H, Okabe TH. Production of titanium powder by the calciothermic reduction of titanium concentrates or ore using the preform reduction process. *Mater Trans*. 2007;48(8):2244–2251.
- [27] Reed TB. *Free energy of formation of binary compounds: an atlas of charts for high-temperature chemical calculations*. Cambridge (MA): The MIT Press; 1972.
- [28] Kroll W. The production of ductile titanium. *Trans Electrochem Soc*. 1940;78:35–47.
- [29] Mo W. *Titanium metallurgy*. Beijing: Metallurgical Industry Press; 1998.
- [30] Seetharaman S. *Treatise on process metallurgy, volume 3: industrial processes*. Amsterdam: Elsevier; 2013.
- [31] Kohli R. *Production of titanium from ilmenite: a review*. Berkeley (CA): Lawrence Berkeley Lab; 1981.
- [32] Doblin C, Chryss A, Monch A. Titanium powder from the TiRO™ process. *Key Eng Mater*. 2012;520:95–100.
- [33] Deura TN, Wakino M, Matsunaga T, et al. Titanium powder production by TiCl_4 gas injection into magnesium through molten salts. *Metall Mater Trans B*. 1998;29:1167–1174.
- [34] Van Vuuren DS, Oosthuizen SJ, Heydenrych MD. Titanium production via metallothermic reduction of TiCl_4 in molten salt: problems and products. *J South Afr Inst Min Metall*. 2011;111:141–148.
- [35] Hansen DA, Gerdemann SJ. Producing titanium powder by continuous vapor-phase reduction. *JOM*. 1998;50:56–58.
- [36] Hunter MA. Metallic titanium. *J Am Chem Soc*. 1910;32(3):330–336.
- [37] Gambogi J, Gerdemann SJ. Titanium metal: extraction to application. In: *Review of extraction, processing, properties & applications of reactive metals*. Hoboken (NJ): John Wiley & Sons; 1999. p. 175–210.
- [38] Crowley G. How to extract low-cost titanium: a new process for titanium extraction and production promises to cut costs and expand applications. *Adv Mater Process*. 2003;161(11):25–27.
- [39] Wei C, Yamamoto Y, Peter WH. Investigation of pressing and sintering processes of CP-Ti powder made by Armstrong process. *Key Eng Mater*. 2010;436:123–130.
- [40] Gerdemann SJ, Oden LL, White JC. Continuous production of titanium powder. *Proceedings of the 1997 Materials Week ‘Titanium Extraction and Processing’*; 1997 Sep 14–18; Indianapolis, IN. TMS, p. 49–54.
- [41] Alexander PP. Production of titanium hydride. US Patent 2,427,338 A. 1947.
- [42] Moll JH, Yolton CF. Production of titanium powder. In: Lee PW, et al., editors. *ASM handbook, volume 7: powder metal technologies and applications*. Materials Park (OH): ASM International; 1990. p. 160–166.
- [43] Borok BA. Obtaining powders of alloys and steels by a complex reduction of oxide mixtures by CaH_2 . *Trans Cent Res Inst Ferrous Metall*. 1965;43:69–80.
- [44] Froes FH. The production of low-cost titanium powders. *JOM*. 1998;50(9):41–43.
- [45] Okabe TH, Kakihira T, Abiko T. Production of titanium powder through an electronically mediated reaction. *Proceedings of the TMS 2004 Symposium on ‘Electrochemical Measurements and Processing of Materials’*; 2004 Mar 14–18; Charlotte, NC. TMS. p. 67–74.
- [46] Henrie TA, Dolezal H, Kleespies EK. Magnesium reduction of titanium oxides in a hydrogen atmosphere. US Patent 3,140,170 A. 1964.
- [47] Bolívar R, Friedrich B. Synthesis of titanium via magnesiothermic reduction of TiO_2 (pigment). *Proceedings of the EMC 2009*; 2009 Jun 28–Jul 1; Innsbruck, Austria. TMS. p. 1–17.
- [48] Nersisyan H, Lee J, Won C. Combustion of TiO_2 –Mg and TiO_2 –Mg–C systems in the presence of NaCl to synthesize nanocrystalline Ti and TiC powders. *Mater Res Bull*. 2003;38(7):1135–1146.

- [49] Jacob K, Gupta S. Calciothermic reduction of TiO_2 : a diagrammatic assessment of the thermodynamic limit of deoxidation. *JOM*. 2009;61(5):56–59.
- [50] Murray JL, Wriedt HA. The O–Ti (oxygen-titanium) system. *J Phase Equilib*. 1987;8(2):148–165.
- [51] Fang ZZ, Middlemas S, Guo J, et al. A new, energy-efficient chemical pathway for extracting Ti metal from Ti minerals. *J Am Chem Soc*. 2013;135(49):18248–18251.
- [52] Zhang Y, Fang ZZ, Xia Y, et al. A novel chemical pathway for energy efficient production of Ti metal from upgraded titanium slag. *Chem Eng J*. 2016;286:517–527.
- [53] Fang ZZ, Zhang Y, Xia Y, et al. Methods of producing a titanium product. US Patent Application 14/935,245. 2016.
- [54] Zhang W, Zhu Z, Cheng CY. A literature review of titanium metallurgical processes. *Hydrometallurgy*. 2011;108(3–4):177–188.
- [55] Middlemas S, Fang ZZ, Fan P. Life cycle assessment comparison of emerging and traditional titanium dioxide manufacturing processes. *J Clean Prod*. 2015;89:137–147.
- [56] Zhang Y, Fang ZZ, Sun P, et al. Thermodynamic destabilization of Ti–O solid solution by H_2 and deoxygenation of Ti using Mg. *J Am Chem Soc*. 2016;138(22):6916–6919.
- [57] McCracken CG, Robison JW, Motchenbacher CA. Manufacture of HDH low oxygen titanium-6aluminum-4vanadium (Ti-6-4) powder incorporating a novel powder de-oxidation step. Proceedings of the Euro PM2009; 2009 Oct 12–14; Copenhagen, Denmark. European Powder Metallurgy Association. p. 187–192.
- [58] Oh JM, Lee BK, Suh CY, et al. Preparation method of Ti powder with oxygen concentration of <1000 ppm using Ca. *Powder Metall*. 2012;55(5):402–404.
- [59] Fisher RL. Deoxidation of titanium and similar metals using a deoxidant in a molten metal carrier. US Patent 4,923,531. 1990.
- [60] Okabe TH, Oishi T, Ono K. Preparation and characterization of extra-low-oxygen titanium. *J Alloys Compd*. 1992;184(1):43–56.
- [61] Zak Fang Z, Xia Y, Sun P, et al. Molten salt de-oxygenation of metal powders. US Patent Application 62/086,524. 2016.
- [62] Xia Y, Fang ZZ, Sun P, et al. The effect of molten salt on oxygen removal from titanium and its alloys using calcium. *J Mater Sci*. 2017;52(7):4120–4128.
- [63] Okabe TH, Hamanaka Y, Taninouchi Y. Direct oxygen removal technique for recycling titanium using molten MgCl_2 salt. *Faraday Discuss*. 2016;190:109–126.
- [64] Fray DJ. Novel methods for the production of titanium. *Int Mater Rev*. 2008;53(6):317–325.
- [65] Fray DJ, Farthing TW, Chen Z. Removal of oxygen from metal oxides and solid solutions by electrolysis in a fused salt. US Patent Application 10/778,529. 2004.
- [66] Chen GZ. Forming metal powders by electrolysis. In: Chang I, et al., editors. *Advances in powder metallurgy: properties, processing and application*. Cambridge: Woodhead Publishing; 2013.
- [67] Schwandt C, Fray DJ. Determination of the kinetic pathway in the electrochemical reduction of titanium dioxide in molten calcium chloride. *Electrochim Acta*. 2005;51(1):66–76.
- [68] Schwandt C, Doughty GR, Fray DJ. The FFC-Cambridge process for titanium metal winning. *Key Eng Mater*. 2010;436:13–25.
- [69] Alexander DTL, Schwandt C, Fray DJ. The electro-deoxidation of dense titanium dioxide precursors in molten calcium chloride giving a new reaction pathway. *Electrochim Acta*. 2011;56(9):3286–3295.
- [70] Ma M, Wang D, Wang W, et al. Extraction of titanium from different titania precursors by the FFC Cambridge process. *J Alloys Compd*. 2006;420(1–2):37–45.
- [71] Bhagat R, Jackson M, Inman D, et al. Production of Ti–W alloys from mixed oxide precursors via the FFC Cambridge process. *J Electrochem Soc*. 2009;156(1):E1–E7.
- [72] Benson LL, Mellor I, Jackson M. Direct reduction of synthetic rutile using the FFC process to produce low-cost novel titanium alloys. *J Mater Sci*. 2016;51:4250–4261.
- [73] Mellor I, Grainger L, Rao K, et al. Titanium powder production via the metalysis process. In: Qian M, et al., editors. *Titanium powder metallurgy: science, technology and applications*. Oxford: Butterworth-Heinemann; 2015. p. 51–67.
- [74] Bhagat R, Jackson M, Inman D, et al. The production of Ti–Mo alloys from mixed oxide precursors via the FFC Cambridge process. *J Electrochem Soc*. 2008;155(6):E63–E69.
- [75] Yan XY, Fray DJ. Electrosynthesis of NbTi and Nb_3Sn superconductors from oxide precursors in CaCl_2 -based melts. *Adv Funct Mater*. 2005;15(11):1757–1761.
- [76] Hu D, Chen GZ. Electrochemical near-net-shape production via the FFC Cambridge process – dedication to the special session for the 2012 max bredig award. *ECS Trans*. 2013;50:29–37.
- [77] Rao K, Bertolini M, Mellor I, et al. Development of new generation FFC pilot plant for production of low cost titanium and titanium alloys. Ti-2011 Proceedings of the 11th World Conference on ‘Titanium’; 2011 Jun 19–24; Beijing, China. Science Press. p. 181–184.
- [78] Hussain P, Bhagat R, Dashwood R. Precursor preparation for Ti–Al–V–Y alloy via FFC Cambridge process. Proceedings of the 2nd International Conference on ‘Materials Engineering and Automatic Control’; 2013; Shandong, China. p. 62–67.
- [79] Jackson B, Jackson M, Dye D, et al. Production of NiTi via the FFC Cambridge process. *J Electrochem Soc*. 2008;155(12):E171–E177.
- [80] Jackson BK, Dye D, Inman D, et al. Characterization of the FFC Cambridge process for NiTi production using in situ X-ray synchrotron diffraction. *J Electrochem Soc*. 2010;157(4):E57–E63.
- [81] Mohandas KS, Fray DJ. FFC Cambridge process and removal of oxygen from metal-oxygen systems by molten salt electrolysis: an overview. *Trans Indian Inst Met*. 2004;57(6):579–592.
- [82] Bertolini M, Shaw L, England L, et al. The FFC Cambridge process for production of low cost titanium and titanium powders. *Key Eng Mater*. 2010;436:75–83.
- [83] Suzuki RO, Ono K, Teranuma K. Calciothermic reduction of titanium oxide and in-situ electrolysis in molten CaCl_2 . *Metall Mater Trans B*. 2003;34(3):287–295.

- [84] Suzuki RO, Fukui S. Reduction of TiO_2 in molten CaCl_2 by Ca deposited during CaO electrolysis. *Mater Trans.* 2004;45(5):1665–1671.
- [85] Suzuki RO. Calciothermic reduction of TiO_2 and in situ electrolysis of CaO in the molten CaCl_2 . *J Phys Chem Solids.* 2005;66(2-4):461–465.
- [86] Ono K, Suzuki R. A new concept for producing Ti sponge: calciothermic reduction. *JOM.* 2002;54(2):59–61.
- [87] Withers JC, Cardarelli F, Laughlin J, et al. Recent improvements for electrowinning titanium metal from composite anodes. International Round Table on ‘Titanium Production in Molten Salts’; 2008 Mar 2–4; Cologne, Germany.
- [88] Withers JC, Loutfy RO. Thermal and electrochemical process for metal production. US Patent 7985326 B2. 2011.
- [89] Withers JC. Production of titanium powder by an electrolytic method and compaction of the powder. In: Qian M, et al., editors. *Titanium powder metallurgy: science, technology and applications*. Oxford: Butterworth-Heinemann; 2015.
- [90] Gardarelli F. Method for electrowinning of titanium metal or alloy from titanium oxide containing compound in the liquid state. US Patent 7,504,017 B2. 2009.
- [91] Pal U. A lower carbon footprint process for production of metals from their oxide sources. *JOM.* 2008;60(2):43–47.
- [92] Suput M, Delucas R, Pati S, et al. Solid oxide membrane technology for environmentally sound production of titanium. *Miner Process Extr Metall Rev.* 2008;117(2):118–122.
- [93] Jewell D, Jia S, Lurtanjek M, et al. Titanium metal production via oxycarbide electorefining. Proceedings of the 28th Annual ITA Conference Titanium 2012; 2012 Oct 7–10; Atlanta, GA. International Titanium Association.
- [94] Wang Q, Song J, Wu J, et al. A new consumable anode material of titanium oxycarbonitride for the USTB titanium process. *Phys Chem Chem Phys.* 2014;16(17):8086–8091.
- [95] San-Martin A, Manchester FD. The H–Ti (hydrogen-titanium) system. *Bull Alloy Phase Diagr.* 1987;8:30–42.
- [96] Drozdenko VA, Petrunko AM, Andreev AE, et al. Manufacture of cost-effective titanium powder from magnesium reduced sponge. US Patent 6,638,336 B1. 2003.
- [97] Kasparov SA, Klevtsov AG, Cheprasov AI, et al. Semi-continuous magnesium-hydrogen reduction process for manufacturing of hydrogenated, purified titanium powder. US Patent 8,007,562 B2. 2011.
- [98] Klevtsov A, Nikishin A, Shuvalov J, et al. Continuous and semi-continuous process of manufacturing titanium hydride using titanium chlorides of different valency. US Patent 8,388,727 B2. 2013.
- [99] Moxson VS, Duz VA, Klevtsov AG, et al. Method of manufacturing pure titanium hydride powder and alloyed titanium hydride powders by combined hydrogen-magnesium reduction of metal halides. US Patent 9,067,264 B2. 2015.
- [100] Gu DD, Meiners W, Wissenbach K, et al. Laser additive manufacturing of metallic components: materials, processes and mechanisms. *Int Mater Rev.* 2012;57(3):133–164.
- [101] Murr LE, Gaytan SM, Ramirez DA, et al. Metal fabrication by additive manufacturing using laser and electron beam melting technologies. *J Mater Sci Technol.* 2012;28(1):1–14.
- [102] German RM. Progress in titanium metal powder injection molding. *Mater.* 2013;6(8):3641–3662.
- [103] Yolton CF. Gas atomized titanium and titanium aluminum alloys. Proceedings of the PM in Aerospace and Defense Technology Conference; 1989 Nov 2–3; Seattle, Washington. Metal Powder Industries Federation; p. 123–131.
- [104] Yolton CF. Method for producing titanium particles. US Patent 5,084,091. 1992.
- [105] Yolton CF, Froes FH. Conventional titanium powder production. In: Ma Q, et al., editors. *Titanium powder metallurgy: science, technology and applications*. Oxford: Butterworth-Heinemann; 2015. p. 149–162.
- [106] Heidloff AJ, Rieken JR, Anderson IE, et al. Advanced gas atomization processing for Ti and Ti alloy powder manufacturing. *JOM.* 2010;62(5):35–41.
- [107] Heidloff AJ, Rieken JR, Anderson IE, et al. Advancements in Ti alloy powder production by close-coupled gas atomization. Proceedings of the 2011 International Conference on ‘Powder Metallurgy and Particulate Materials’; 2011 May 18–21; San Francisco, CA. Metal Powder Industries Federation.
- [108] Hohman M, Ludwig N. System for the production of powders from metals. US Patent 5,284,329 A. 1994.
- [109] Franz H, Plochl L, Schimansky FP. Recent advances of titanium alloy powder production by ceramic-free inert gas atomization. Proceedings of the 24th Annual ITA conference Titanium 2008; 2008; Las Vegas, Nevada. International Titanium Association.
- [110] Hanusiak WM, McBride DR. Titanium powder production apparatus and method. US Patent Application 13/414,769. 2013.
- [111] Roberts PR, Airey JJ, Blout JE, et al. Method and apparatus for producing fine metal powder. US Patent 4,824,489 A. 1989.
- [112] Roberts PR. The production of PREP titanium powder. Proceeding of the 1989 PM Conference ‘Advances in Powder Metallurgy’; 1989; San Diego, CA. Metal Powder Industries Federation. p. 427–438.
- [113] Entezarian M, Allaire F, Tsantrizos P, et al. Plasma atomization: a new process for the production of fine, spherical powders. *JOM.* 1996;48(6):53–55.
- [114] Tsantrizos PG, Allaire F, Entezarian M. Method of production of metal and ceramic powders by plasma atomization. US Patent 5,707,419. 1998.
- [115] ASTM International. ASTM F2885 standard specification for metal injection molded titanium-6aluminum-4vanadium components for surgical implant applications. 2011.
- [116] ASTM International. ASTM F1580 standard specification for titanium and titanium-6 aluminum-4 vanadium alloy powders for coatings of surgical implants. 2012.
- [117] SAE International. Aerospace Material Specification (AMS) 4998 E: titanium alloy powder, 6Al-4V. 2013.
- [118] ASTM International. ASTM F2924: standard specification for additive manufacturing titanium-6 aluminum-4 vanadium with powder bed fusion. 2014.
- [119] ASTM International. ASTM F3001: standard specification for additive manufacturing titanium-6 aluminum-4 vanadium ELI (extra low interstitial) with powder bed fusion. 2014.
- [120] TEKNA. [cited 2017 Jul 26]. Available from: <http://www.tekna.com/spheroidization-systems>

- [121] Boulos M. Plasma power can make better powders. *Met Powder Rep.* **2004**;59(5):16–21.
- [122] Araci K, mangabhai D, Akhtar K. Production of titanium by the Armstrong process. In: Qian M, et al., editors. *Titanium powder metallurgy: science, technology and applications*. Oxford: Butterworth-Heinemann; **2015**. p. 149–162.
- [123] Gai G, Yang Y, Jin L, et al. Particle shape modification and related property improvements. *Powder Technol.* **2008**;183(1):115–121.
- [124] Sun YY, Gulizia S, Oh CH, et al. Manipulation and characterization of a novel titanium powder precursor for additive manufacturing applications. *JOM.* **2015**;67(3):564–572.
- [125] Sun P, Fang ZZ, Xia Y, et al. A novel method for production of spherical Ti-6Al-4V powder for additive manufacturing. *Powder Technol.* **2016**;301:331–335.
- [126] Fang ZZ, Xia Y, Sun P, et al. Production of substantially spherical metal powders. US Patent 9,421,612. 2016.
- [127] Majima K, Hirata T, Shouji K. Effects of purity of titanium powder and porosity on static tensile properties of sintered titanium specimens. *J Jpn Inst Met.* **1987**;51:1194–1200.
- [128] Yan M, Tang HP, Qian M. Scavenging of oxygen and chlorine from powder metallurgy (PM) titanium and titanium alloys. In: Qian M, et al., editors. *Titanium powder metallurgy*. Oxford: Butterworth-Heinemann; **2015**. p. 253–276.
- [129] Froes FH. Enhanced fatigue behaviour in blended elemental PM titanium alloys. *Met Powder Rep.* **1987**;42:217–218.
- [130] Qian M. Cold compaction and sintering of titanium and its alloys for near-net-shape or preform fabrication. *Int J Powder Metall.* **2010**;46:29–44.
- [131] Mo W, Deng GZ, Luo FC. *Titanium metallurgy*. Beijing: Publishing Press of Metallurgical Industry; **2007**.
- [132] Bickerdike RL, Sutcliffe DA. The tensile strength of titanium at various temperatures. *Powder Metall.* **1951**;9:153–159.
- [133] Watanabe T, Horikoshi Y. The sintering phenomenon of titanium powders- a discussion. *Int J Powder Metall Powder Technol.* **1976**;12:209–214.
- [134] Lütjering G, Williams JC. *Titanium*. Berlin: Springer; **2007**.
- [135] Lampman S. Wrought titanium and titanium alloys. In: *ASM handbook, volume 2: properties and selection: nonferrous alloys and special purpose materials*. Materials Park (OH): ASM International; **1990**. p. 592–633.
- [136] Fitzpatrick GA, Broughton T. Diffusion bonding aeroengine components. *Defence Sci J.* **1988**;38:477–486.
- [137] Lee YT, Peters M, Wirth G. Effects of thermomechanical treatment on microstructure and mechanical properties of blended elemental Ti-6Al-4V compacts. *Mater Sci Eng A.* **1988**;102:105–114.
- [138] Eylon D, Vogt RG, Froes FH. Property improvement of low chlorine titanium alloy blended elemental powder compacts by microstructure modification. *Proceedings of the 1986 Annual PM Conference 'Progress in Powder Metallurgy'*; 1986; Metal Powder Industries Federation. p. 625–634.
- [139] Kroll W. Verformbare legierungen des titans. *Zeitschrift fuer Metallkunde.* **1937**;29:189–192.
- [140] Dean RS, Long JR, Wartman FS, et al. Preparation and properties of ductile titanium. *Trans AIME.* **1946**;166:369–381.
- [141] *Titanium: past, present, and future*. Washington (DC): National Materials Advisory Board; **1983**.
- [142] Saito T. The automotive application of discontinuously reinforced TiB-Ti composites. *JOM.* **2004**;56:33–36.
- [143] Arensbarger DS, Pugin VS, Fedorchenko IM. Properties of electrolytic and reduced titanium powders and sinterability of porous compacts from such powders. *Sov Powder Metall Met Ceram.* **1968**;7:362–367.
- [144] Kanto Yakin Kogyo Co. [cited 2017 Apr 6]. Available from: <http://www.oxynton.com/>
- [145] Heaney DF, German RM. Advances in the sintering of titanium powders. *Proceedings Euro PM2004*; 2004; Vienna, Austria. European Powder Metallurgy Association.
- [146] Yang YF, Luo SD, Schaffer GB, et al. Sintering of Ti-10V-2Fe-3Al and mechanical properties. *Mater Sci Eng A.* **2011**;528:6719–6726.
- [147] Ivasishin OM, Savvakina DG. The impact of diffusion on synthesis of high-strength titanium alloys from elemental powder blends. *Key Eng Mater.* **2010**;436:113–121.
- [148] Eylon D. Faceted fracture in beta annealed titanium alloys. *Metall Trans A.* **1979**;10:311–317.
- [149] Brooks CR. Heat treatment, structure and properties of nonferrous alloys. Metals Park (OH): American Society for Metals; **1982**; 420.
- [150] Boyer R, Welsch G, Collings EW. *Materials properties handbook – titanium alloys*. Materials Park (OH): ASM International; **1994**.
- [151] Yang L. Effect of particle size on oxygen content and porosity of sintered Ti-6Al-4V [Master's thesis]. Salt Lake City (UT): University of Utah; **2015**.
- [152] Abkowitz S, Siergiej JM, Regan RD. Titanium P/M preforms, parts and composites. *Mod Dev Powder Metall.* **1971**;4:501–511.
- [153] Samarov V, Seliverstov D, Froes FHS. Fabrication of near-net-shape cost-effective titanium components by use of prealloyed powders and hot isostatic pressing. In: Qian M, et al., editor. *Titanium powder metallurgy*. Oxford (UK): Butterworth-Heinemann; **2015**. p. 313–336.
- [154] Wirth G, Grundhoff KJ, Smarsly W. Correlations between post-HIP treatment, resulting microstructure, and fatigue behavior of prealloyed Ti-6Al-4V powder compacts. *Sampe Quart.* **1986**;17:34–39.
- [155] Witt RH, Highberger WT. Hot isostatic pressing of near-net titanium structural parts. *Proceedings of the Symposium 109th AIME Annual Meeting 'Powder Metallurgy of Titanium Alloys'*; 1980 Feb 26–28; Los Vegas, NV. Metallurgical Society of AIME; p. 255–265.
- [156] Schwenker SW, Eylon D, Froes EH. Influence of foreign particles on fatigue behavior of Ti-6Al-4V prealloyed powder compacts. *Metall Trans A.* **1986**;17:271–280.
- [157] Zhang K, Mei J, Wain N, et al. Effect of hot-isostatic pressing parameters on the microstructure and properties of powder Ti-6Al-4V hot-isostatically-pressed samples. *Metall Mater Trans A.* **2010**;41:1033–1045.
- [158] Abkowitz S, Abkowitz S, Fisher H. Titanium alloy components manufacture from blended elemental powder and the qualification process. In: Qian M,

- et al., editors. Titanium powder metallurgy. Oxford (UK): Butterworth-Heinemann; 2015. p. 299–312.
- [159] Dawson DB, Ulithny MG. Characterization of the tensile properties of hot isostatically pressed Ti-6Al-6V-2Sn. Proceedings of the Symposium 109th AIME Annual Meeting 'Powder Metallurgy of Titanium Alloys'; 1980 Feb 26–28; Los Vegas, NV. The Metallurgical Society of AIME; p. 115–126.
- [160] Kim Y, Kim E-P, Song Y-B, et al. Microstructure and mechanical properties of hot isostatically pressed Ti-6Al-4V alloy. *J Alloys Compd.* 2014;603:207–212.
- [161] Kao WH, Eylon D, Yolton CF, et al. Effect of temporary alloying by hydrogen (hydrovac) on the vacuum hot pressing and microstructure of titanium alloy powder compacts. *Progr Powder Metall.* 1981;37:289–301.
- [162] Devillard J, Herteman JP. Evaluation of Ti-6Al-4V powder compacts fabricated by the PSV process. Proceedings of the Symposium 109th AIME Annual Meeting 'Powder Metallurgy of Titanium Alloys'; 1980 Feb 26–28; Los Vegas, NV. The Metallurgical Society of AIME; p. 59–70.
- [163] Bolzoni L, Meléndez IM, Ruiz-Navas EM, et al. Microstructural evolution and mechanical properties of the Ti-6Al-4V alloy produced by vacuum hot-pressing. *Mater Sci Eng A.* 2012;546:189–197.
- [164] Bolzoni L, Ruiz-Navas EM, Neubauer E, et al. Mechanical properties and microstructural evolution of vacuum hot-pressed titanium and Ti-6Al-7Nb alloy. *J Mech Behav Biomed Mater.* 2012;9:91–99.
- [165] Bolzoni L, Ruiz-Navas EM, Neubauer E, et al. Inductive hot-pressing of titanium and titanium alloy powders. *Mater Chem Phys.* 2012;131:672–679.
- [166] Bolzoni L, Ruiz-Navas EM, Gordo E. Influence of vacuum hot-pressing temperature on the microstructure and mechanical properties of Ti-3Al-2.5V alloy obtained by blended elemental and master alloy addition powders. *Mater Chem Phys.* 2012;137:608–616.
- [167] Bolzoni L, Ruiz-Navas EM, Gordo E. Processing of elemental titanium by powder metallurgy techniques. *Mater Sci Forum.* 2013;765:383–387.
- [168] Boeing. BMS7-393B: titanium 6Al-4V, pressed and sintered powder compacts. 2012.
- [169] Dale JR. Sustainability manufacturing within the PM industry. *Int J Powder Metall.* 2011;47:23–25.
- [170] Ivasishin OM, Savvakina DG, Gumenyak MM, et al. Role of surface contamination in titanium PM. *Key Eng Mater.* 2012;520:121–132.
- [171] Duz VA, Ivasishin OM, Moxson VS, et al. Cost-effective titanium alloy powder compositions and method for manufacturing flat or shaped articles from these powders. U.S. Patent 7,993,577. 2011.
- [172] Wang H, Lefler M, Fang ZZ, et al. Titanium and titanium alloy via sintering of TiH₂. *Key Eng Mater.* 2010;436:157–163.
- [173] Robertson IM, Schaffer GB. Comparison of sintering of titanium and titanium hydride powders. *Powder Metall.* 2010;53:12–19.
- [174] Paramore JD. Relationship between processing, structure, and properties of titanium alloys produced by hydrogen sintering and phase transformation (HSPT) [PhD thesis]. Salt Lake City (UT): University of Utah; 2015.
- [175] Ivasishin OM, Savvakina DG, Froes F, et al. Synthesis of alloy Ti-6Al-4V with low residual porosity by a powder metallurgy method. *Powder Metall Met Ceram.* 2002;41:382–390.
- [176] McQuillan AD. An experimental and thermodynamic investigation of the hydrogen-titanium system. *Proc R Soc A.* 1950;204:309–323.
- [177] Carreño-Morelli E, Bidaux JE, Rodríguez-Arbaizar M, et al. Production of titanium grade 4 components by powder injection moulding of titanium hydride. *Powder Metall.* 2014;57:89–92.
- [178] Gregory E. Fabrication of niobium superconductor alloys. U.S. Patent 3,472,705. 1969.
- [179] Greenspan J, Rizzitano F, Scala E. Metal matrix composites by decomposition sintering of titanium hydride. 1970, 24.
- [180] Greenspan J, Rizzitano F, Scala E. Titanium powder metallurgy by decomposition sintering of the hydride. Proceedings of the 2nd International Conference 'Titanium, Science and Technology'; 1973 May 2–5; Cambridge, MA. Plenum Press; p. 365–379.
- [181] Obara K, Nishino Y, Matsumoto S. Process for producing a sintered article of a titanium alloy. U.S. Patent 3,950,166. 1976.
- [182] Yolton CF, Froes FH. Method for producing powder metallurgy articles. US Patent 4,219,357. 1980.
- [183] Pankevich AP, Chertovich AF, Libenson GA. Production of titanium specimens from a titanium hydride powder by hot pressing. *Sov Powder Metall Met Ceram.* 1986;25:89–92.
- [184] Li Y, Chou XM, Yu L. Dehydrogenation debinding process of MIM titanium alloys by TiH₂ powder. *Powder Metall.* 2006;49:236–239.
- [185] Taddei E, Henriques V. Densification and microstructural behaviour on the sintering of blended elemental Ti-35Nb-7Zr-5Ta alloy. *Mater Sci Forum.* 2006;530-531:341–346.
- [186] Carreño-Morelli E, Krstev W. Powder injection moulding of titanium from TiH₂. Proceedings Euro PM2009; 2009; Copenhagen, Denmark. European Powder Metallurgy Association; p. 1–6.
- [187] Zhang JM, Yi JH, Gan GY, et al. Research on dehydrogenation and sintering process of titanium hydride for manufacture titanium and titanium alloy. *Adv Mater Res.* 2013;616:1823–1829.
- [188] Robertson IM, Schaffer GB. Swelling during sintering of titanium alloys based on titanium hydride powder. *Powder Metall.* 2010;53:27–33.
- [189] Dunlop J, Rensing H. Method for making tungsten-titanium sputtering targets and product. U.S. Patent 5,234,487. 1993.
- [190] Gladden T. Process for the manufacture by sintering of a titanium part and a decorative article made using a process of this type. U.S. Patent 5,441,695. 1995.
- [191] Bertheville B, Neudemberger M, Bidaux J-E. Powder sintering and shape-memory behaviour of NiTi compacts synthesized from Ni and TiH₂. *Mater Sci Eng A.* 2004;384:143–150.
- [192] Ivasishin OM, Anokhin VM, Demidik AN, et al. Cost-effective blended elemental powder metallurgy of titanium alloys for transportation application. *Key Eng Mater.* 2000;188:55–62.
- [193] Ivasishin OM, Savvakina DG, Moxson VS, et al. Titanium powder metallurgy for automotive components. *Mater Technol.* 2002;17(1):20–25.
- [194] Ivasishin OM, Bondareva KA, Bondarchuk VI, et al. Fatigue resistance of powder metallurgy Ti-6Al-4V alloy. *Strength Mater.* 2004;36:225–230.

- [195] Ivasishin OM, Savvakina DG, Bielov IS, et al. Microstructure and properties of titanium alloys synthesized from hydrogenated titanium powders. *Proceedings MS&T 2005*; 2005; Pittsburgh, PA. TMS. p. 151–158.
- [196] Ivasishin OM, Eylon D, Bondarchuk VI, et al. Diffusion during powder metallurgy synthesis of titanium alloys. *Defect Diffus Forum*. 2008;277:177–185.
- [197] Savvakina DH, Humenyak MM, Matviichuk MV, et al. Role of hydrogen in the process of sintering of titanium powders. *Mater Sci*. 2012;47:651–661.
- [198] Ivasishin OM. Cost-effective manufacturing of titanium parts with powder metallurgy approach. *Mater Forum*. 2005;29:1–8.
- [199] Moxson V, Ivanov E. Method for manufacturing fully dense metal sheets and layered composites from reactive alloy powders. U.S. Patent 7,566,415. 2009.
- [200] Moxson V, Duz V. Process of direct powder rolling of blended titanium alloys, titanium matrix composites, and titanium aluminides. U.S. Patent 7,311,873. 2007.
- [201] Ivasishin O, Moxson V. Low-cost titanium hydride powder metallurgy. In: Qian M, et al., editors. *Titanium powder metallurgy*. Oxford: Butterworth-Heinemann; 2015. p. 117–148.
- [202] Fang ZZ, Sun P, Wang H. Hydrogen sintering of titanium to produce high density fine grain titanium alloys. *Adv Eng Mater*. 2012;14:383–387.
- [203] Sachdev AK, Kulkarni K, Fang ZZ, et al. Titanium for automotive applications: challenges and opportunities in materials and processing. *JOM*. 2012;64:553–565.
- [204] Sun P, Fang ZZ, Koopman M. A comparison of hydrogen sintering and phase transformation (HSPT) processing with vacuum sintering of CP-Ti. *Adv Eng Mater*. 2013;15:1007–1013.
- [205] Dunstan M, Paramore J, Koopman M, et al. HSPT: a low-cost method for producing titanium alloys with exceptional mechanical properties. *Proc. 2014 World Congress on 'Powder Metallurgy and Particulate Materials'*; 2014 May 18–22; Orlando, FL. Metal Powder Industries Federation; p. 9127–9136.
- [206] Fang ZZ, Sun P, Paramore J, et al. Powder metallurgy methods for the production of fine and ultrafine grain Ti and Ti alloys. U.S. Patent Application 14/152,787. 2014.
- [207] Paramore JD, Fang ZZ, Sun P. Hydrogen sintering of titanium and its alloys. In: Qian M, et al., editors. *Titanium powder metallurgy*. Oxford: Butterworth-Heinemann; 2015. p. 163–182.
- [208] Paramore JD, Fang ZZ, Sun P, et al. A powder metallurgy method for manufacturing Ti-6Al-4V with wrought-like microstructures and mechanical properties via hydrogen sintering and phase transformation (HSPT). *Scr Mater*. 2015;107:103–106.
- [209] Paramore JD, Sun P, Dunstan M, et al. Microstructure and mechanical properties of titanium alloys produced by hydrogen sintering and phase transformation (HSPT). 13th World Conference on Titanium; 2015; San Diego, CA. John Wiley & Sons.
- [210] Sun P, Fang ZZ, Koopman M, et al. An experimental study of the (Ti-6Al-4V)-xH phase diagram using in situ synchrotron XRD and TGA / DSC techniques. *Acta Mater*. 2015;84:29–41.
- [211] Sun P, Fang ZZ, Koopman M, et al. Phase transformations and formation of ultra-fine microstructure during hydrogen sintering and phase transformation (HSPT) processing of Ti-6Al-4V. *Metall Mater Trans A*. 2015;46:5546–5560.
- [212] Sun P. Powder metallurgy titanium by the hydrogen sintering and phase transformation (HSPT) process [PhD thesis]. Salt Lake City (UT): University of Utah; 2015.
- [213] Paramore JD, Fang ZZ, Dunstan M, et al. Hydrogen-enabled microstructure and fatigue strength engineering of titanium alloys. *Sci Rep*. 2017;7:41444.
- [214] Cao F, Ravi Chandran KS, Kumar P, et al. New powder metallurgical approach to achieve high fatigue strength in Ti-6Al-4V alloy. *Metall Mater Trans A*. 2016;47:2335–2345.
- [215] Buchelnikov VD, Louzguine-Luzgin DV, Xie G, et al. Heating of metallic powders by microwaves: experiment and theory. *J Appl Phys*. 2008;104:1–11.
- [216] Sato M, Fukusima H, Ozeki F, et al. Experimental investigation of mechanism of microwave heating in powder metals. 29th International Conference on 'Infrared and Millimeter Waves' and 12th International Conference on 'Terahertz Electronics'; 2004 Sep 27–Oct 1; Karlsruhe, Germany. IEEE; p. 831–832.
- [217] Luo S, Yang Y, Schaffer G, et al. Characteristics of microwave sintering of titanium powder compacts. Ti-2011 Proceedings of the 11th World Conference on 'Titanium'; 2012; Beijing, China. Science Press. p. 1826–1829.
- [218] Luo SD, Guan CL, Yang YF, et al. Microwave heating, isothermal sintering, and mechanical properties of powder metallurgy titanium and titanium alloys. *Metall Mater Trans A*. 2013;44:1842–1851.
- [219] Bruce RW, Fliflet AW, Huey HE, et al. Microwave sintering and melting of titanium powder for low-cost processing. *Key Eng Mater*. 2010;436:131–140.
- [220] Kutty MG, Bhaduri S, Bhaduri SB. Gradient surface porosity in titanium dental implants: relation between processing parameters and microstructure. *J Mater Sci Mater Med*. 2004;15:145–150.
- [221] Luo SD, Bettles CJ, Yan M, et al. Microwave sintering of titanium. *Key Eng Mater*. 2010;436:141–147.
- [222] Marcelo T, Mascarenhas JMG, Oliveira FAC. Microwave sintering: a novel approach to powder technology. *Mater Sci Forum*. 2010;636–637:946–951.
- [223] Hashiguchi T, Sueyoshi H. Effect of atmosphere on microwave heating of titanium powder. *Powder Metall*. 2011;54:537–542.
- [224] Luo SD, Yan M, Schaffer GB, et al. Sintering of titanium in vacuum by microwave radiation. *Metall Mater Trans A*. 2011;42:2466–2474.
- [225] Fliflet AW, Miller SL, Imam MA. Evaluation of microwave-sintered titanium and titanium alloy powder compacts. *Ceram Trans*. 2012;234:81–92.
- [226] Yu P, Stephani G, Luo SD, et al. Microwave-assisted fabrication of titanium hollow spheres with tailored shell structures for various potential applications. *Mater Lett*. 2012;86:84–87.
- [227] Imam MA, Fliflet AW. Sintering of metal and alloy powders by microwave/millimeter-wave heating. U.S. Patent 8,431,071 B2. 2013.
- [228] Luo SD, Yang YF, Schaffer GB, et al. Novel fabrication of titanium by pure microwave radiation of titanium hydride powder. *Scr Mater*. 2013;69:69–72.
- [229] Luo SD, Qian M, Imam MA. Microwave sintering of titanium and titanium alloys. In: Qian M, et al., editors. *Titanium powder metallurgy*. Oxford: Butterworth-Heinemann; 2015. p. 237–251.

- [230] Nakamori Y, Orimo SI, Tsutaoka T. Dehydriding reaction of metal hydrides and alkali borohydrides enhanced by microwave irradiation. *Appl Phys Lett*. 2006;88:2–5.
- [231] Matsuo M, Nakamori Y, Yamada K, et al. Effects of microwave irradiation on the dehydriding reaction of the composites of lithium borohydride and microwave absorber. *Appl Phys Lett*. 2007;90:232907.
- [232] ASTM International. ASTM B348-10: standard specification for titanium and titanium alloy bars and billets. 2010.
- [233] Tokita M. Trends in advanced SPS spark plasma sintering systems and technology. *J Soc Powder Technol Jpn*. 1993;30:790–804.
- [234] Groza JR, Zavaliangos A. Sintering activation by external electrical field. *Mater Sci Eng A*. 2000;287:171–177.
- [235] Eriksson M, Shen Z, Nygren M. Fast densification and deformation of titanium powder. *Powder Metall*. 2005;48:231–236.
- [236] Ertorer O, Topping TD, Li Y, et al. Nanostructured Ti consolidated via spark plasma sintering. *Metall Mater Trans A*. 2011;42:964–973.
- [237] Yang Y, Imai H, Kondoh K, et al. Comparison of spark plasma sintering of elemental and master alloy powder mixes and prealloyed Ti-6Al-4V powder. *Int J Powder Metall*. 2014;50:41–47.
- [238] Yang Y, Qian M. Spark plasma sintering and hot pressing of titanium and titanium alloys. In: Qian M, et al., editors. *Titanium powder metallurgy*. Oxford: Butterworth-Heinemann; 2015. p. 219–235.
- [239] Naeser C, Zirm F. Steel strip experimentally rolled from powder. *Stahl Eisen*. 1950;70:995–1004.
- [240] Knopp WV. Roll compacting of metal powders. In: *ASM handbook, volume 7: powder metal technologies and applications*. Materials Park (OH): ASM International; 1998. p. 389–395.
- [241] Park NK, Lee CH, Kim JH, et al. Characteristics of powder-rolled and sintered sheets made from HDH Ti powders. *Key Eng Mater*. 2012;520:281–288.
- [242] Cantin GMD, Kean PL, Stone NA, et al. Innovative consolidation of titanium and titanium alloy powders by direct rolling. *Powder Metall*. 2011;54:188–192.
- [243] Peter WH, Muth T, Chen W, et al. Titanium sheet fabricated from powder for industrial applications. *JOM*. 2012;64:566–571.
- [244] Mangabhai D, Araci K, Akhtar MK, et al. Processing of titanium powder into consolidated parts & sheet. *Key Eng Mater*. 2013;551:57–66.
- [245] Cantin GMD, Gibson MA. Titanium sheet fabrication from powder. In: Qian M, et al., editors. *Titanium powder metallurgy*. Oxford: Butterworth-Heinemann; 2015. p. 383–403.
- [246] Liu Y, Chen LFF, Tang HPP, et al. Design of powder metallurgy titanium alloys and composites. *Mater Sci Eng A*. 2006;418:25–35.
- [247] Qiu JW, Liu Y, Liu YB, et al. Microstructures and mechanical properties of titanium alloy connecting rod made by powder forging process. *Mater Des*. 2012;33:213–219.
- [248] Yamaguchi T, Hotta A, Shibata Y, et al. Process for forging titanium-based material, process for producing engine valve, and engine valve. U.S. Patent 6,599,467. 2003.
- [249] Froes FHS, Gungor MN, Imam MA. Cost-affordable titanium: the component fabrication perspective. *JOM*. 2007;59:28–31.
- [250] El-Soudani SM, Yu KO, Crist EM, et al. Optimization of blended-elemental powder-based titanium alloy extrusions for aerospace applications. *Metall Mater Trans A*. 2013;44:899–910.
- [251] Guo H, Zhao Z, Duan C, et al. The powder sintering and isothermal forging of Ti-10V-2Fe-3Al. *JOM*. 2008;60:47–49.
- [252] Barnes JE, Peter W, Blue CA. Evaluation of low cost titanium alloy products. *Mater Sci Forum*. 2009;618–619:165–168.
- [253] Yamamoto Y, Peter WH, Sabau AS, et al. Low cost titanium near-net-shape manufacturing using Armstrong process CP-Ti and Ti-6Al-4V powders. *Proceedings of the 2010 International Conference on 'Powder Metallurgy and Particulate Materials'*; 2010; Princeton, NJ. Metal Powder Industries Federation. p. 324–333.
- [254] Chen W, Yamamoto Y, Peter WH, et al. Cold compaction study of Armstrong process® Ti-6Al-4V powders. *Powder Technol*. 2011;214:194–199.
- [255] Chen W, Yamamoto Y, Peter WH, et al. The investigation of die-pressing and sintering behavior of ITP CP-Ti and Ti-6Al-4V powders. *J Alloys Compd*. 2012;541:440–447.
- [256] Reddoch TW. Pneumatic isostatic forging. In: *ASM handbook, volume 7: powder metal technologies and applications*. Materials Park (OH): ASM International; 1998. p. 638–641.
- [257] Hodge ES, Tavenner RF. Method for pneumatic isostatic processing of a workpiece. U.S. Patent 5,816,090. 1998.
- [258] Hodge ES, Reddoch TW, Viswanathan S. A new casting defect healing technology. 1997.
- [259] Kaneko Y, Ameyama K, Saito K, et al. Injection molding of Ti powder. *J Jpn Soc Powder Powder Metall*. 1988;35:646–650.
- [260] Ameyama K, Kaneko Y, Iwasaki H, et al. Application of metal injection molding to titanium and titanium hydride powders. *J Soc Mater Sci Jpn*. 1990;39:120–125.
- [261] Kaneko Y, Ameyama K, Sakaguchi S. Application of injection molding to Ti-5 wt.%Co and Ti-6 wt.%Al-4 wt.%V mixed powders. *J Jpn Soc Powder Powder Metall*. 1990;37:605.
- [262] Froes FH. Advances in titanium metal injection molding. *Powder Metall Met Ceram*. 2007;46:303–310.
- [263] German RM. Status of metal powder injection molding of titanium. *Int J Powder Metall*. 2010;46:11–17.
- [264] Wen G, Cao P, Gabbitis B, et al. Development and design of binder systems for titanium metal injection molding: an overview. *Metall Mater Trans A*. 2013;44:1530–1547.
- [265] Ebel T, Friederici V, Imgrund P, et al. Metal injection molding of titanium. In: Qian M, et al., editors. *Titanium powder metallurgy*. Oxford: Butterworth-Heinemann; 2015. p. 337–360.
- [266] German RM. Powder-processing linkages to properties for complex titanium shapes by injection molding. In: Qian M, et al., editor. *Titanium powder metallurgy*. Oxford: Butterworth-Heinemann; 2015. p. 361–382.
- [267] Kyogoku H, Toda T, Shinohara K. Sintering behavior of titanium compacts with the addition of TiH powder by using injection moldings. *J Jpn Soc Powder Powder Metall*. 1993;40:439–443.

- [268] Kusaka K, Kohno T, Kondo T, et al. Tensile behavior of sintered titanium by MIM process. *J Jpn Soc Powder Powder Metall.* **1995**;42:383–387.
- [269] Kato K. Effect of sintering temperature on density and tensile properties of titanium compacts by metal injection molding. *J Jpn Soc Powder Powder Metall.* **1999**;46:865–869.
- [270] Thavanayagam G, Pickering KL, Swan JE, et al. Analysis of rheological behaviour of titanium feedstocks formulated with a water-soluble binder system for powder injection moulding. *Powder Technol.* **2015**;269:227–232.
- [271] Nyberg E, Miller M, Simmons K, et al. Microstructure and mechanical properties of titanium components fabricated by a new powder injection molding technique. *Mater Sci Eng C.* **2005**;25:336–342.
- [272] Carreño-Morelli E, Krstev W, Romeira B, et al. Titanium parts by powder injection moulding of TiH₂-based feedstocks. *PIM Int.* **2010**;4:60–63.
- [273] Cha B, Jang JM, Lee W, et al. Development of recycled TiH₂ feedstock for powder injection molding. *Adv Mater Res.* **2011**;291–294:3024–3027.
- [274] Park SJ, Wu Y, Heaney DF, et al. Rheological and thermal debinding behaviors in titanium powder injection molding. *Metall Mater Trans A.* **2009**;40:215–222.
- [275] Yan M, Qian M, Kong C, et al. Impacts of trace carbon on the microstructure of as-sintered biomedical Ti-15Mo alloy and reassessment of the maximum carbon limit. *Acta Biomater.* **2014**;10:1014–1023.
- [276] Zhao D, Ebel T, Yan M, et al. Trace carbon in biomedical beta-titanium alloys: recent progress. *JOM.* **2015**;67:2236–2243.
- [277] Zhao D, Chang K, Ebel T, et al. Titanium carbide precipitation in Ti-22Nb alloy fabricated by metal injection moulding. *Powder Metall.* **2014**;57:2–4.
- [278] Zhao D, Chang K, Ebel T, et al. Sintering behavior and mechanical properties of a metal injection molded Ti-Nb binary alloy as biomaterial. *J Alloys Compd.* **2015**;640:393–400.
- [279] Ferri OM, Ebel T, Bormann R. High cycle fatigue behaviour of Ti-6Al-4V fabricated by metal injection moulding technology. *Mater Sci Eng A.* **2009**;504:107–113.
- [280] Muterlle PV, Molinari A, Perina M, et al. Influence of shot peening on tensile and high cycle fatigue properties of Ti6Al4V alloy produced by MIM. *Proceedings of the PM2010*; 2010 Oct 10–14; Florence, Italy. European Powder Metallurgy Association. p. 791–796.
- [281] Takekawa J, Sakurai N. Effect of the processing conditions on density, strength and microstructure of Ti-12Mo alloy fabricated by PIM process. *J Jpn Soc Powder Powder Metall.* **1999**;46:877–881.
- [282] Bidaux JE, Closuit C, Rodriguez-Arbaizar M, et al. Processing of a low modulus Ti-Nb biomaterial by metal injection molding (MIM). *PIM Int.* **2012**;6:72–75.
- [283] Zhao D, Chang K, Ebel T, et al. Microstructure and mechanical behavior of metal injection molded Ti-Nb binary alloys as biomedical material. *J Mech Behav Biomed Mater.* **2013**;28:171–182.
- [284] Kafkas F, Ebel T. Metallurgical and mechanical properties of Ti-24Nb-4Zr-8Sn alloy fabricated by metal injection molding. *J Alloys Compd.* **2014**;617:359–366.
- [285] Shimizu T, Kitajima A, Kato K, et al. Supercritical debinding and its application to PIM of Ti-Al intermetallic compounds. *Proceedings of the 2000 PM World Congress*; 2000 Nov 12–16; Kyoto, Japan. Japan Society of Powder and Powder Metallurgy. p. 292–295.
- [286] Terauchi S, Teraoka T, Shinkuma T, et al. Development of production technology by metallic powder injection molding for TiAl-type intermetallic compound with high efficiency. *15th International Plansee Seminar*; 2001; Reutte, Austria. Plansee Holding AG. p. 610–624.
- [287] Gerling R, Schimansky F-P. Prospects for metal injection moulding using a gamma titanium aluminide based alloy powder. *Mater Sci Eng A.* **2002**;329–331:45–49.
- [288] Gerling R, Aust E, Limberg W, et al. Metal injection moulding of gamma titanium aluminide alloy powder. *Mater Sci Eng A.* **2006**;423:262–268.
- [289] Kim YC, Lee S, Ahn S, et al. Application of metal injection molding process to fabrication of bulk parts of TiAl intermetallics. *J Mater Sci.* **2007**;42:2048–2053.
- [290] Zhang H, He X, Qu X, et al. Microstructure and mechanical properties of high Nb containing TiAl alloy parts fabricated by metal injection molding. *Mater Sci Eng A.* **2009**;526:31–37.
- [291] Carreño-Morelli E, Rodríguez-Arbaizar M, Amherd A, et al. Porous titanium processed by powder injection moulding of titanium hydride and space holders. *Powder Metall.* **2014**;57:93–96.
- [292] Tuncer N, Bram M, Laptev A, et al. Study of metal injection molding of highly porous titanium by physical modeling and direct experiments. *J Mater Process Technol.* **2014**;214:1352–1360.
- [293] Laptev AM, Daudt NF, Guillon O, et al. Increased shape stability and porosity of highly porous injection-molded titanium parts. *Adv Eng Mater.* **2015**;17:1579–1587.
- [294] Gibson I, Rosen DW, Stucker B. *Additive manufacturing technologies*. Boston (MA): Springer; **2010**; 459.
- [295] Dutta B, Froes FHS. The additive manufacturing (AM) of titanium alloys. In: Qian M, et al., editors. *Titanium powder metallurgy*. Oxford: Butterworth-Heinemann; **2015**. p. 447–468.
- [296] Frykholm R, Andersson B-G, Carlström R. Digital Metal® – a technology for production of small precision components at high productivity with different type of metals. *Proceedings of the AMPM 2015*; 2015 May 17–20; San Diego, CA. Metal Powder Industries Federation.
- [297] SAE International. AMS4999A: titanium alloy direct deposited products, 6Al-4V, annealed. 2011.
- [298] Koike M, Greer P, Owen K, et al. Evaluation of titanium alloys fabricated using rapid prototyping technologies—electron beam melting and laser beam melting. *Materials.* **2011**;4:1776–1792.
- [299] Wang F, Williams S, Colegrove P, et al. Microstructure and mechanical properties of wire and arc additive manufactured Ti-6Al-4V. *Metall Mater Trans A.* **2013**;44:968–977.
- [300] Frazier WE. Metal additive manufacturing – a review. In: *ASM handbook volume 20: materials selection and design*. Materials Park (OH): ASM International; **2015**.

- [301] Mellor I, Doughty G, Piper M, et al. A novel process for the production of titanium powders bespoke to additive manufacturing. Proceedings of the 13th World Conference on Titanium; San Diego, CA. TMS; 2016. p. 145–150.
- [302] Withers JC, Shapovalov V, Storm R, et al. There is low cost titanium componentry today. Key Eng Mater. 2013;551:11–15.
- [303] Sherman RG, Kessler HD. Investigation of the heat treatability of the 6% aluminum-4% vanadium titanium based alloy. Trans ASM. 1956;48:657–676.
- [304] Eylon D, Pierce CM. Effect of microstructure on notch fatigue properties of Ti-6Al-4V. Metall Trans A. 1976;7:111–121.
- [305] Williams JC, Starke EA. The role of thermomechanical processing in tailoring the properties of aluminum and titanium alloys. In: G. Krauss, editor. Deformation, processing, and structure. Materials Park (OH): ASM International; 1984. p. 306–314.
- [306] Andersen PJ, Svoyatytsky VM, Froes FH, et al. Fracture behavior of blended elemental P/M titanium alloys. Mod Dev Powder Metall. 1981;13:537–549.
- [307] Smith PR, Cooke CM, Patel A, et al. Evaluation of Ti-6Al-4V powder metallurgy plate. National PM Conf. Proceedings of the ‘Progress in Powder Metallurgy’; 1982; MPIF. p. 339–359.
- [308] Smith PR, Froes FH, Cooke CM. Evaluation of blended elemental Ti-6Al-4V plate in three microstructural conditions. 28th National SAMPE Symposium and Exhibition; 1983; Anaheim, CA; p. 406–421.
- [309] Abkowitz S, Kardys GJ, Fujishiro S, et al. Titanium alloy shapes from elemental blend powder and tensile and fatigue properties of low chloride compositions. Proceedings of the Symposium AIME Annual Meeting ‘Titanium Net Shape Technologies’; 1984; Los Angeles, CA. The Metallurgical Society of AIME; p. 107–120.
- [310] Park J, Toaz MW, Ro DH, et al. Blended elemental powder metallurgy of titanium alloys. Proceedings of the Symposium AIME Annual Meeting ‘Titanium Net Shape Technologies’; 1984; Los Angeles, CA. The Metallurgical Society of AIME; p. 95–105.
- [311] Abkowitz S, Rowell D. Superior fatigue performance of P/M Ti-6Al-4V. JOM. 1986;38:36–39.
- [312] Weiss I, Eylon D, Toaz MW, et al. Effect of isothermal forging on microstructure and fatigue behavior of blended elemental Ti-6Al-4V powder compacts. Metall Mater Trans A. 1986;17:549–559.
- [313] Hagiwara M, Kaieda Y, Kawabe Y, et al. Fatigue property enhancement of α - β titanium alloys by blended elemental P/M approach. ISIJ Int. 1991;31:922–930.
- [314] Moxson VS, Sjoblom P, Trzcinski MJ. Ti-6Al-4V properties achieved via extra low chlorine titanium powder. Proceedings of the PM1992; 1992; San Francisco, CA. Metal Powder Industries Federation. p. 125–140.
- [315] Froes FH, Suryanarayana C. Powder processing of titanium alloys. Rev Particulate Mater. 1993;1:223–275.
- [316] Fujita T, Ogawa A, Ouchi C, et al. Microstructure and properties of titanium alloy produced in the newly developed blended elemental powder metallurgy process. Mater Sci Eng A. 1996;213:148–153.
- [317] Haase C, Lapovok R, Ng HP, et al. Production of Ti-6Al-4V billet through compaction of blended elemental powders by equal-channel angular pressing. Mater Sci Eng. A. 2012;550:263–272.
- [318] Hagiwara M, Kaieda Y, Kawabe Y, et al. Property enhancement of alpha-beta titanium alloys by blended elemental P/M method. Proceedings of the 7th International Conference on Titanium ‘Titanium 92, Science and Technology’; 1992 Jun 29–Jul 2; San Diego, CA. TMS; p. 887–894.
- [319] Hall IW, Hammond C. Fracture toughness, strength and microstructure in α + β titanium alloys. Proceedings of the 3rd International Conference on Titanium ‘Titanium and Titanium alloys’; 1976 May 18–21; Moscow, USSR. Springer. p. 601–613.
- [320] Eylon D, Froes FH, Heggie DG, et al. Influence of thermomechanical processing on low cycle fatigue of prealloyed Ti-6Al-4V powder compacts. Metall Trans A. 1983;14:2497–2505.
- [321] Peebles RE, Parsons LD. Study of production methods of aerospace quality titanium alloy powder. Proceedings of the Symposium. AIME Annual Meeting ‘Titanium Net Shape Technologies’; 1984; Los Angeles, CA. The Metallurgical Society of AIME. p. 21–28.
- [322] Levin L, Vogt RG, Eylon D, et al. Fatigue resistance improvement of Ti-6Al-4V by thermochemical treatment. Titan Sci Technol. 1985;DMG:2107–2114.
- [323] Adachi S, Wagner L, Lütjering G. Influence of mean stress on fatigue strength of Ti 6Al-4V. Strength Met Alloys. 1986;3:2117–2122.
- [324] Chronister DJ, Scott SW, Stickle DR, et al. Induction skull melting of titanium and other reactive alloys. JOM. 1986;38:51–54.
- [325] Muller C, Gysler A, Lütjering G. Correlation between microstructure and fracture toughness of titanium alloys. Proceedings of the 7th International Conference on ‘The Strength of Metals and Alloys’; 1986 Aug 12–16; Montreal, Canada. Pergamon Press. p. 1175–1180.
- [326] Cooke CM, Eylon D, Froes FH. Effect of powder particle size on fatigue properties of Ti-6Al-4V compacts. Proceedings of the 8th Annual Technology Meeting of Int. Metall. Soc. ‘Microstructural Science’; 1987; Seattle, WA. ASM International. p. 303–314.
- [327] Sheinker AA, Chanani GR, Bohlen JW. Evaluation and application of prealloyed titanium P/M parts for air-frame structures. Int J Powder Metall. 1987;23:171–176.
- [328] Moll JH, Yolton CF. Review and status of titanium materials produced from spherical prealloyed powder. Proceedings of the 1988 International PM Conference ‘Modern Developments in Powder Metallurgy’; 1988; Orlando, FL. Metal Powder Industries Federation. p. 709–726.
- [329] Abouelmagd G, Büchkremer HP, El-Magd E, et al. Mechanical properties of a TiAl6V4 alloy processed by powder metallurgy. J Mater Process Technol. 1993;37:583–597.
- [330] Bozic D, Sekulic D, Stasic J, et al. The influence of microstructural characteristics and contaminants on the mechanical properties and fracture topography of low cost Ti6Al4V alloy. Int J Mater Res. 2008;99:1268–1274.
- [331] Xu L, Guo R, Bai C, et al. Effect of hot isostatic pressing conditions and cooling rate on microstructure and properties of Ti-6Al-4V Alloy from atomized powder. J Mater Sci Technol. 2014;30:1289–1295.
- [332] Bowen AW, Stubbington CA. The effect of heat treatment on the fatigue strength of Ti-6Al-4V. Third International Conference on Titanium; 1976 May 18–21; New York, NY. Plenum; 1982. p. 1989–2001.

- [333] Bowen AW. The effect of testing direction on the fatigue and tensile properties of a Ti-6Al-4V bar. Proceedings of the 2nd International Conference 'Titanium Science and Technology'; 1973 May 2–5; Cambridge, MA. Plenum Press. p. 1271–1281.
- [334] Hall IW, Hammond C. The relationship between crack propagation characteristics and fracture toughness in $\alpha + \beta$ alloys. Proceedings of the 2nd International Conference 'Titanium Science and Technology'; 1973; Cambridge, MA. Plenum Press. p. 1365–1376.
- [335] Thompson AW, Williams JC, Frandsen JD, et al. The effect of microstructure on fatigue crack propagation rate in Ti-6Al-4V. Proceedings of the 3rd International Conference on Titanium; 1976; Moscow, USSR. Springer. p. 691–704.
- [336] Yoder GR, Cooley LA, Crooker TW. Observations on microstructurally sensitive fatigue crack growth in a widmanstätten Ti-6Al-4V alloy. *Metall Trans A*. **1977**;8:1737–1743.
- [337] Margolin H, Mahajan Y. Void formation, void growth and tensile fracture in Ti-6Al-4V. *Metall Trans A*. **1978**;9:781–791.
- [338] Bose K, Birla NC, Goel DB. Some studies on tensile properties and fracture behaviour of Ti-6Al-4V alloy at room and cryogenic temperatures. *Trans Indian Inst Met*. **1983**;36:181–188.
- [339] Ivasishin OM, Savvakina DG, Bielov IS, et al. BEPM synthesis of Ti-6Al-4V alloy using hydrogenated titanium. Proceedings of the Euro PM 2005; 2005; Prague, Czech Republic. European Powder Metallurgy Association. p. 115–120.
- [340] Ivasishin OM, Savvakina DG, Moxson VS, et al. Production of titanium components from hydrogenated titanium powder: optimization of parameters. Proceedings of the 11th World Conference on 'Titanium'; 2007; p. 757–760.
- [341] Joshi VV, Lavender C, Moxon V, et al. Development of Ti-6Al-4V and Ti-1Al-8V-5Fe alloys using low-cost TiH_2 powder feedstock. *J Mater Eng Perform*. **2013**;22:995–1003.
- [342] Yang YF, Luo SD, Qian M. The effect of lanthanum boride on the sintering, sintered microstructure and mechanical properties of titanium and titanium alloys. *Mater Sci Eng A*. **2014**;618:447–455.
- [343] Herteman JP, Eylon D, Froes FH. Mechanical properties of advanced titanium powder metallurgy compacts. Proceedings of the 5th International Conference on 'Titanium'; 1984 Sep 10–14; Munich, Germany. DGM. p. 303–310.
- [344] Kumar P, Chandran KSR. Strength–ductility property maps of powder metallurgy (PM) Ti-6Al-4V alloy: a critical review of processing–structure–property relationships. *Metall Mater Trans A*. **2017**;48:1–19.
- [345] Miura H, Itoh Y, Uematsu T, et al. The influence of density and oxygen content on the mechanical properties of injection molded Ti-6Al-4V alloys. Proceedings of the 2010 International Conference on 'Powder Metallurgy and Particulate Materials'; 2010; Metal Powder Industries Federation. p. 46–53.
- [346] Ebel T, Milagres Ferri O, Limberg W, et al. Metal injection moulding of titanium and titanium-aluminides. *Key Eng Mater*. **2012**;520:153–160.
- [347] Yan M, Dargusch MS, Ebel T, et al. A transmission electron microscopy and three-dimensional atom probe study of the oxygen-induced fine microstructural features in as-sintered Ti-6Al-4V and their impacts on ductility. *Acta Mater*. **2014**;68:196–206.
- [348] Yan M, Xu W, Dargusch MS, et al. Review of effect of oxygen on room temperature ductility of titanium and titanium alloys. *Powder Metall*. **2014**;57:251–257.
- [349] Yoshimura T, Imai H, Thirerujirapong T, et al. Cost effective pure titanium with high mechanical response by oxide dispersion strengthening. *Mater Trans*. **2009**;50(12):2751–2756.
- [350] Jones TL, Kondoh K, Mimoto T, et al. The development of a Ti-6Al-4V alloy via oxygen solid solution strengthening for aerospace & defense applications. *Key Eng Mater*. **2013**;551:118–126.
- [351] Sun B, Li S, Imai H, et al. Fabrication of high-strength Ti materials by in-process solid solution strengthening of oxygen via P/M methods. *Mater Sci Eng A*. **2013**;563:95–100.
- [352] Yan M, Liu Y, Liu YB, et al. Simultaneous gettering of oxygen and chlorine and homogenization of the β phase by rare earth hydride additions to a powder metallurgy Ti-2.25Mo-1.5Fe alloy. *Scr Mater*. **2012**;67:491–494.
- [353] Yan M, Liu Y, Schaffer GB, et al. In situ synchrotron radiation to understand the pathways for the scavenging of oxygen in commercially pure Ti and Ti-6Al-4V by yttrium hydride. *Scr Mater*. **2013**;68:63–66.
- [354] Yan M, Luo SD, Schaffer GB, et al. Impurity (Fe, Cl, and P)-induced grain boundary and secondary phases in commercially pure titanium (CP-Ti). *Metall Mater Trans A*. **2013**;44:3961–3969.
- [355] Yan M, Dargusch MS, Kong C, et al. In situ synchrotron radiation study of TiH_2 -6Al-4V and Ti-6Al-4V: accelerated alloying and phase transformation, and formation of an oxygen-enriched $\text{Ti}_4\text{Fe}_2\text{O}$ Phase in TiH_2 -6Al-4V. *Metall Mater Trans A*. **2015**;46:41–45.
- [356] Friedman GI. Titanium powder metallurgy. *Int J Powder Metall*. **1970**;6:43–54.
- [357] Kumar P. Processing and mechanical behavior of powder metallurgy (PM) Ti-6Al-4V alloy made from TiH_2 [Thesis]. Salt Lake City (UT): University of Utah; **2016**.
- [358] Kumar P, Ravi Chandran KS, Sun P, et al. Effects of powder processing and sintering on tensile ductility of PM Ti-6Al-4V alloy made by hydrogen sintering of titanium hydride powders. Proceedings of the 13th World Conference on 'Titanium'; 2015; San Diego, CA. John Wiley & Sons. p. 1355–1360.
- [359] Kumar P, Ravi Chandran KS, Cao F, et al. The nature of tensile ductility as controlled by extreme-sized pores in powder metallurgy Ti-6Al-4V alloy. *Metall Mater Trans A*. **2016**;47:2150–2161.
- [360] Eylon D, Vogt RG, Froes FH. Fatigue improvement of prealloyed and blended elemental titanium powder metallurgy compacts by microstructure modifications. Proceedings of the International PM Conference 'Modern Developments in Powder Metallurgy'; 1985; Metal Powder Industry. p. 563–575.
- [361] Yan Y, Nash GL, Nash P. Effect of density and pore morphology on fatigue properties of sintered Ti-6Al-4V. *Int J Fatigue*. **2013**;55:81–91.
- [362] Chait R, DeSisto TS. The influence of grain size on the high cycle fatigue crack initiation of a metastable β Ti alloy. *Metall Mater Trans A*. **1977**;8:1017–1020.
- [363] Cao F, Kumar P, Koopman M, et al. Understanding competing fatigue mechanisms in powder metallurgy

- Ti-6Al-4V alloy: role of crack initiation and duality of fatigue response. *Mater Sci Eng A*. [2015](#);630:139–145.
- [364] Cao F, Ravi Chandran KS. Fatigue performance of powder metallurgy (PM) Ti-6Al-4V alloy: a critical analysis of current fatigue data and metallurgical approaches for improving fatigue strength. *JOM*. [2016](#);68:735–746.
- [365] Lee YT. Deformation behaviour of thermomechanically treated PREP Ti-6Al-4V alloy. *Zeitschrift fuer Metallkunde*. [1987](#);78:49–57.
- [366] ASTM International. ASTM E466-15: standard practice for conducting force controlled constant amplitude axial fatigue tests of metallic materials. 2015.